CSE 407
Mobile and Wireless Communication System

Text Book:

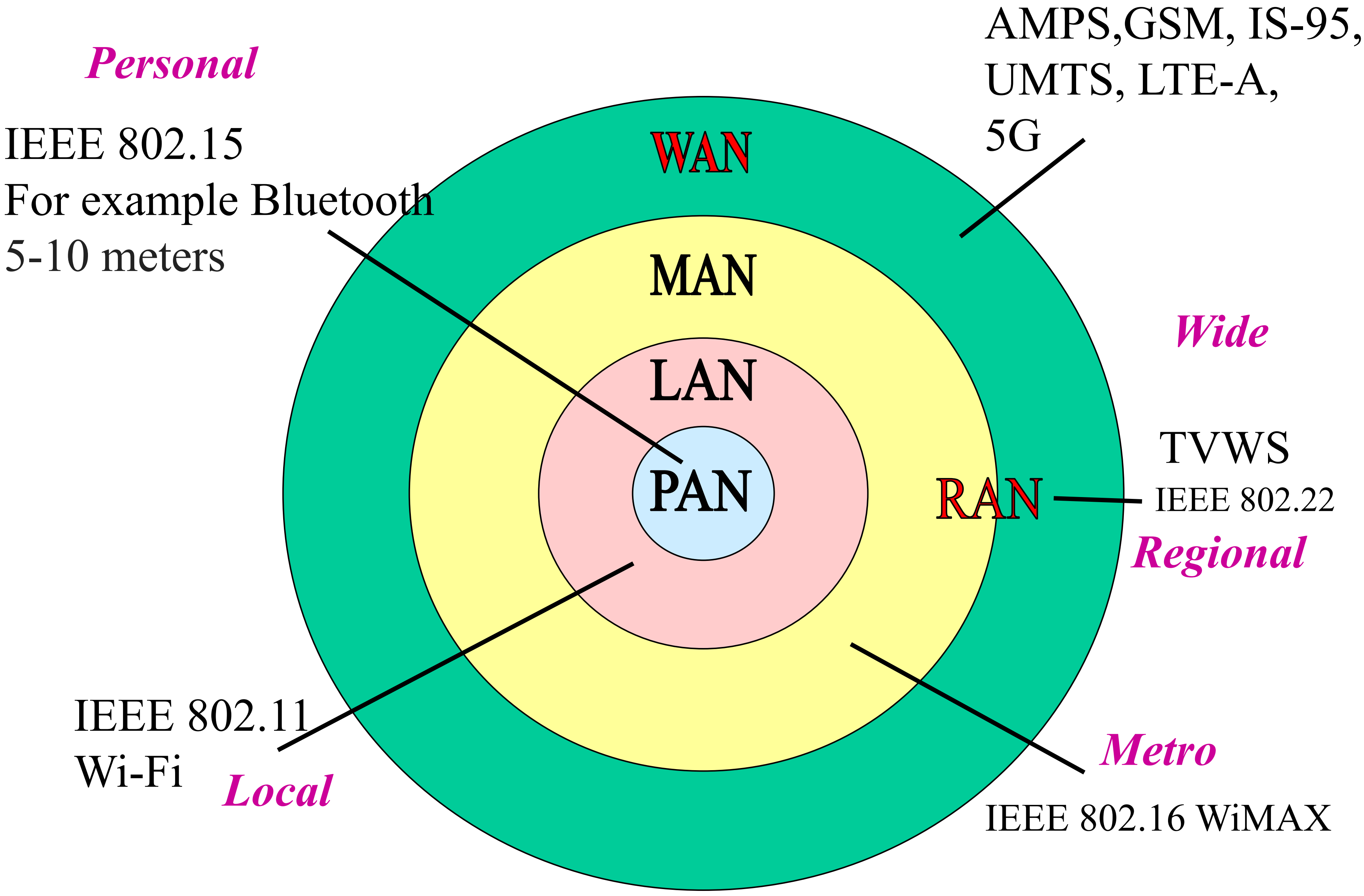
Computer and Communication Networks, Nader F. Mir, Pearson Education

5G Wireless Systems: Simulation and Evaluation Techniques (Wireless Networks), Yang Yang, Jing Xu, Guang Shi, Cheng-Xiang Wang, Springer

Fundamentals of WiMAX: Understanding Broadband Wireless Networking, Jeffry G. Andrew, Arunabha Ghosh, Rias Muhamad, Pearson Education

Reference Book:

An Introduction to LTE: LTE, LTE‑Advanced, SAE and 4G Mobile Communications, Christopher Cox, John Wiley & Sons, Ltd

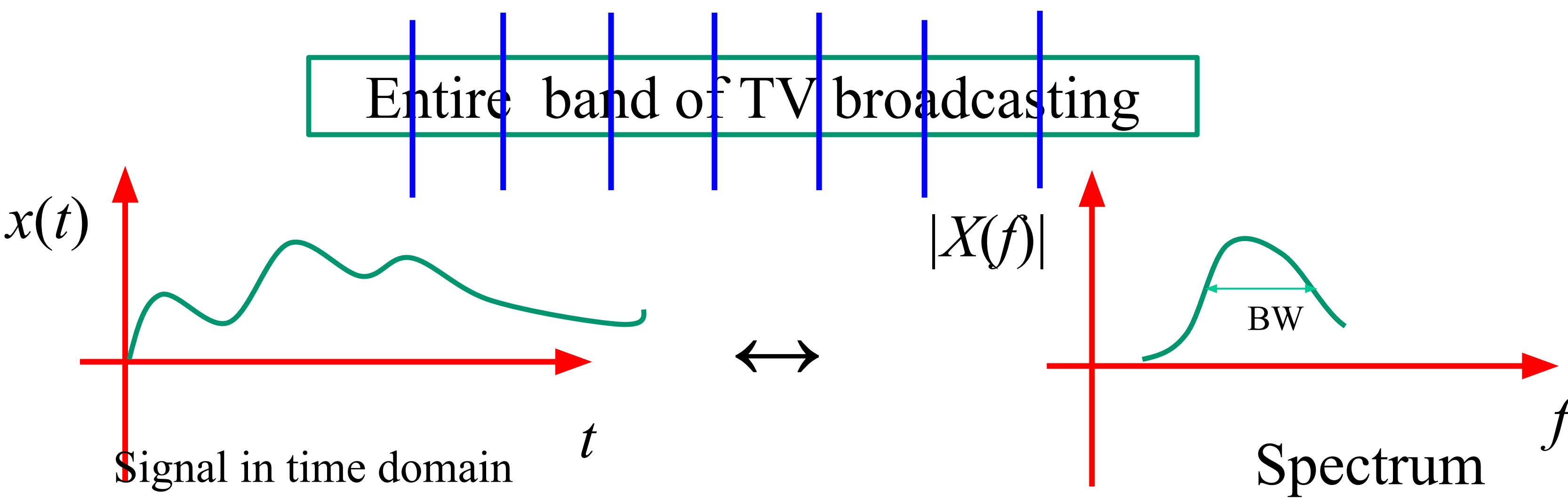


IEEE 802 LAN/MAN Standards Committee. TV White Spaces (**TVWS**) are frequencies made available for unlicensed use.

IEEE 802.22 WRAN

- ✔ **IEEE 802.22**, is a standard for **wireless regional area network (WRAN),** which uses white spaces (vacant TV channels) in the television (TV) frequency spectrum (in the VHF and UHF bands).

- ✔ The development of the IEEE 802.22 WRAN standard is aimed at using **Cognitive Radio** (CR) techniques to allow sharing of geographically unused spectrum allocated to the television broadcast service.



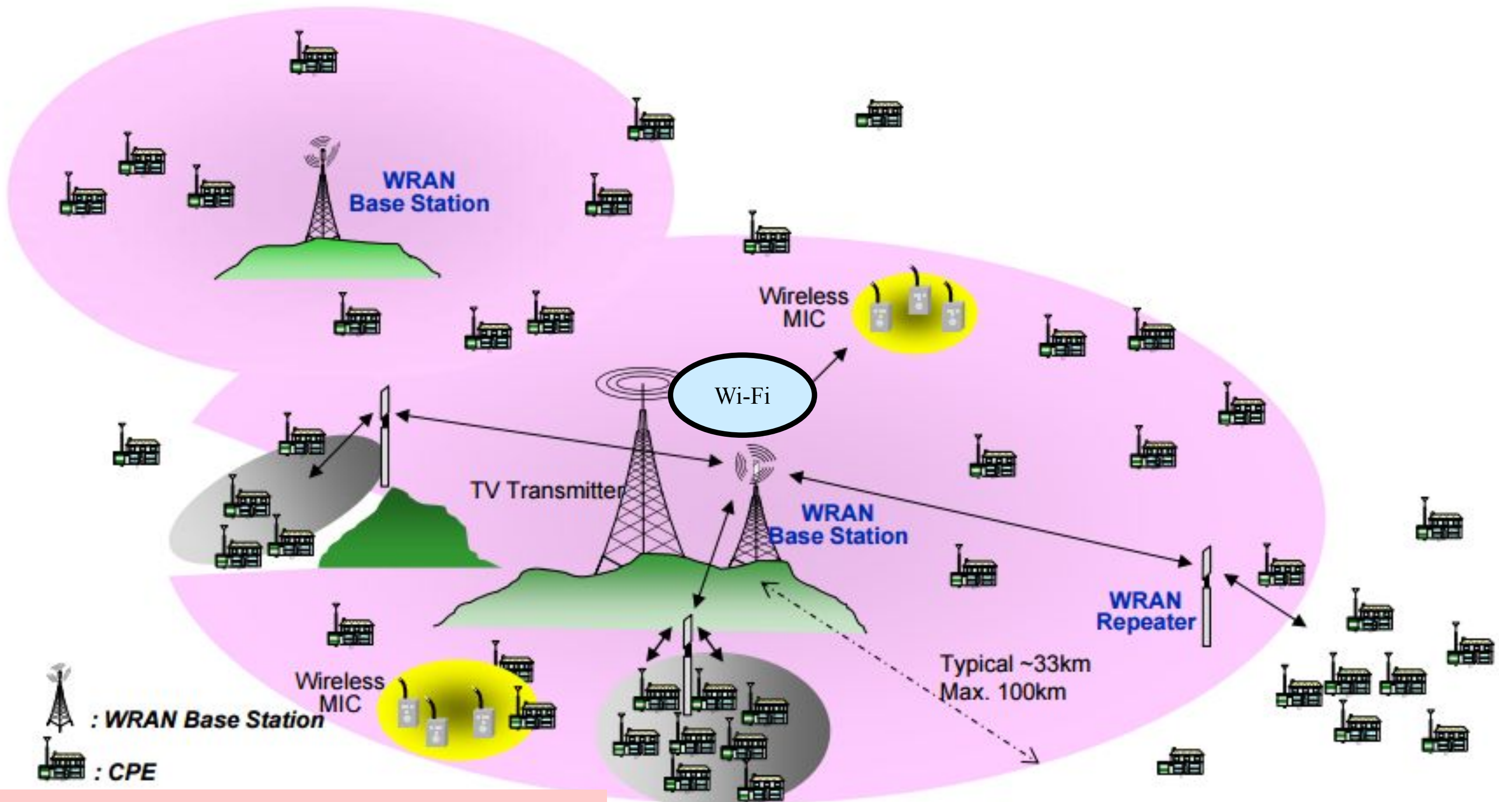
IEEE 802.22 WRAN

- **Visual Explanation:**
- The left graph shows a **signal in the time domain** $x(t)$.
- The right graph shows the corresponding **spectrum in the frequency domain** $|X(f)|$.
- The **entire TV broadcasting band** is marked with vertical lines, showing available frequencies.
- The shaded area under the spectrum graph (labeled **BW**) represents a bandwidth that could be used by WRAN, as it's **not currently occupied**.

IEEE 802.22 WRAN

- **Purpose of the Standard:**
- The goal of IEEE 802.22 is to enable **broadband access** in **rural and remote areas**.
- It uses **Cognitive Radio (CR)** technology, which:
  - Detects unused frequency bands.
  - Allows **dynamic spectrum access** by sharing TV spectrum **without interference**.
- **Cognitive Radio Concept:**
- Cognitive radios scan the spectrum to **identify unused channels**.
- Then, they **dynamically use** those channels for wireless communication, **avoiding interference** with primary users (TV broadcasters).

IEEE 802.22 WRANs are designed to operate in the TV broadcast bands while assuring that no harmful interference is caused to the other networks: digital TV and analog TV broadcasting, low power licensed/unlicensed devices such as wireless microphones (MIC), WLAN etc.



Customer-Premises Equipment (CPE)

Diagram Explanation:

- The illustrated network architecture includes:
- **TV Transmitters** – traditional broadcast sources.
- **WRAN Base Stations** – central nodes in a WRAN system.
- **CPE (Customer-Premises Equipment)** – end-user equipment that connects to the base station (highlighted in red).
- **Wi-Fi Access Points** – show co-existence with other networks like WLAN.
- **Wireless MICs** – indicate other vulnerable services that need protection from interference.
- **Modulated Signal & Modem** – indicate the signal flow and end-user reception.

Cognitive Radio Network (CRN)

In cognitive radio network there are two types of users:

(a) Primary users (licensed user/ paid user)

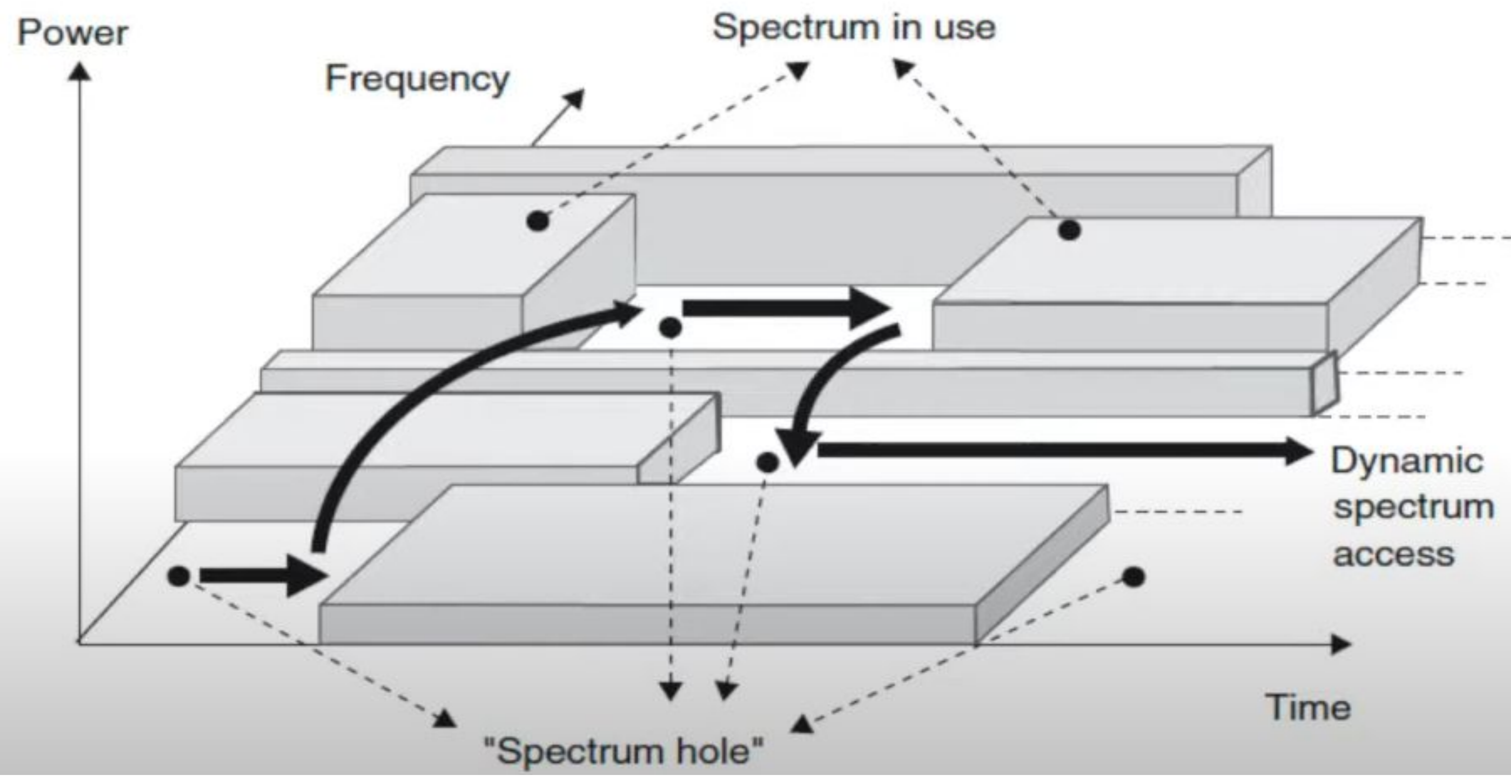
(b) Secondary users (unlicensed user/ nonpaying user).

The PU can access any channel in case of its availability, but SU users have lower priority and can access a channel only when it is unused by PUs. A secondary user (SU) searches for free channels and communicates using free channels when available. When PU arrives to use that channel, SU instantly release the channel.

Cognitive Radio Network (CRN)

- A key requirement for cognitive radio is spectrum sensing, i.e., detecting the presence of other users
- **Spectrum sensing** : the cognitive radio has to identify which parts of the time–frequency plane are not used by primary users

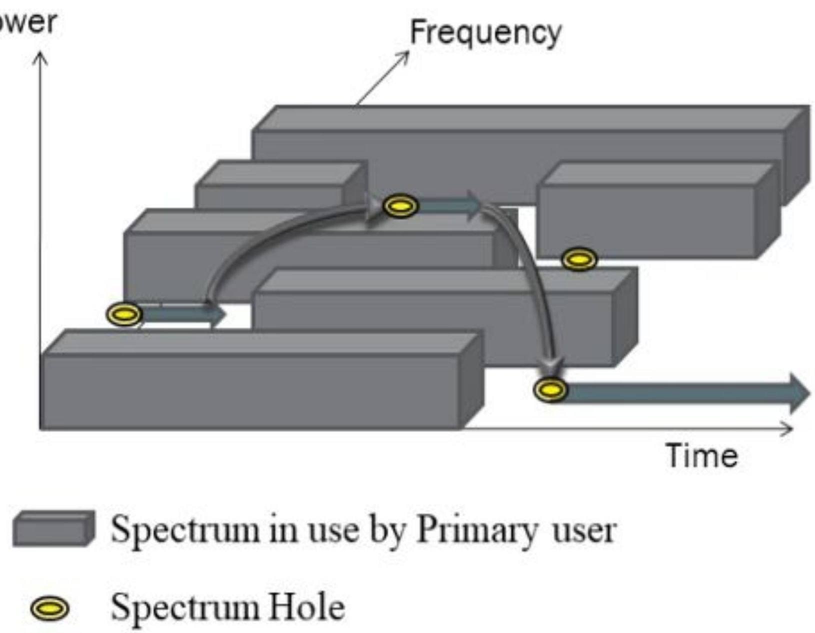
Cognitive Radio Network (CRN)



Cognitive Radio Network (CRN)

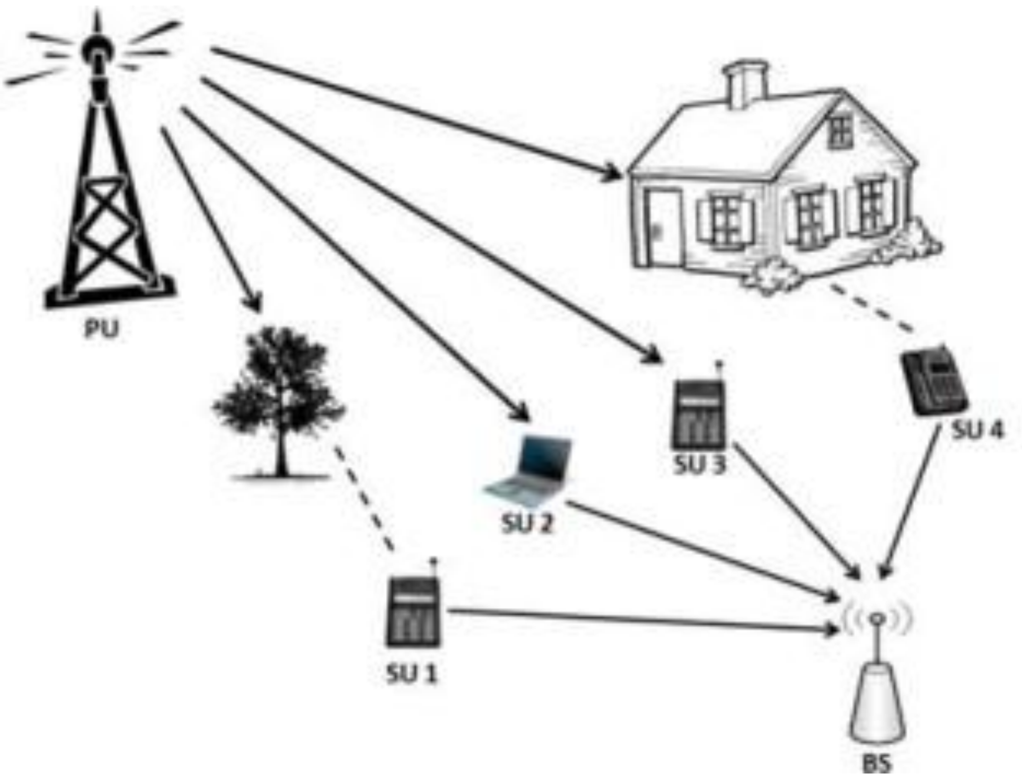
SPECTRUM HOLE / WHITE SPACE

- A spectrum hole is a band of frequencies assigned to a primary user, but, at a particular time and specific geographic location, the band is not being utilized by that user.



The spectrum sensing of SU are analyzed using two hypothesis like below,

H_0: PU does not exist.
H_1: PU does exist



Cognitive radio network (CRN)

The received signal for the *i*-th SU can be expressed as follows:

$$y(t) = \begin{cases} n(t); & H_0 \\ s(t) + n(t); & H_1 \end{cases}$$

, where $s(t)$ is the transmitted signal and $n(t)$ is the noise.

$$y(t) = \begin{cases} r(t) < \tau; & H_0 \\ r(t) \geq \tau; & H_1 \end{cases}$$

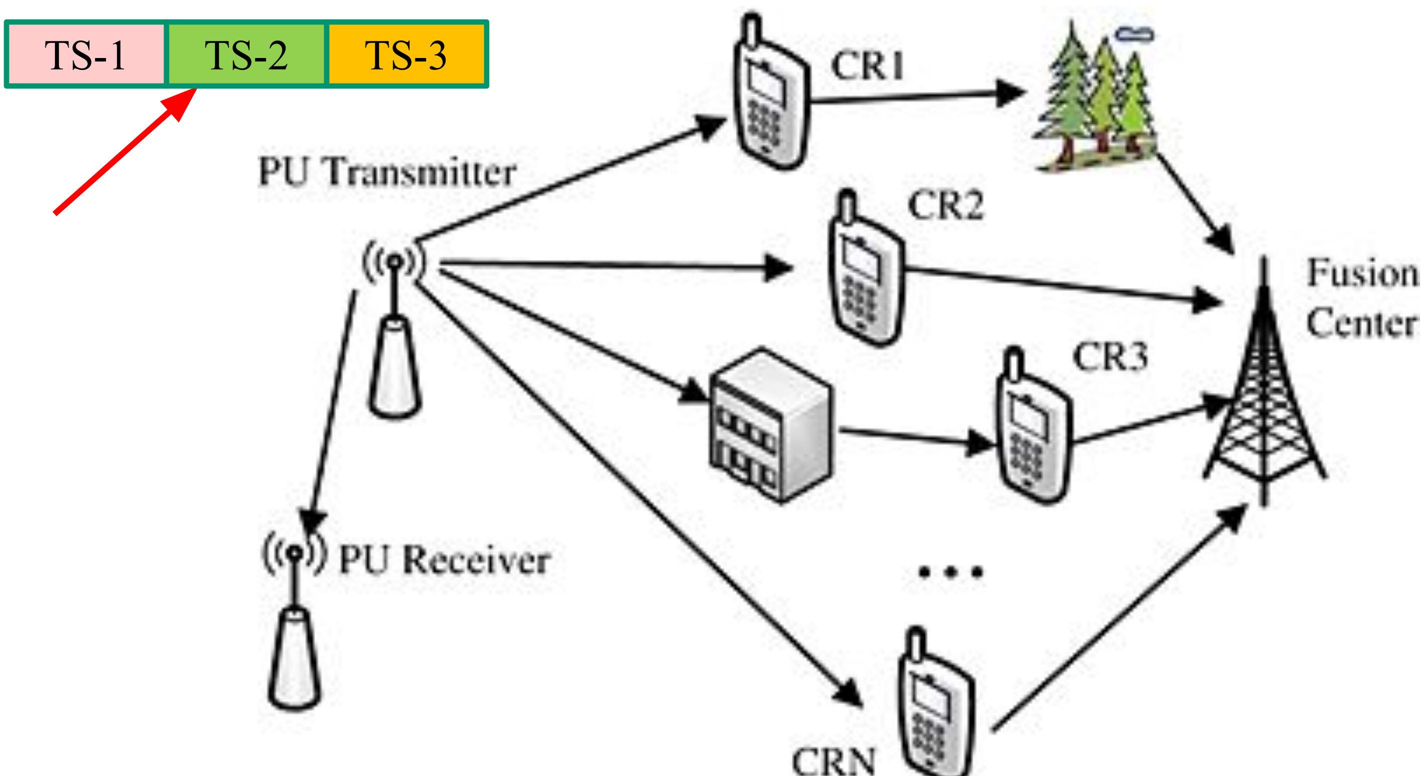
- ✔ Consider a situation when there is no PU in a physical channel but the SU senses the presence of PU, the phenomenon is called **false alarm**.

- ✔ Under presence of a PU on physical channel, if the received signal strength is found lower than the threshold value, then the phenomenon is called **misdetection**.

$$y(t) = \begin{cases} r(t) \geq \tau; & \text{False Alarm} \\ r(t) < \tau; & \text{Misdetection} \end{cases}$$

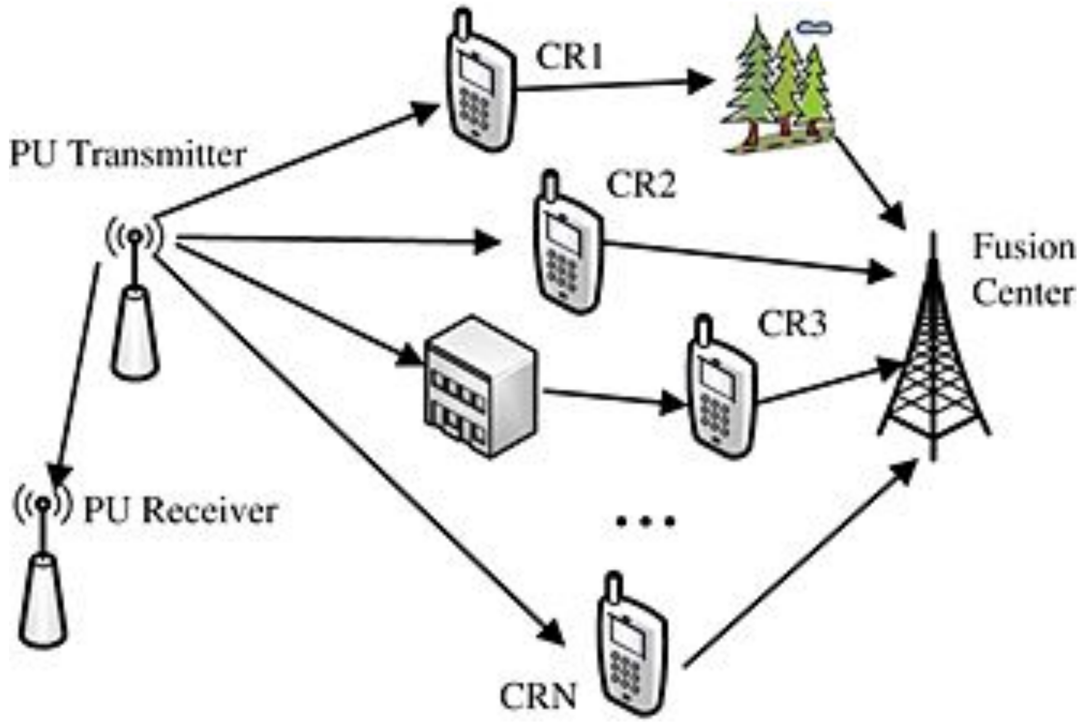
Cooperative Spectrum Sensing

✔ In cooperative spectrum sensing scenario where N CRs can be coordinated to enhance the performance of spectrum sensing as a whole. In cooperative spectrum sensing, each CR detects the PU independently, as shown in Figure below.



- ✔ Since the detection results of CR1 and CR3 are inaccurate because of the shadowing brought by the high buildings, we let the other CRs in a better environment, such as CR2 and CR*N*, help CR1 and CR3 for improving their detection performance by sharing sensing information.

- ✔ The fusion center functions as a base station, which combines the detection results from all the CRs to get the final decision on the presence of the PU by a specific fusion rule, such as AND Logic, OR Logic, Majority scheme etc.



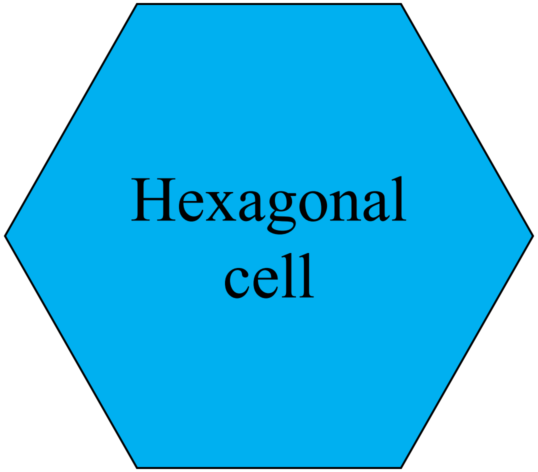
WAN (Mobile Cellular Network)

A wide area network (WAN) provides long-distance transmission of data, image, audio, and video information over large geographic areas that may comprise a country, a continent, or even the whole world. Mobile cellular network is the wireless WAN.

- Break the metropolitan area into small areas.
- Each area is approximated with a hexagonal ***cell***.
- A ***Radio Base Station*** **(RBS)** is located at the center of each cell.
- Each cell is assigned only a fraction of the total number of channels.
- Cells that are sufficiently far apart can reuse the same carrier frequency.

Cell and cell cluster

Concept of mobile cellular communication system has arises from cell. Cell is the unit coverage of mobile cellular network. The hexagonal cells (like honey cell) are arranged in a continuous fashion to cover the entire service are.

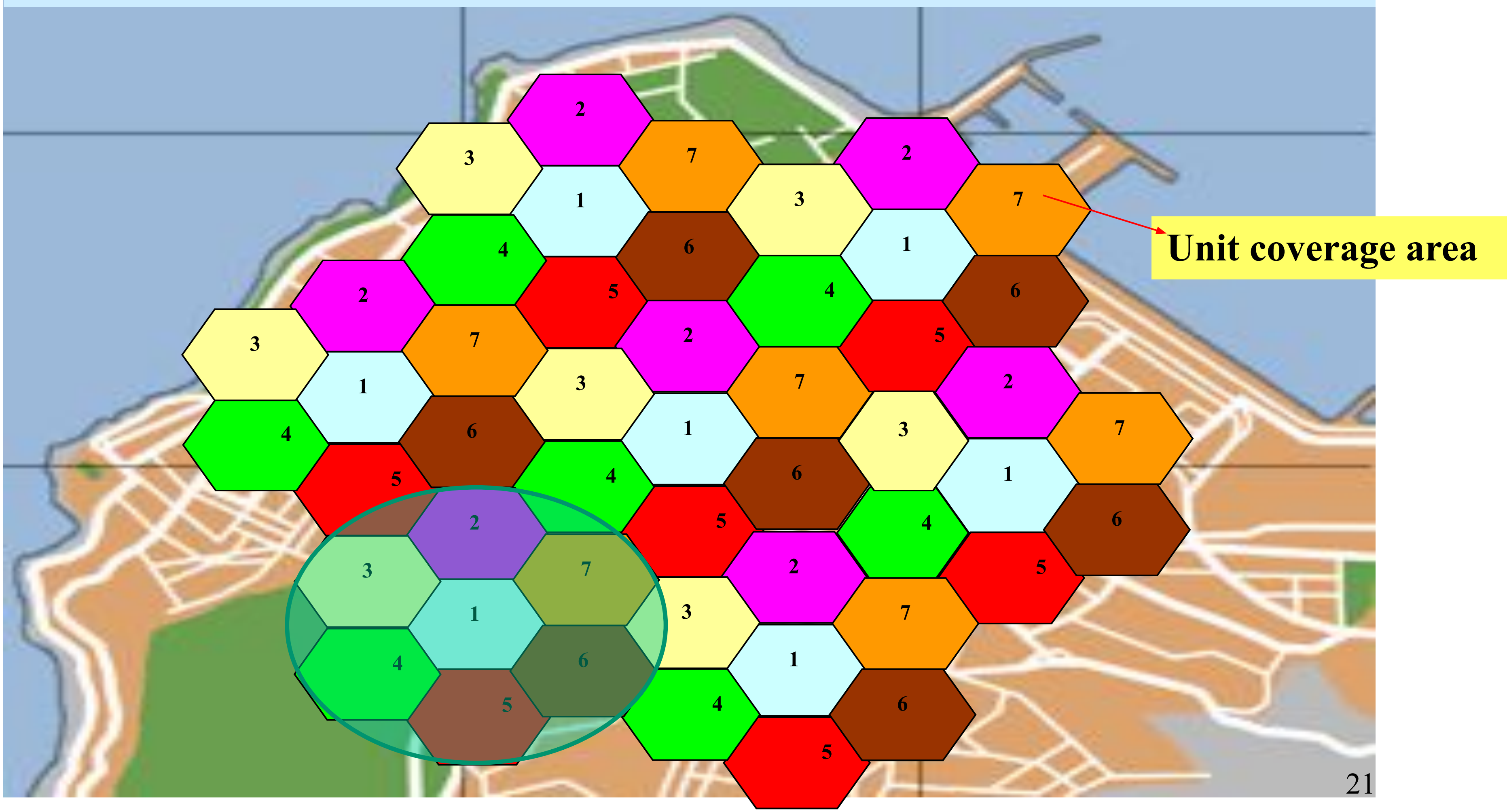


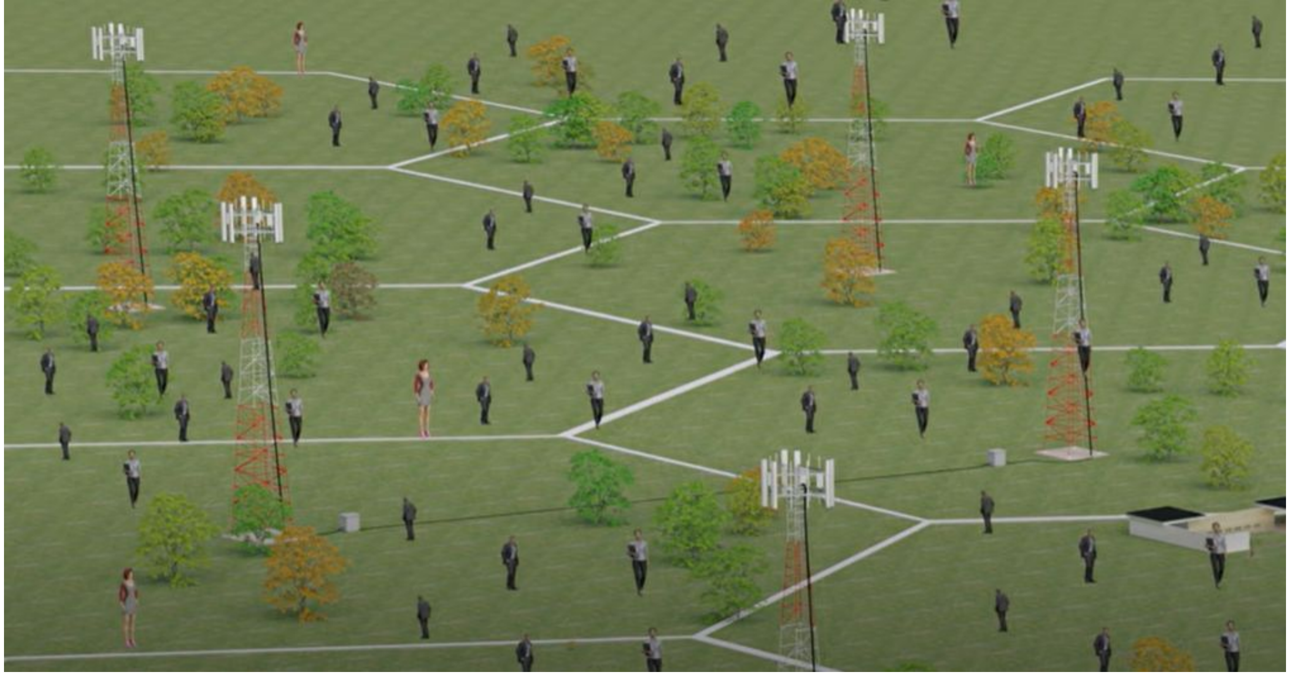
Cell cluster/ pattern

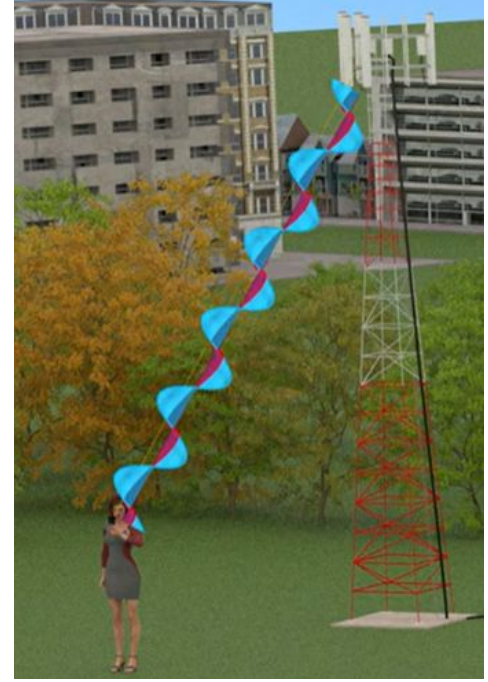
For example, licensed BW =10 MHz (up or down link)

In GSM separation between adjacent channels is 200 KHz

Number of channe*ls*, $n = (10*10^6)/(200*10^3) = 10000/200 = 50$

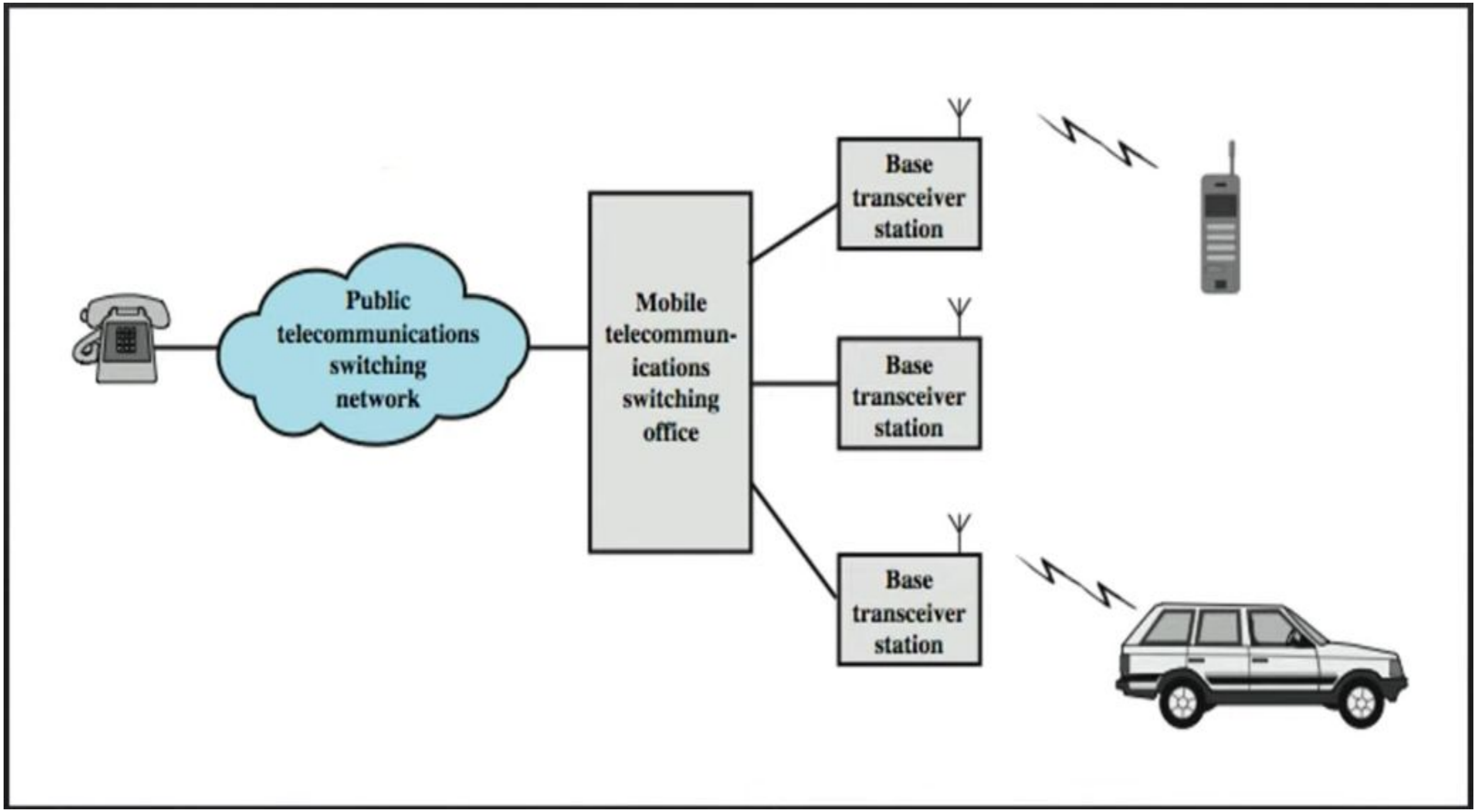






- ✔ Every user under the coverage of the network is within a particular cell under a RBS.
- ✔ Several RBS/BTS are connected to a BSC (Base Station Controller)
- ✔ Several BSC are connected to MSC (Mobile Switching Center)

- ✔ User Device ↔ RBS (radio access)
- ✔ RBS ↔ BSC (resource management)
- ✔ BSC ↔ MSC (switching and routing)



Overview of cellular system

RBS (Radio Base Station) or BTS (Base Transceiver Station):

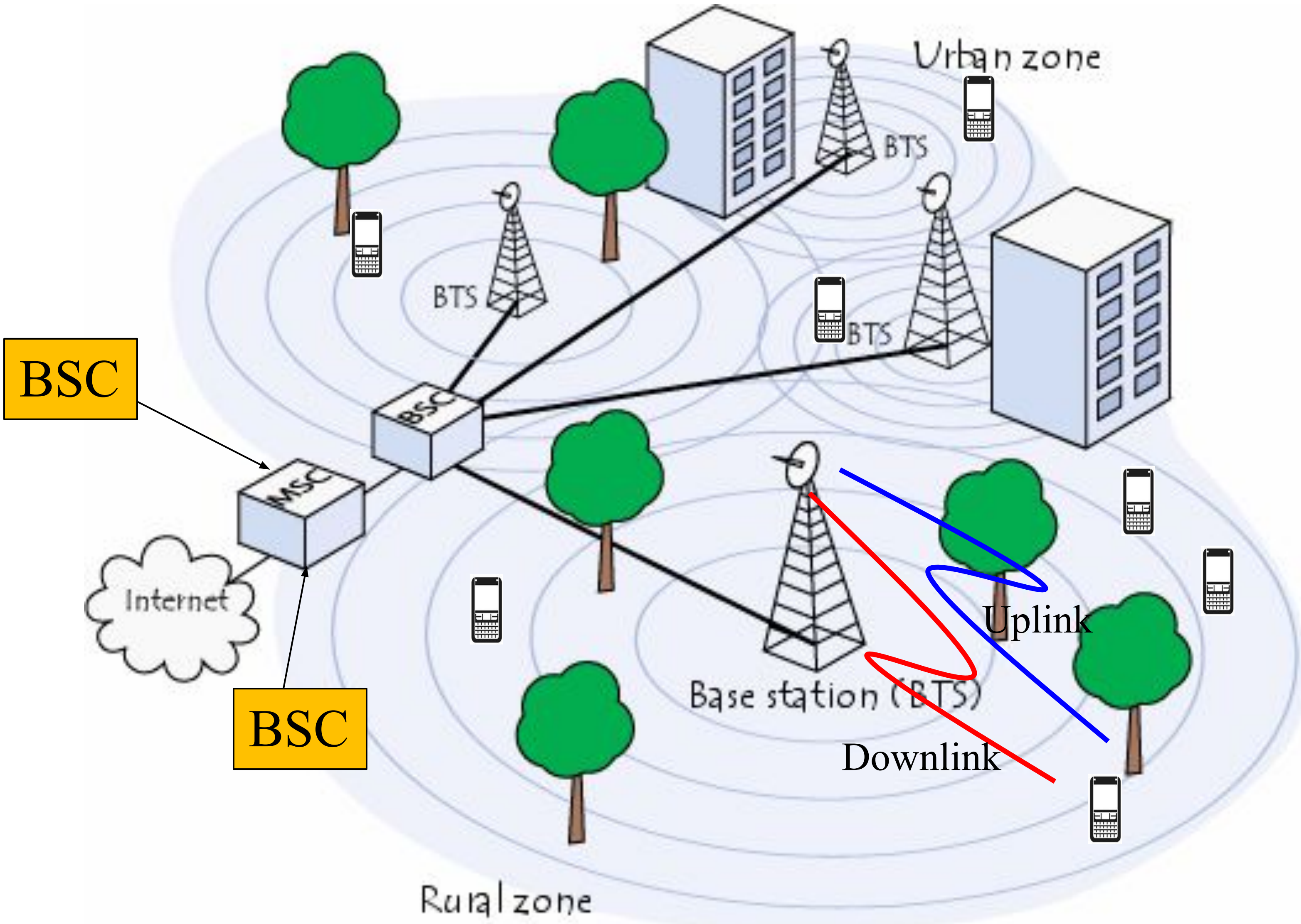
- Each cell in the network is served by an RBS/BTS.
- It provides the radio interface for communication between the mobile device and the network.
- Located in cell towers or rooftop sites.

BSC (Base Station Controller):

- Manages multiple RBSs.
- Controls handovers, frequency assignment, and power levels
- Acts as a middle manager between the RBS and MSC.

MSC (Mobile Switching Center):

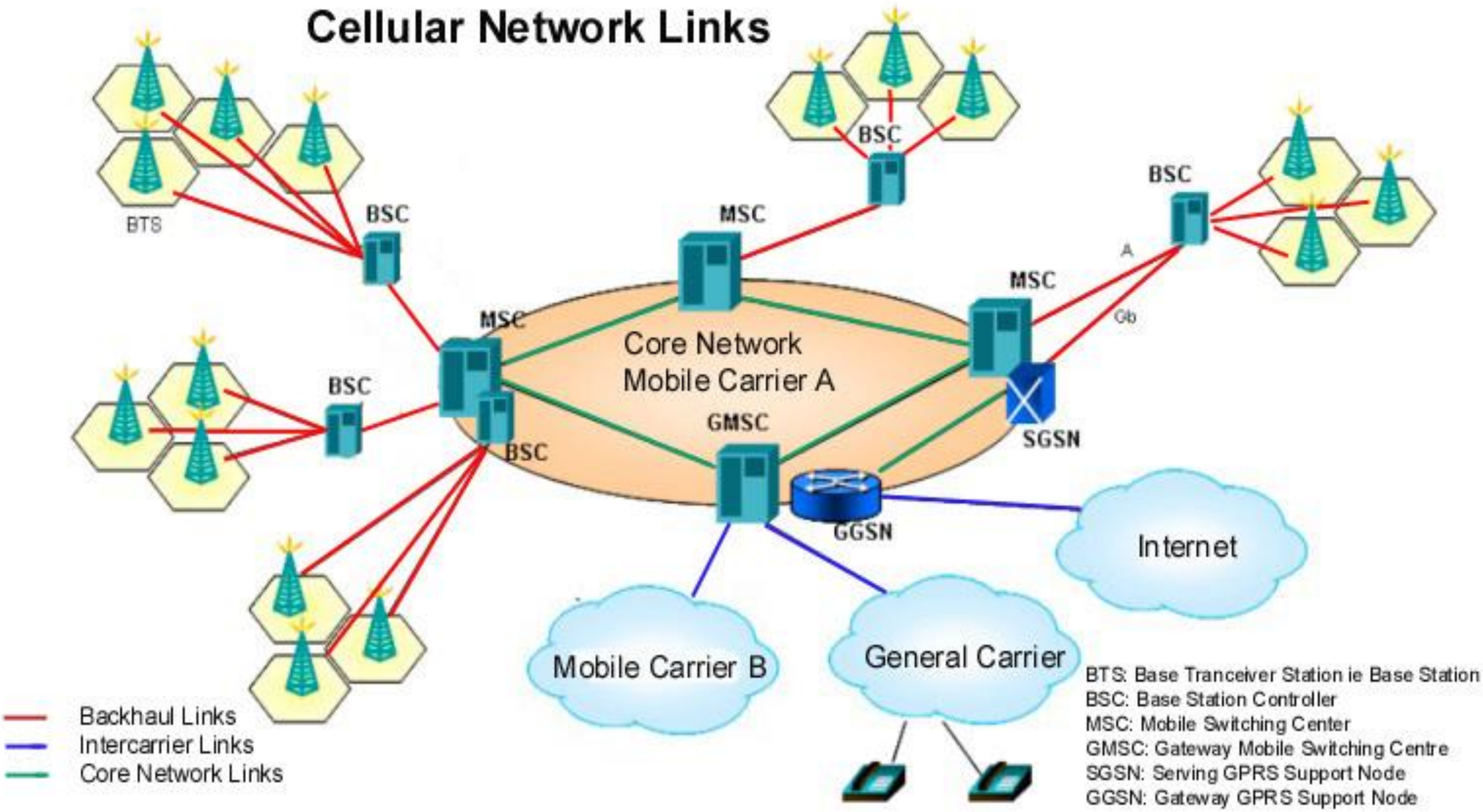
- Connects several BSCs.
- Handles call routing, SMS, mobility management, handover across BSCs, and connection to other networks (like PSTN or the internet).



Mobile cellular network

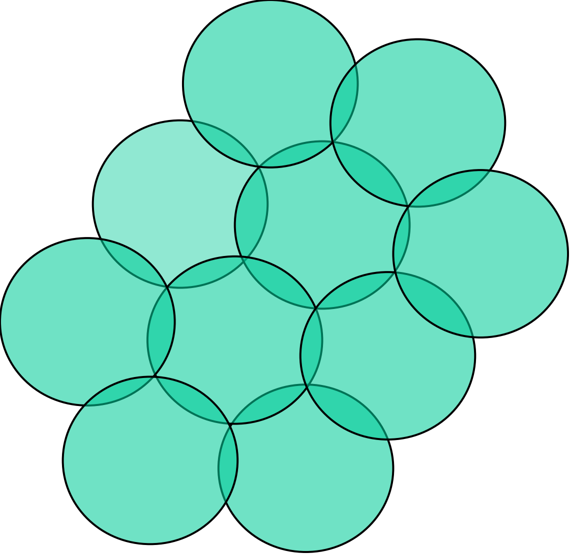
BTS→ Base Transreceiver Station
BSC → Base Station Controller
MSC → Mobile Switching Center

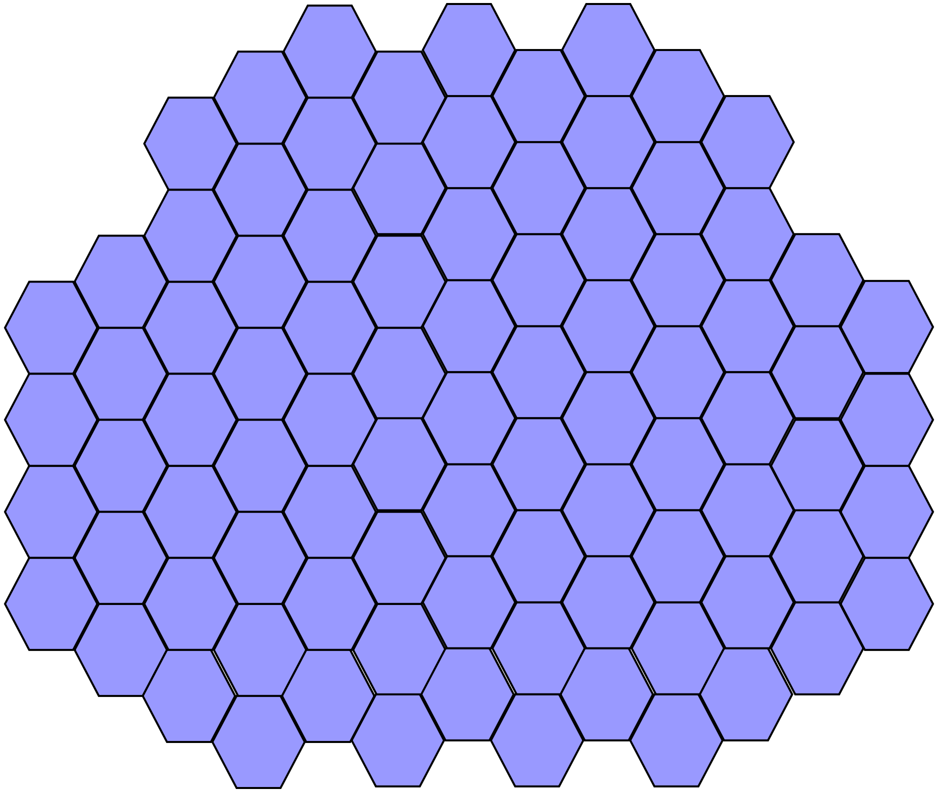
RBS → BTS → Node B (NB) →evolve Node B (eNB) →Next generation Node B(gNB)



Mobile cellular network

- **Link Types:**
- **Red lines = Backhaul Links**
  Connect BTS → BSC → MSC (used to carry traffic/data from cell towers back to the core network)
- **Green lines = Core Network Links**
  Connect internal elements of **Mobile Carrier A**'s core (MSC, GMSC, SGSN, GGSN)
- **Blue lines = Intercarrier Links**
  Connect **Mobile Carrier A** to **other carriers** (Carrier B, General Carrier, and Internet)





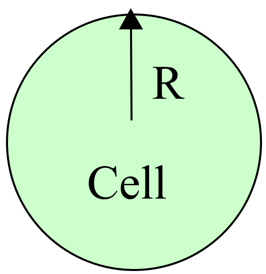
Hexagonal Cells in a continuous fashion

Why Hexagonal Cell?

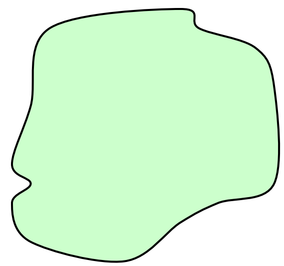
- ✔ Hexagonal-shaped communication cells are artificial and that such a shape cannot be generated in the real world. Engineers draw hexagonal-shaped cells on a layout to simplify the planning and design of a cellular system because it approaches a circular shape that is the ideal power coverage area. The circular shapes have overlapped areas which make the drawing unclear.

- ✔ The hexagonal shaped cells fit the planned area nicely with no gap and on overlap between the cells. The real cell has irregular shape because of natural, and man made obstacle in service area.

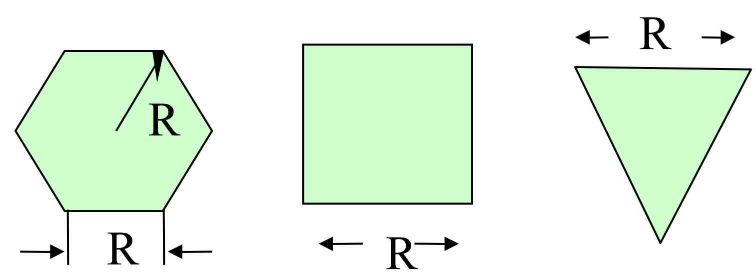
Cell Shape



(a) Ideal cell

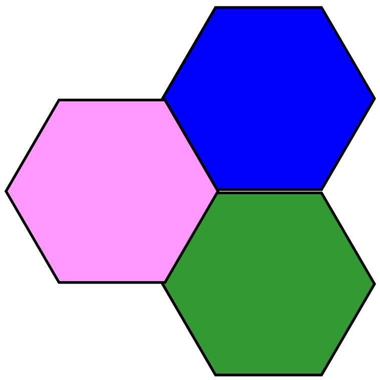


(b) Actual cell

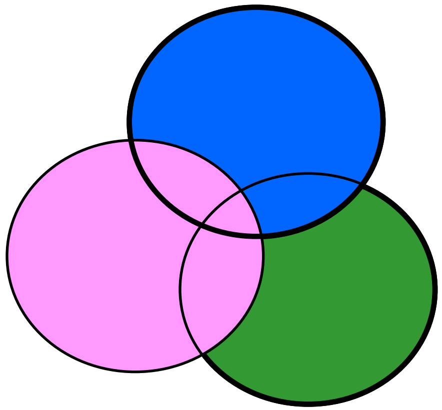


(c) Different cell models

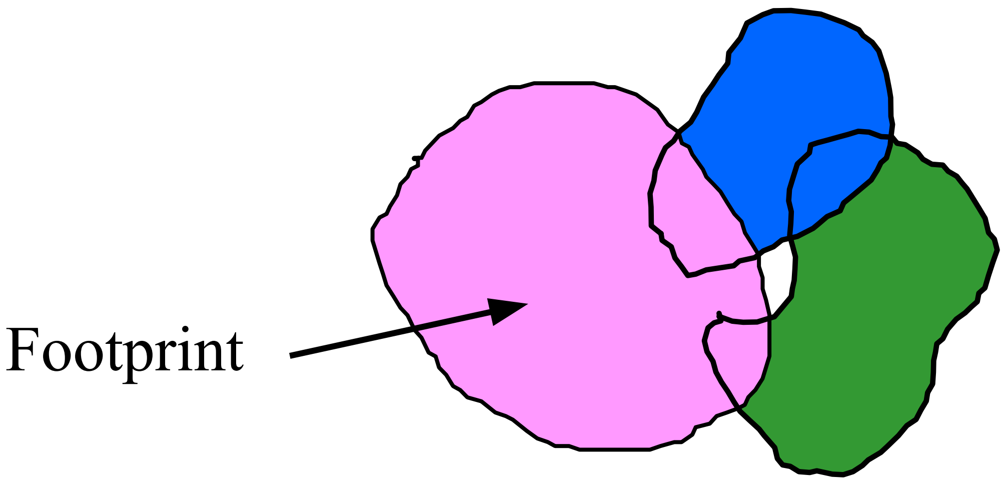
Why hexagonal cell is used by a network planar?



Factitious



Idealized Coverage



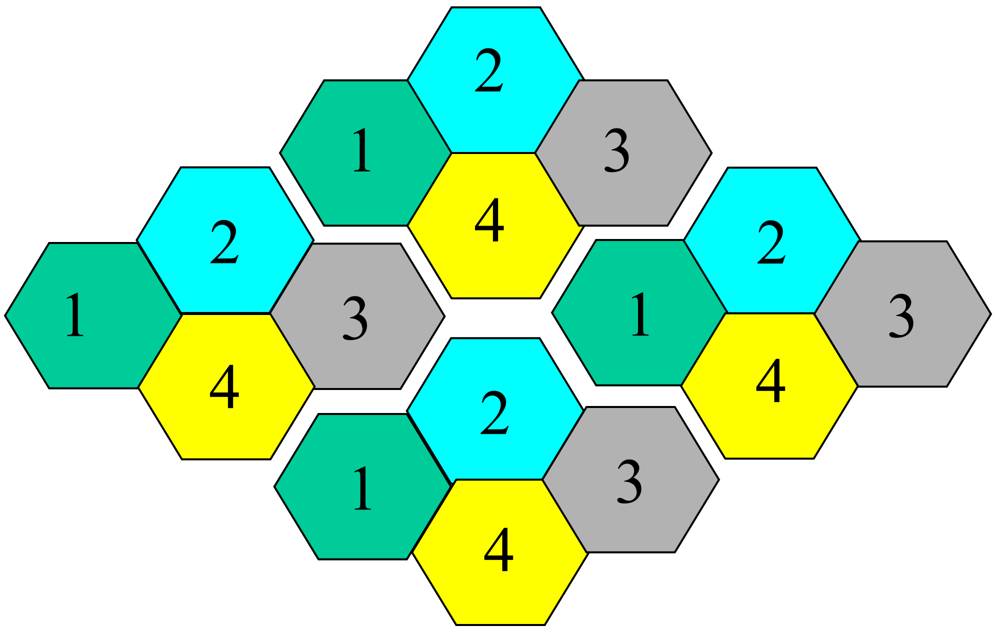
Reality!

Cell Cluster/ Pattern

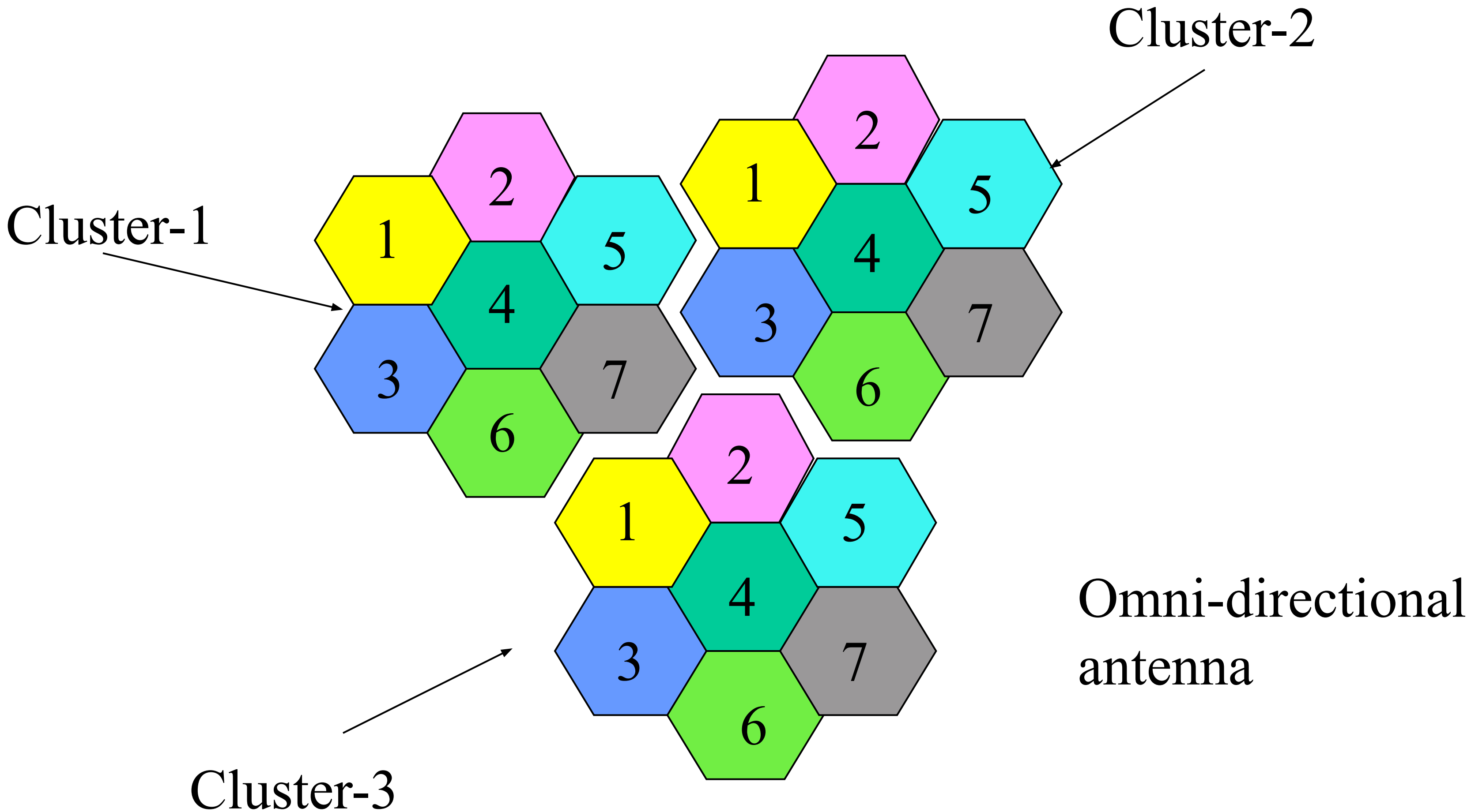
- ✔ Cell is the unit coverage area of WAN. Cells in a cellular network are generally "grouped" together into cell clusters. Cellular networks are generally designed as a repeated cluster pattern.

- ✔ The number of cells in a cluster (typically 4,7, 12 or 21) is a trade-off between the traffic capacity in the cluster and its interference with the adjacent cells and co-channel cells (where the same frequencies will be re-used).

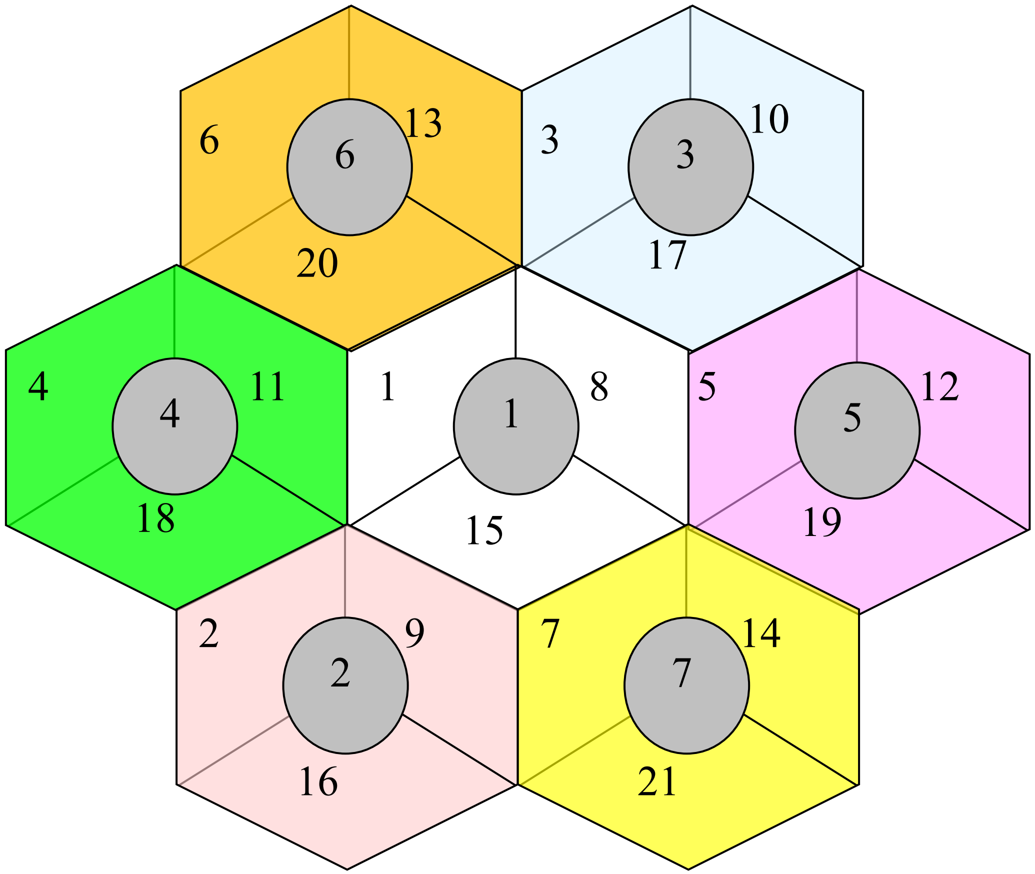
Cell Pattern $N = 4$



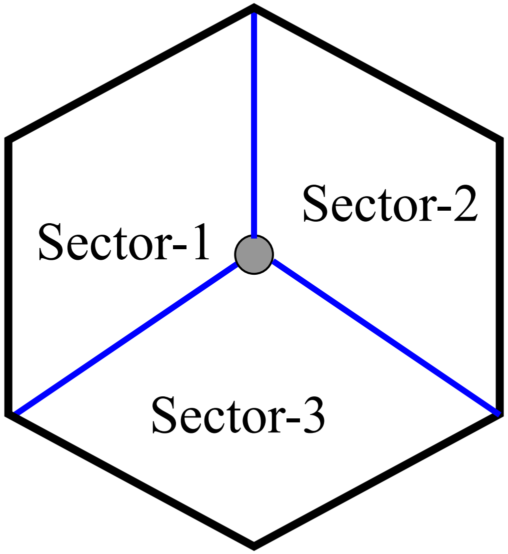
Cell Pattern $N = 7$



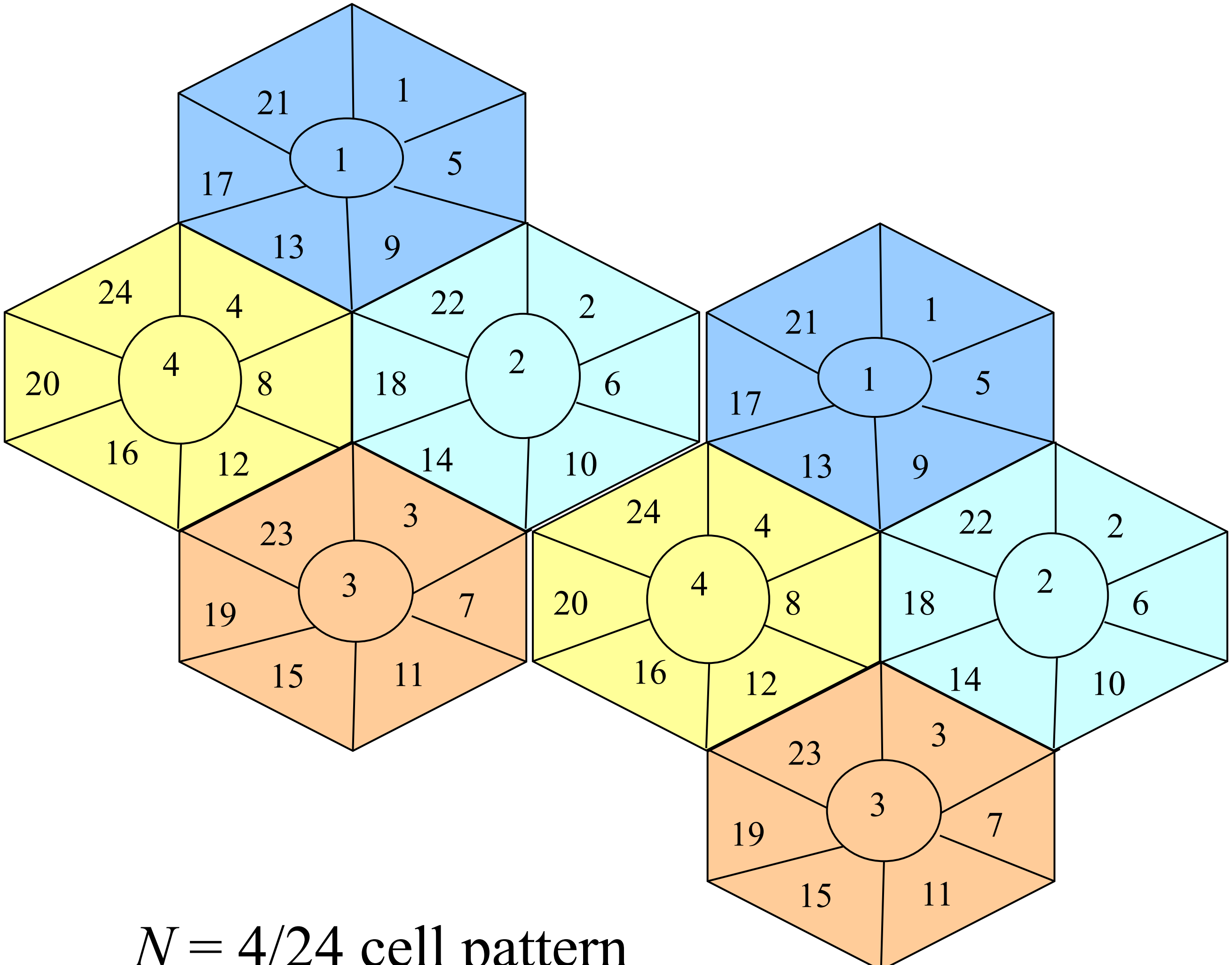
In cell sectoring scheme a cell is divided into three sectors of 120 deg each like fig. below, where each sector is considered as an OMNI cell. Each sector contains one or more control channels and several voice channels. Three directional antennas are used at the center of the cell with 120 deg apart to illuminate each sector.



$N = 7/21$ cell pattern

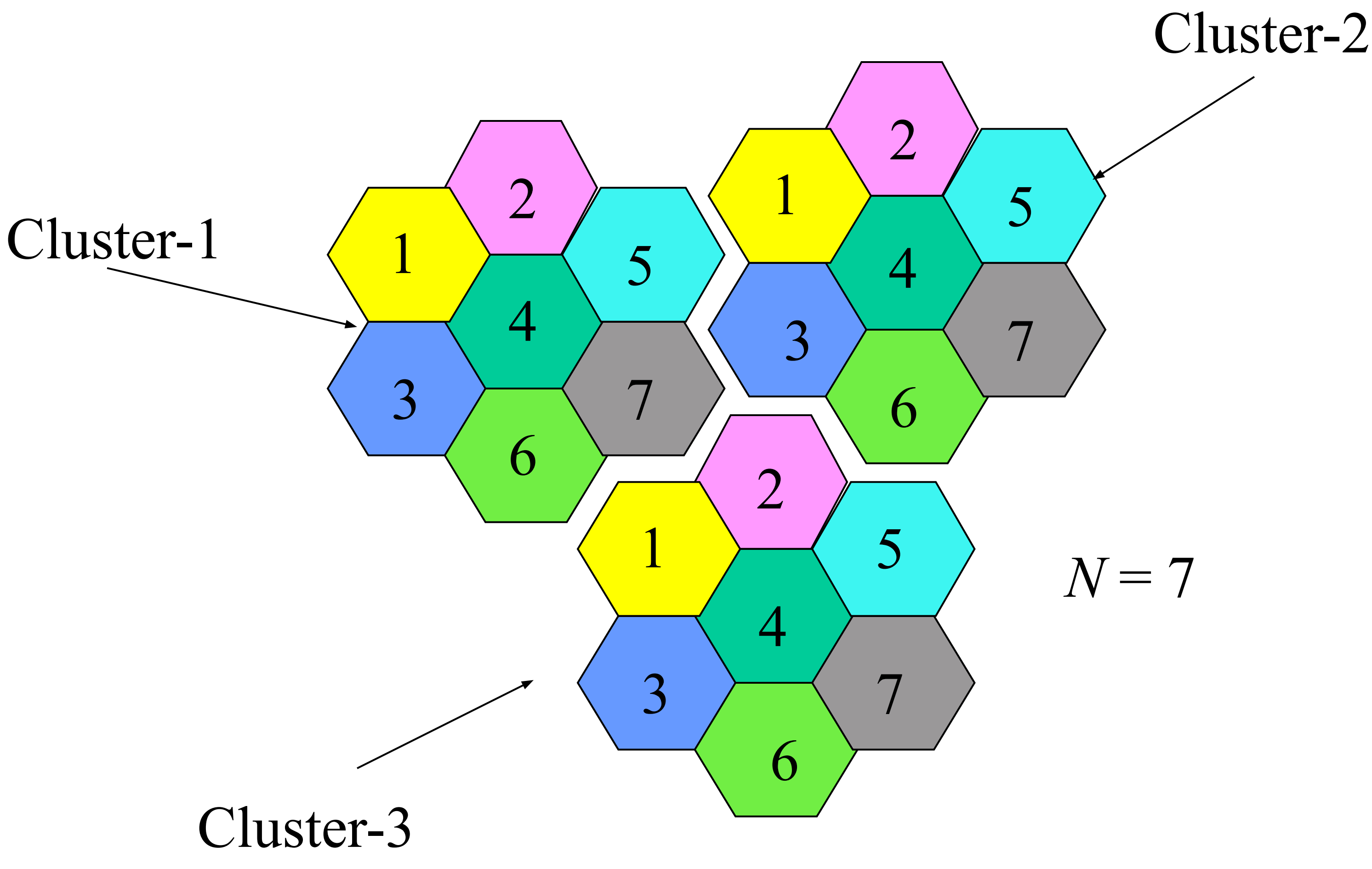


Directional antenna

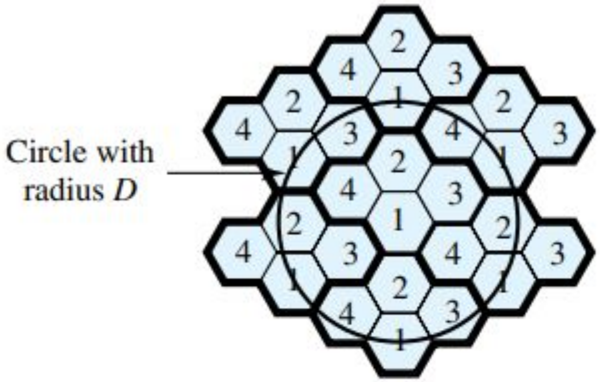


N = 4/24 cell pattern

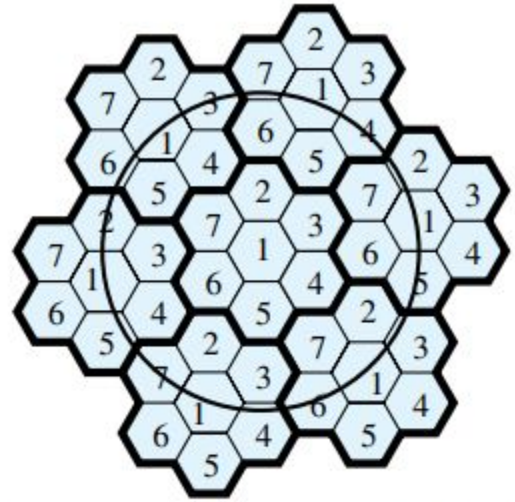
Frequency Reuse



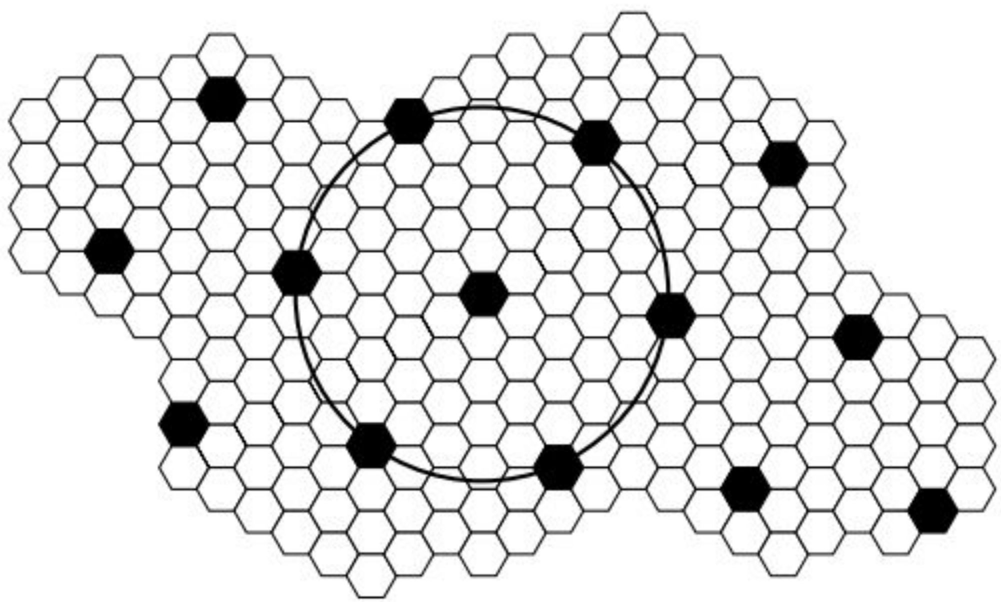
$N = 7$



(a) Frequency reuse pattern for $N = 4$



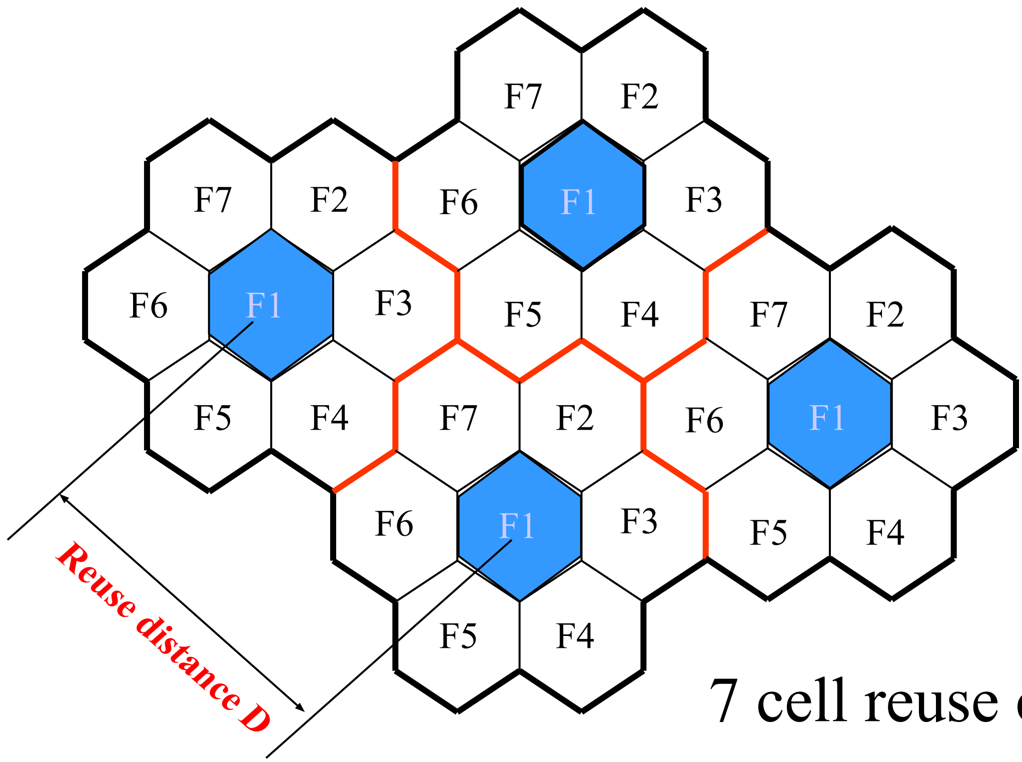
(b) Frequency reuse pattern for $N = 7$



(c) Black cells indicate a frequency reuse for $N = 19$

Frequency Reuse Distance

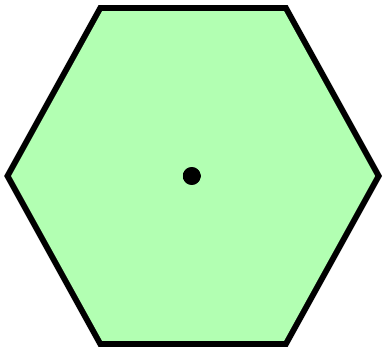
The minimum distance which allows the same frequency to be reused will depend on many factors such as the number of co-channel cells in the vicinity of the center cell, the type of geographic terrain contour, antenna height and the transmitted power at each cell site.



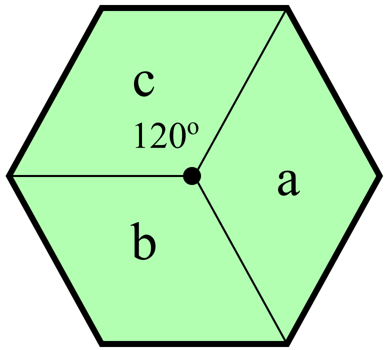
7 cell reuse cluster

Cell Sectoring by Antenna Design

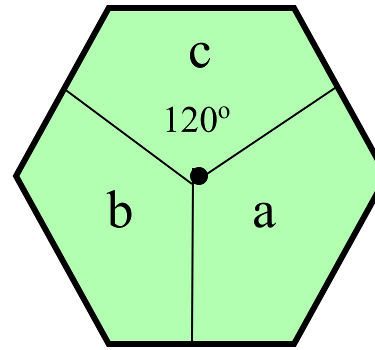
To reduce interference from surrounding cells each BTS use sectored antennas, where the signal radiation of each antenna is directional.



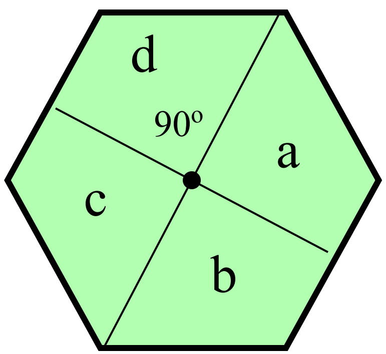
(a) Omni



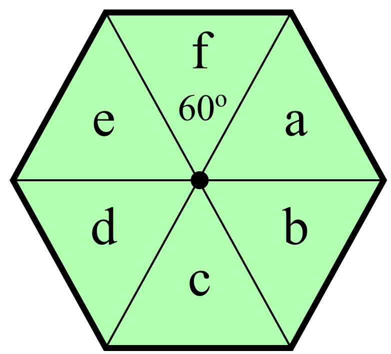
(b) 120° sector



(c) 120° sector (alternate)



(d) 90° sector



(e) 60° sector

Interferences In Mobile Cellular Network

- Co-site Channel Interferences
- Adjacent Channel Interferences
- Co-channel Channel Interferences

Co-site Channel Interferences

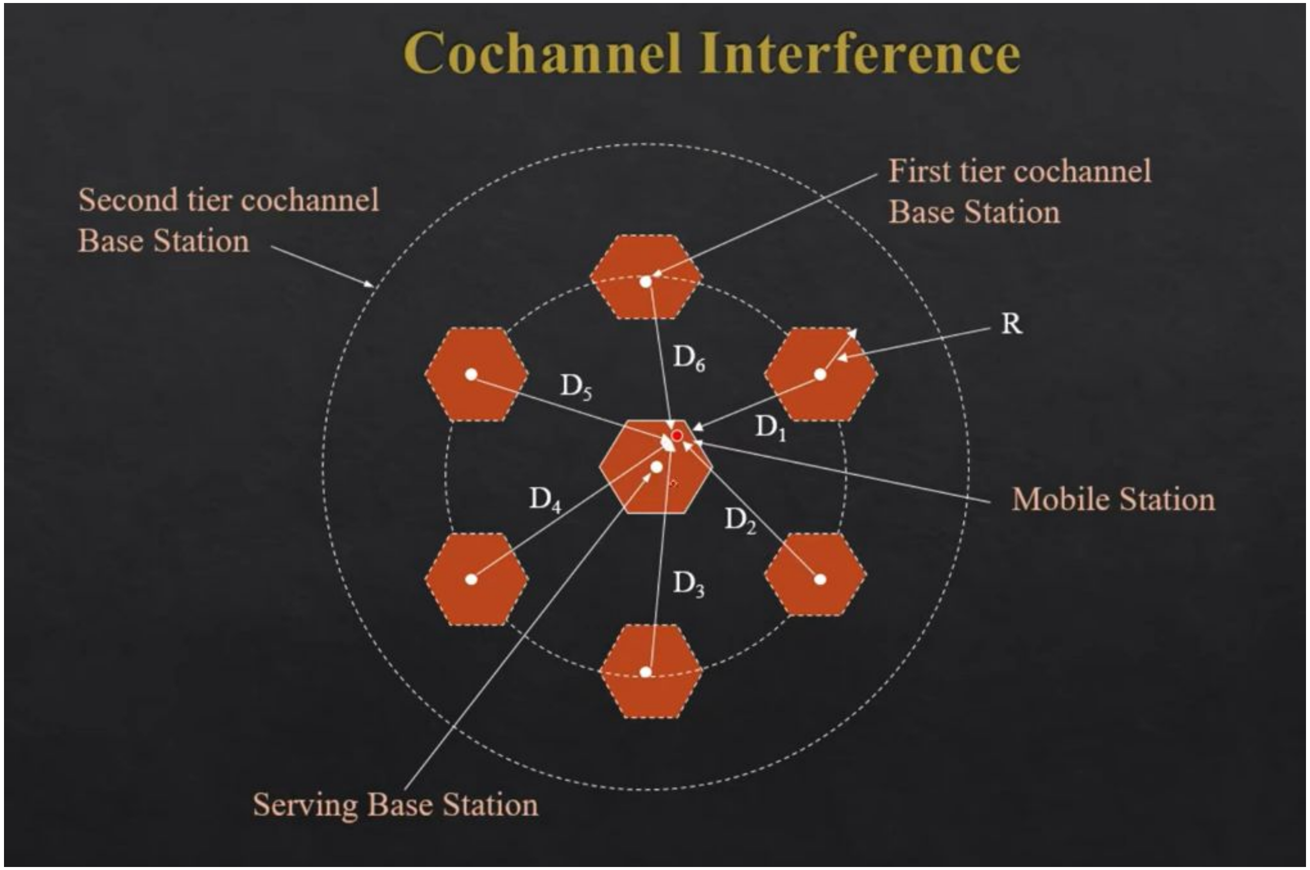
Mutual interferences created by the users inside a cell is called co-site channel interferences. Since users are very closed in cell therefore separation between carries within a cell should be wide enough to avid interferences. At the same time uplink and down link carriers of an user are separated by 45MHz.

Co-channel Interferences

Interference caused by users in different cells using the same frequency channel is called co-channel interference.

Even though co-channel cells are geographically separated, they reuse the same frequency, so interference can still occur, especially if the distance between them isn't large enough.

Co-channel and adjacent channel interferences can be reduced using sectored cells.


Cochannel Interference
Second tier cochannel
Base Station
First tier cochannel
Base Station
R
D6
D5
D1
Mobile Station
D4
D2
D3
Serving Base Station

Adjacent Channel Interferences

Interferences created by the users of two adjacent cells are called adjacent channel interferences. Because of small distance between adjacent cells an wide band gap should be maintained between carries of adjacent cell to avoid interferences.

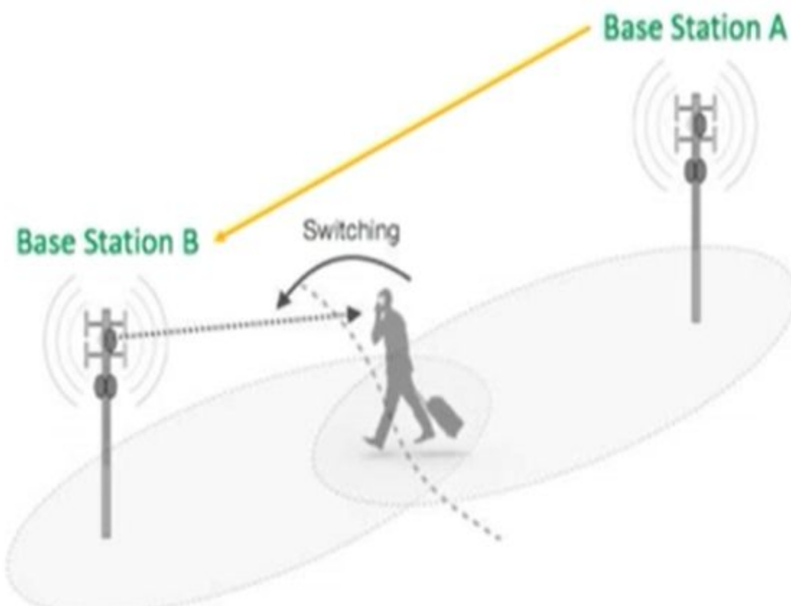
Handover (Handoff)

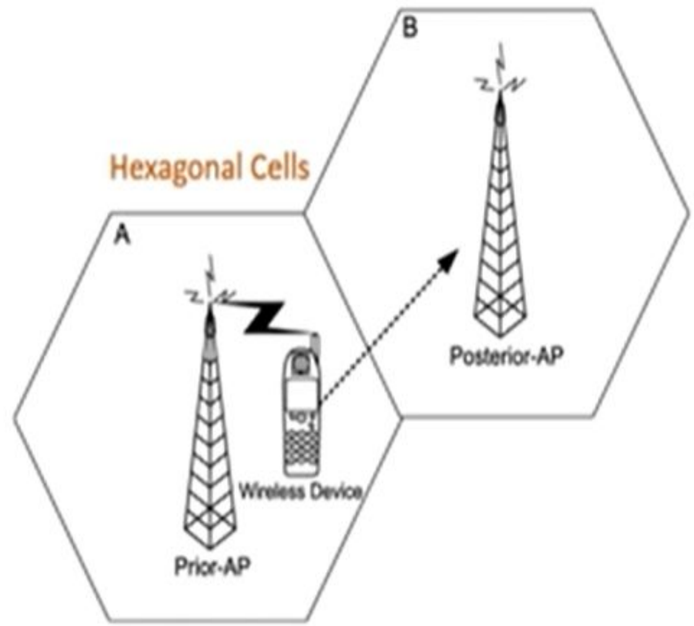
Hand-off is the process of switching from one traffic channel to another by the user in midst of a communication. Hand-off takes place when a user crosses the cell boundary. It Provides continuity of communication across cells. Handoffs must be performed successfully and as infrequently as possible, and be imperceptible to the users.

Process

- ✔ Received signal weakens as mobile moves out of cell (Minimum usable signal level is normally set to be between -90 dBm and -100 dBm).
- ✔ A slightly save signal level is used as a threshold at which a handoff is made.
- ✔ Cell site at some point requests handover to cell with stronger signal strength.
- ✔ MSC switches call to new cell after allocating channels, and informing the two mobiles of the new channels (voice and control channels).

Handoff or Handover

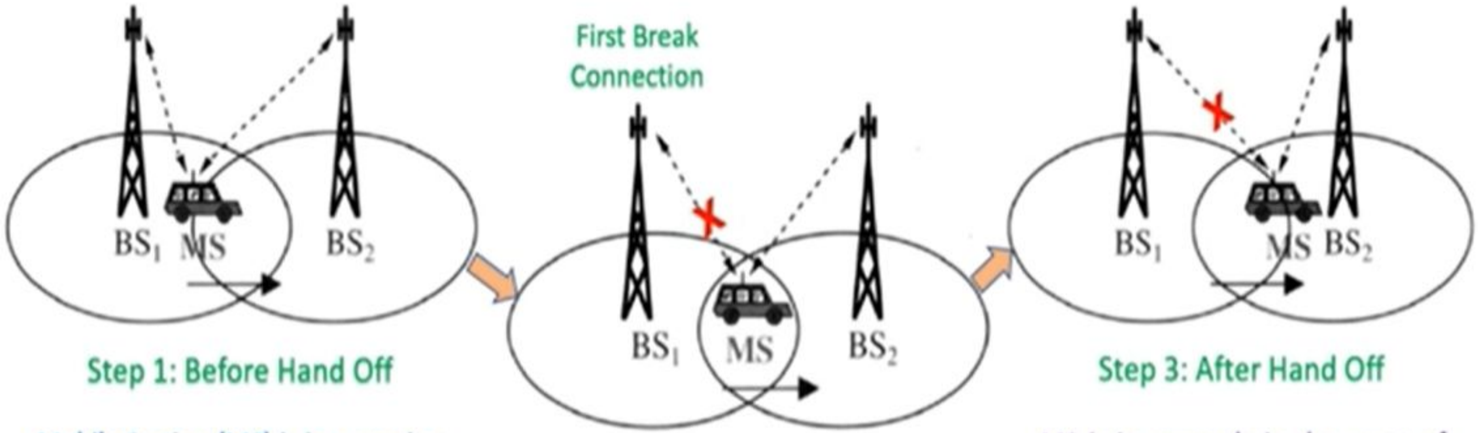




In cellular telecommunications, the terms handover or handoff refers to the process of transferring ongoing call or data connectivity from one Base Station (BS) to other Base Station (BS).

In handoff or handover, when a mobile moves into the different cell while the conversation is in progress then the Mobile Switching Center (MSC) transfer the call to a new channel belonging to the new Base Station (BS) or Access Point (AP).

Hard Handoff



Step 1: Before Hand Off

Mobile Station (MS) is in a moving car which is currently in the range of first Base Station (BS_1) and MS is moving towards second Base Station (BS_2)

Step 2: During Hand Off

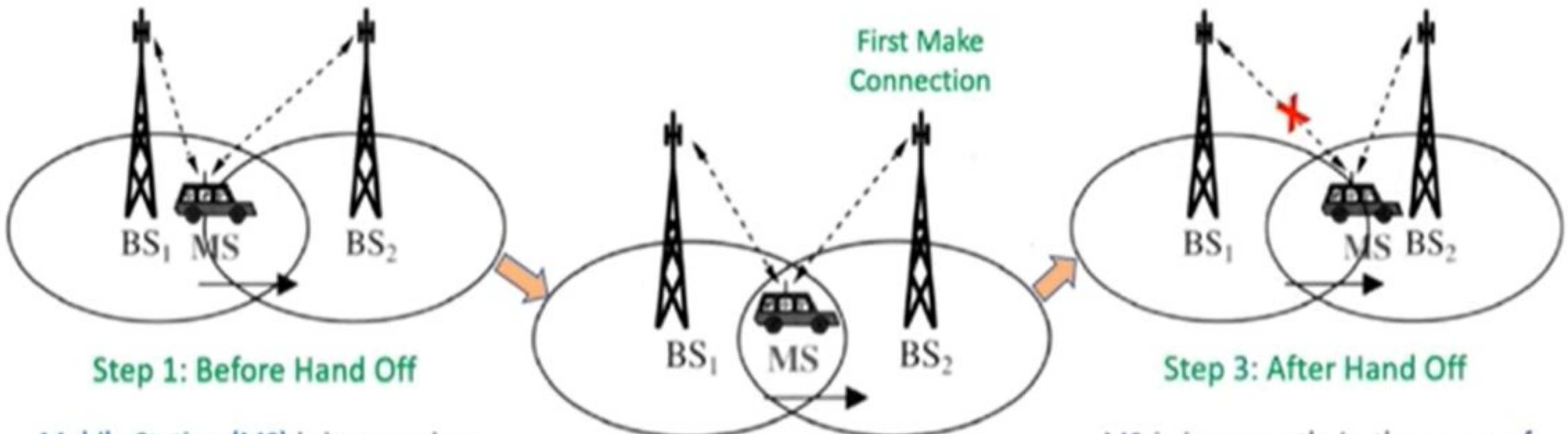
MS reached the boundary of both Base Stations (BS_1 and BS_2)

In hard handoff, MS first break the connection with first Base Station (BS_1) and then make the connection with second Base Stations (BS_2)

Step 3: After Hand Off

MS is in currently in the range of second Base Stations (BS_2)

Soft Handoff



Step 1: Before Hand Off

Mobile Station (MS) is in a moving car which is currently in the range of first Base Station (BS_1) and MS is moving towards second Base Station (BS_2)

Step 2: During Hand Off

MS reached the boundary of both Base Stations (BS_1 and BS_2)

In soft handoff, MS first make the connection with second Base Station (BS_2) and then break the connection with first Base Station (BS_1)

Step 3: After Hand Off

MS is in currently in the range of second Base Stations (BS_2)

Handoff

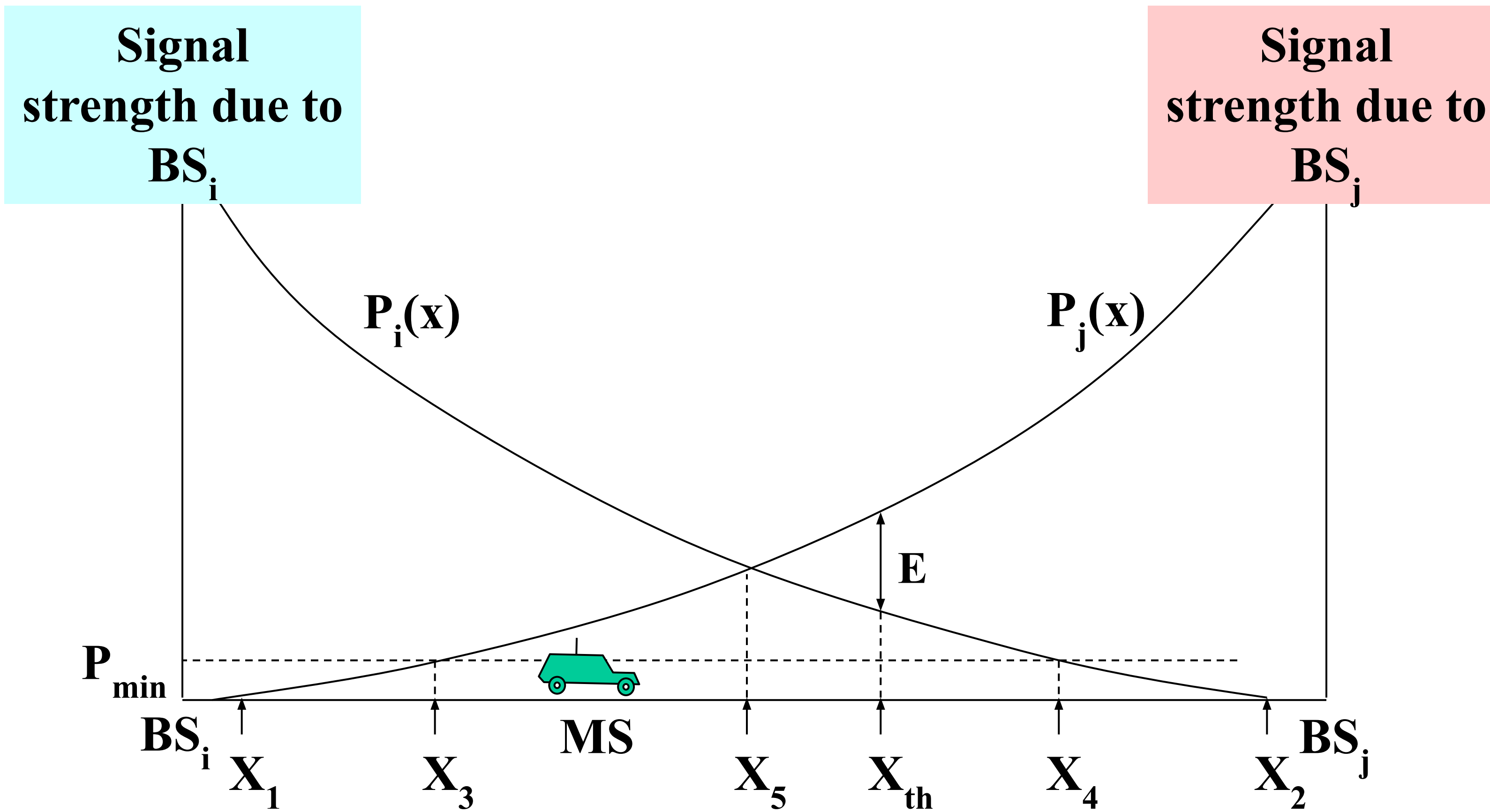
Hard Handoff

Soft Handoff

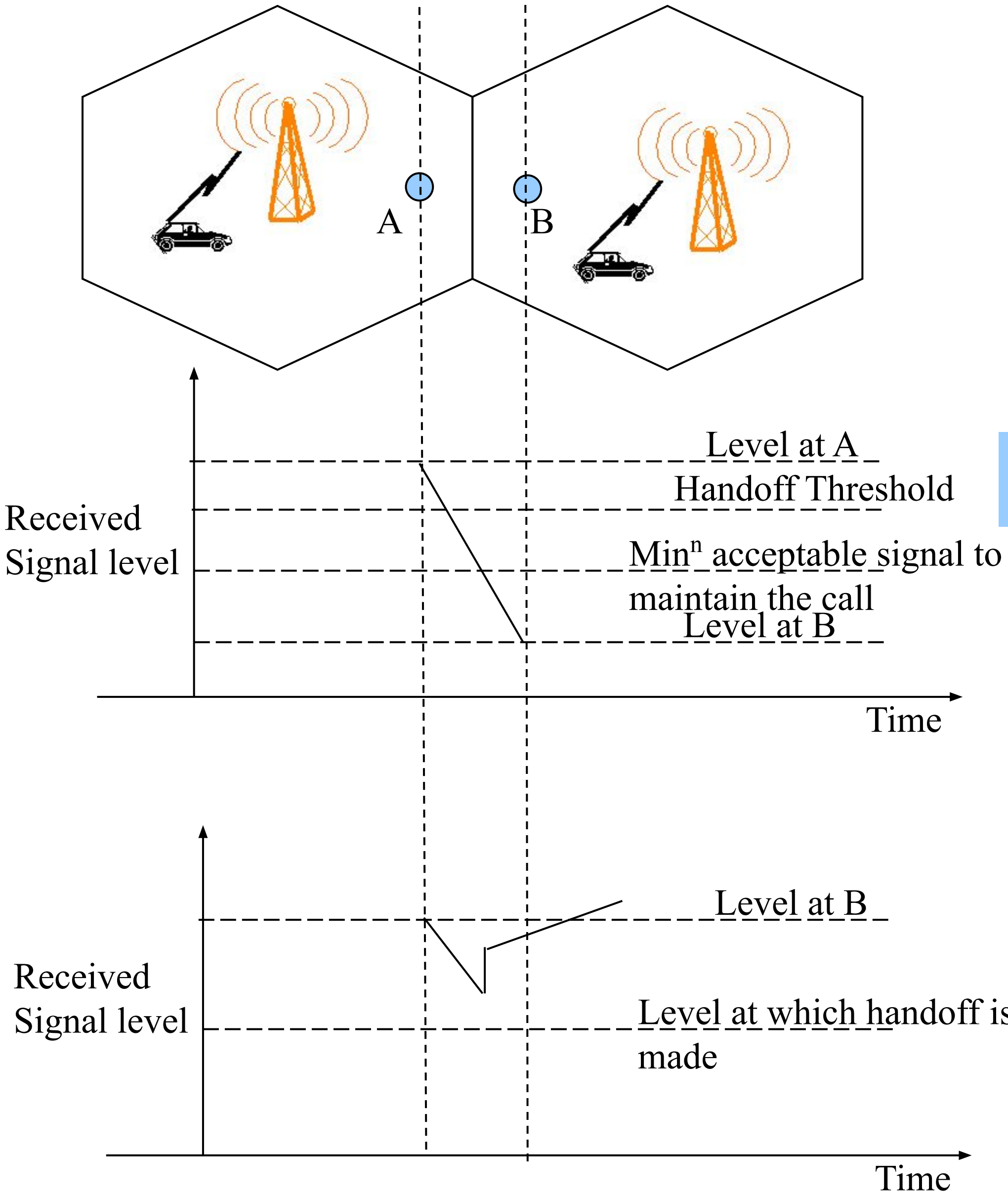
In a hard handoff, an actual break in the connection occurs while switching from one cell to another.	In soft handoff, at least one of the links is kept when radio links are added and removed to the mobile station.
The radio links from the mobile station to the existing cell is broken before establishing a link with the next cell.	This ensures that during the handoff, no break occurs.
It is generally an inter-frequency handoff.	This is generally adopted in co-located sites.
It is a “break before make” policy.	It is a “make before break” policy.
It connects to single Base Station (BS) at a time.	It connects to multiple Base Stations (BS) simultaneously.

Handoff Region

If a mobile unit moves out of range of one cell and into the range of another during a connection, the traffic channel has to change to one assigned to the BS in the new cell (Figure 14.6f). The system makes this change without either interrupting the call or alerting the user.



By looking at the variation of signal strength from either base station it is possible to decide on the optimum area where handoff can take place.



Improper handoff situation

Proper handoff situation

- **Upper Graph: Improper Handoff Situation**
- **What happens:**
  - The mobile moves from cell A to cell B.
  - The signal from **base station A** degrades **below the minimum acceptable signal level before** a handoff is made.
  - The call may drop or become poor in quality.
- **Key indicators:**
  - Handoff occurs **too late**.
  - The signal level at A falls **below threshold** and below the "minimum acceptable signal."
  - **Result:** Communication is compromised.

Lower Graph: Proper Handoff Situation

- **What happens:**
  - The mobile still moves from cell A to B.
  - The handoff is initiated **at the right time** (when signal from A is just around threshold and signal from B is good).
  - The transition is smooth.
- **Key indicators:**
  - Handoff occurs **before** the signal from A drops below acceptable level.
  - The signal level from B is already **strong enough**.
  - **Result:** Seamless communication continues.

Proper Handoff: Ensures uninterrupted service by switching to the new base station before the current signal degrades too much.

Improper Handoff: Delays the switch, risking call drop or poor service.

Network Traffic

Traffic Parameters of a network:

- ✔Average call or packet arrival rate (λ calls/min)
- ✔Average call or packet duration (t_h mim)
- ✔Offered traffic (or traffic intensity)

  ($A = \lambda t_h$ Erlangs)
- ✔Number of channels (n)
- ✔Blocking probability or QoS (B)
- ✔Mean Length of queue
- ✔Mean delay

$A \rightarrow$ Offered Traffic, $B \rightarrow$ Blocking Probability, $n \rightarrow$ Number of channel

Erlang B Traffic Table

Maximum Offered Load Versus B and N

B is in %

N/B	0.01	0.05	0.1	0.5	1.0	2	5	10	15	20	30	40
1	.0001	.0005	.0010	.0050	.0101	.0204	.0526	.1111	.1765	.2500	.4286	.6667
2	.0142	.0321	.0458	.1054	.1526	.2235	.3813	.5954	.7962	1.000	1.449	2.000
3	.0868	.1517	.1938	.3490	.4555	.6022	.8994	1.271	1.603	1.930	2.633	3.480
4	.2347	.3624	.4393	.7012	.8694	1.092	1.525	2.045	2.501	2.945	3.891	5.021
5	.4520	.6486	.7621	1.132	1.361	1.657	2.219	2.881	3.454	4.010	5.189	6.596
6	.7282	.9957	1.146	1.622	1.909	2.276	2.960	3.758	4.445	5.109	6.514	8.191
7	1.054	1.392	1.579	2.158	2.501	2.935	3.738	4.666	5.461	6.230	7.856	9.800
8	1.422	1.830	2.051	2.730	3.128	3.627	4.543	5.597	6.498	7.369	9.213	11.42
9	1.826	2.302	2.558	3.333	3.783	4.345	5.370	6.546	7.551	8.522	10.58	13.05
10	2.260	2.803	3.092	3.961	4.461	5.084	6.216	7.511	8.616	9.685	11.95	14.68
11	2.722	3.329	3.651	4.610	5.160	5.842	7.076	8.487	9.691	10.86	13.33	16.31
12	3.207	3.878	4.231	5.279	5.876	6.615	7.950	9.474	10.78	12.04	14.72	17.95
13	3.713	4.447	4.831	5.964	6.607	7.402	8.835	10.47	11.87	13.22	16.11	19.60
14	4.239	5.032	5.446	6.663	7.352	8.200	9.730	11.47	12.97	14.41	17.50	21.24
15	4.781	5.634	6.077	7.376	8.108	9.010	10.63	12.48	14.07	15.61	18.90	22.89
16	5.339	6.250	6.722	8.100	8.875	9.828	11.54	13.50	15.18	16.81	20.30	24.54
17	5.911	6.878	7.378	8.834	9.652	10.66	12.46	14.52	16.29	18.01	21.70	26.19
18	6.496	7.519	8.046	9.578	10.44	11.49	13.39	15.55	17.41	19.22	23.10	27.84
19	7.093	8.170	8.724	10.33	11.23	12.33	14.32	16.58	18.53	20.42	24.51	29.50
20	7.701	8.831	9.412	11.09	12.03	13.18	15.25	17.61	19.65	21.64	25.92	31.15

Example-1

A city has a population of 3×10^6. A mobile cellular service provider has 500 cells with 15 channels each to serve the people. On an average each user generates 1 call/hour with duration of 1.5 minutes. Determine market penetration of the service provider. Given the GoS (grade of service) of the network is 5%.

(Grade of service is synonymous to blocking probability)

Ans.

Traffic intensity/user = $\lambda.t_h$= (1/60).1.5 = 0.025Erl/user

For each cell, n = 15, B = 5%

From Erlang's table, the offered traffic of a cell, A=10.63Erls.

Number of users/cell = 10.63/0.025 = 425

Total number of users = 500×425 = 212500

Penetration rate = (Number of users/Total population) ×100

$=(212500/(3\times 10^6)) \times 100$

=7.08%

Given:

- Population of the city: $3 \times 10^6 = 3{,}000{,}000$
- Number of cells: 500
- Number of channels per cell (n): 15
- Call generation rate per user (λ): 1 call/hour
- Average call duration (t): 1.5 minutes
- Grade of Service (GoS) / Blocking Probability (B): 5% = 0.05

Step 1: Traffic Intensity per User (in Erlangs)

$$\text{Traffic per user} = \lambda \times t = \left(\frac{1}{60}\right) \times 1.5 = 0.025 \text{ Erlangs/user}$$

Step 2: From Erlang B Table

- For n = 15 channels and B = 0.05, the offered traffic A ≈ 10.63 Erlangs

Step 3: Number of Users per Cell

$$\text{Users per cell} = \frac{10.63}{0.025} = 425.2 \approx 425 \text{ users}$$

- **Step 4: Total Users in Network**

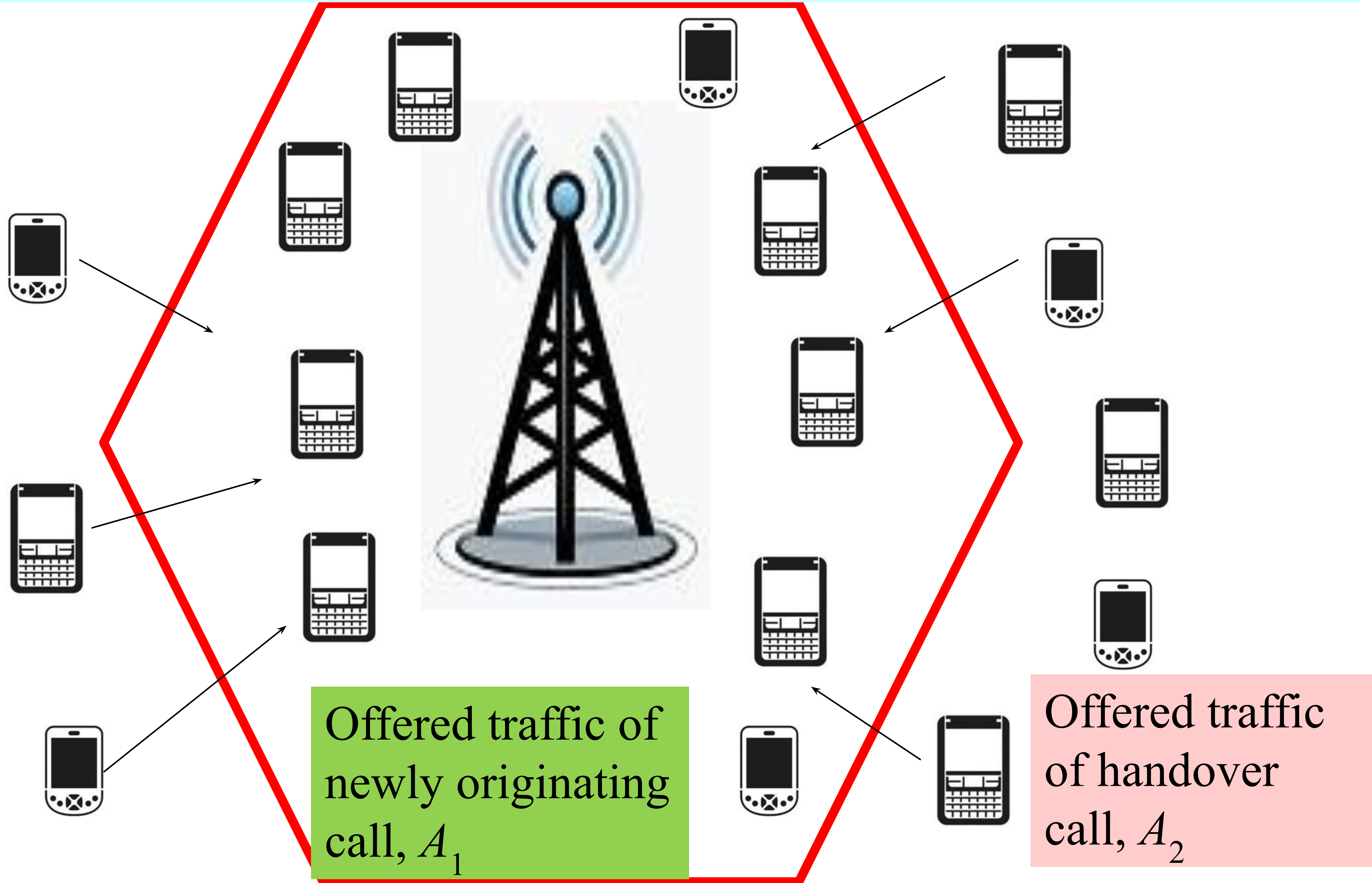
$$\text{Total users} = 500 \text{ cells} \times 425 \text{ users/cell} = 212,500$$

- **Step 5: Market Penetration**

$$\text{Penetration rate} = \left(\frac{212500}{3000000}\right) \times 100 = 7.08\%$$

- This means only **7.08% of the city's population** are subscribers of this cellular service provider.

In a mobile cellular network there are two types of offered traffics: newly originating call and hand over call.



- **Blocking probability**
- **Blocking probability** is the likelihood that a **call attempt is denied access** to the system because **all channels are busy**.
- In telecommunication systems (especially **circuit-switched**, like early mobile networks), resources (e.g., channels) are **limited per cell**.
- If **all available channels are in use**, a **new incoming call** is **blocked**—this is called **call blocking**.

- **Blocking Probability = Grade of Service (GoS)**
- GoS is a measure of service quality, and in many systems, it is synonymous with the blocking probability.
- A GoS of 5% means 5 out of 100 call attempts may be blocked during peak traffic hours.

Blocking probability,

$$B(n, A) = \frac{\frac{A^n}{n!}}{\sum_{i=0}^{n} \frac{A^i}{i!}}$$

$$= \frac{\frac{A^n}{n!}}{1 + A + \frac{A^2}{2!} + \frac{A^3}{3!} + \cdots \quad \cdots \quad \cdots + \frac{A^n}{n!}}$$

Offered traffic (A Erlangs)
Number of channels (n)
Blocking probability B
$A = \lambda t_h$

For two dimensional traffic case,

$$B(n, A_1, A_2) = \frac{\sum_{y=0}^{n} \frac{A_1^y}{y!} \frac{A_2^{n-y}}{(n-y)!}}{\sum_{x_1=0}^{n} \sum_{x_2=0}^{n-x_1} \frac{A_1^{x_1}}{x_1!} \frac{A_2^{x_2}}{x_2!}}$$

Tabular solution of combined traffic for n = 3 channels; where the offered traffic of newly originating call is A_1 and that of handover call is A_2.

Newly originating

A_1

$A_1^3/3!$	3	(0, 3)	(1, 3)	(2, 3)	(3, 3)
$A_1^2/2!$	2	(0, 2)	(1, 2)	(2, 2)	(3, 2)
A_1	1	(0,1)	(1, 1)	(2, 1)	(3, 1)
1	0	(0, 0)	(1,0)	(2,0)	(3,0)
	Channels	0	1	2	3
		1	A_2	$A^2_2/2!$	$A^3_2/3!$

A_2

Handover

Tabular solution of combined traffic for n = 3 channels; where the offered traffic of newly originating call is A_1 and that of handover call is A_2.

$A_1^3/3!$	3	(0, 3)	(1, 3)	(2, 3)	(3, 3)
$A_1^2/2!$	2	(0, 2)	(1, 2)	(2, 2)	(3, 2)
A_1	1	(0,1)	(1, 1)	(2, 1)	(3, 1)
1	0	(0, 0)	(1,0)	(2,0)	(3,0)
x_1 x_2	Channels	0	1	2	3
		1	A_2	$A_2^2/2!$	$A_2^3/3!$

$A_1^3/3!$	3	(0, 3)			
$A_1^2/2!$	2	(0, 2)	(1, 2)		
A_1	1	(0,1)	(1, 1)	(2, 1)	
1	0	(0, 0)	(1,0)	(2,0)	(3,0)
x_1 x_2	Channels	0	1	2	3
		1	A_2	$A_2^2/2!$	$A_2^3/3!$

Tabular solution of combined traffic for n = 3 channels; where the offered traffic of newly originating call is A_1 and that of handover call is A_2.

$A_1^3/3!$	3	$A_1^3/3!$			
$A_1^2/2!$	2	$A_1^2/2!$	$A_2A_1^2/2!$		
A_1	1	A_1	$A_2 A_1$	$A_1A_2^2/2!$	
1	0	1	A_2	$A_2^2/2!$	$A_2^3/3!$
x_1 x_2	n ↑→	0	1	2	3
		1	A_2	$A_2^2/2!$	$A_2^3/3!$

Complete occupied states

$A_1^3/3!$	3	$A_1^3/3!$			
$A_1^2/2!$	2	$A_1^2/2!$	$A_2 A_1^2/2!$		
A_1	1	A_1	$A_2 A_1$	$A_1 A_2^2/2!$	
1	0	1	A_2	$A_2^2/2!$	$A_2^3/3!$
x_1 x_2	Channels	0	1	2	3
		1	A_2	$A_2^2/2!$	$A_2^3/3!$

$$B(n, A) = \frac{\text{Sum of complete occupied states}}{\text{sum of all states}}$$

Example-2

$A_1 = 3$ Erls and $A_2 = 2$ Erls, $n = 8$

The possible probability states and blocking probability

A_1

8	**0.1622**	0.162								
7	**0.433**	0.433	0.867							
6	**1.0125**	1.0125	2.025	2.025						
5	**2.025**	2.025	4.05	4.05	2.7					
4	**3.375**	3.375	6.75	6.75	4.5	2.25				
3	**4.5**	4.5	9	9	6	3	1.2			
2	**4.5**	4.5	9	9	6	3	1.2	0.4		
1	**3**	3	6	6	4	2	0.8	0.26	0.076	
0	**1**	1	2	2	1.33	0.666	0.266	0.08	0.025	0.0063
		1	**2**	**2**	**1.33**	**0.66**	**0.266**	**0.08**	**0.025**	**0.0063**
n→↑		0	1	2	3	4	5	6	7	8

A_2

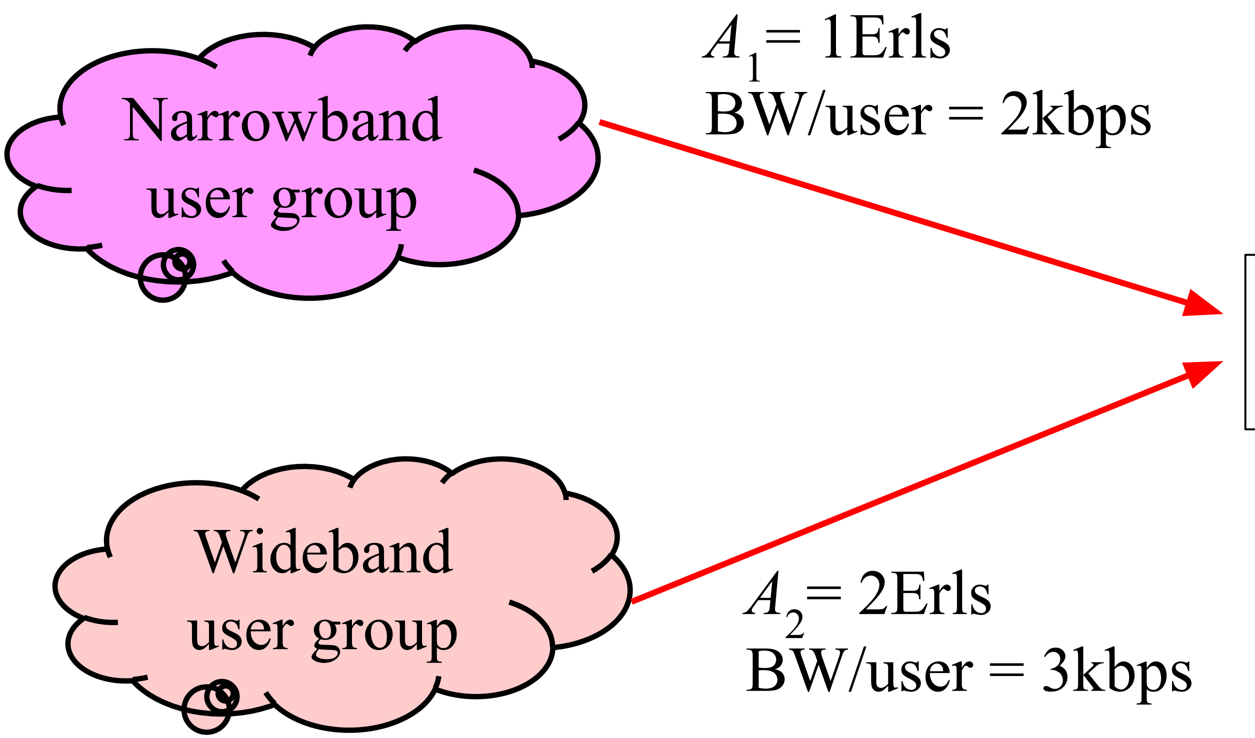
The sum of entire sampling space, S= 138.307

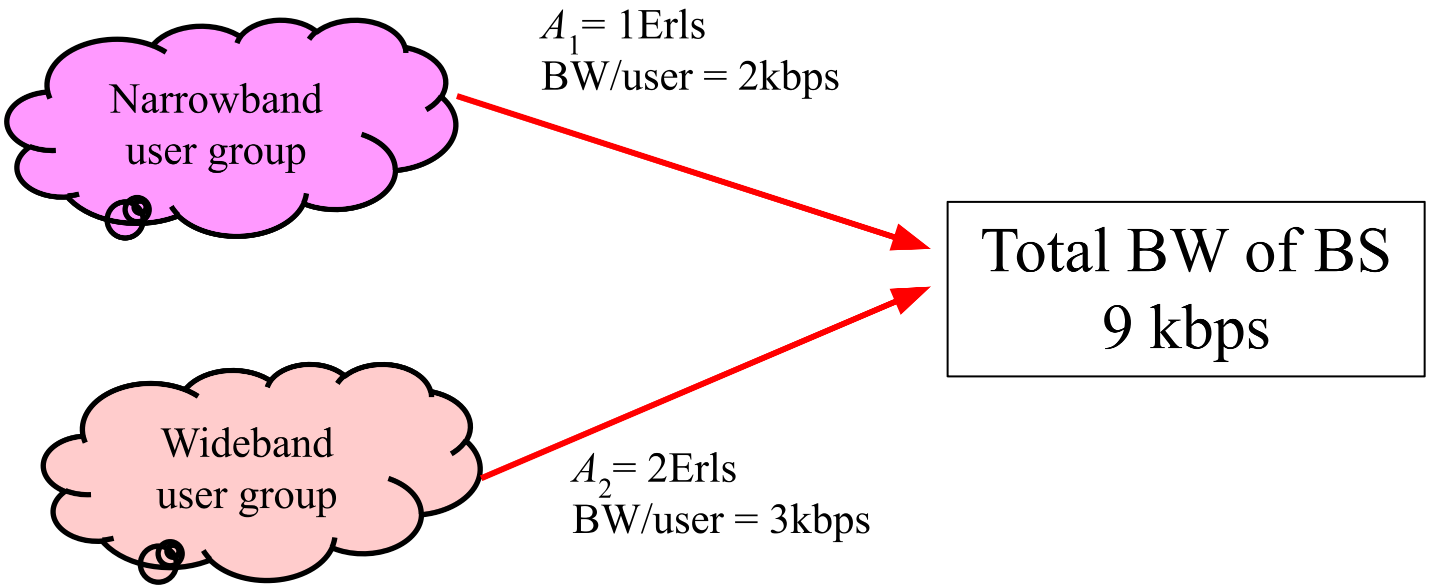
The sum of complete occupied state, O = 9.6863

Blocking probability, $B = O/S$ =9.6863/138.307 = 0. 07 = 7%

Example-3

Let us assume bandwidth of each arrival of A_1 is 2kbps and that of A_2 is 3kbps. Given $A_1 = 1$ and $A_2 = 2$ Erls. If total Bandwidth supported by the network is 9kbps, then probability state will be with the condition: $0 \le BW_1 x_1 + BW_2 x_2 \le 9$. Determine QoS of each traffic.





Possible Probability States

A_1 2kbps

4	$A_1^4/4!$	(0,4)	×(1,4)	×(2,4)	×(3,4)
3	$A_1^3/3!$	(0,3)	(1,3)	×(2,3)	×(3,3)
2	$A_1^2/2!$	(0,2)	(1,2)	×(2,2)	×(3,2)
1	A_1	(0,1)	(1,1)	(2,1)	×(3,1)
0	1	(0,0)	(1,0)	(2,0)	(3,0)
A_1=1		1	A_2	$A_2^2/2!$	$A_2^3/3!$
A_2=2		0	1	2	3

A_2 3kbps

The probability states before normalization

$A_1^4/4!$ (=1/24)	4	1/24			
$A_1^3/3!$ (=1/6)	3	1/6	1/3		
$A_1^2/2!$ (=1/2)	2	1/2	1		
A_1 (=1)	1	1	2	2	
1	0	1	2	2	4/3
$A_1=1$		0	1	2	3
$A_2=2$		1	A_2 (2)	$A_2^2/2!$ (=2)	$A_2^3/3!$(=4/3)

The sum of entire sample space, S = 13.375

A_1

$A^3_1/4!$	4	1/24			
$A_1^3/3!$	3	1/6	1/3		
$A_1^2/2!$	2	1/2	1		
A_1	1	1	2	2	
1	0	1	2	2	4/3
$A_1=1$		0	1	2	3
$A_2=2$		1	A_2	$A^2_2/2!$	$A^3_2/3!$

A_2

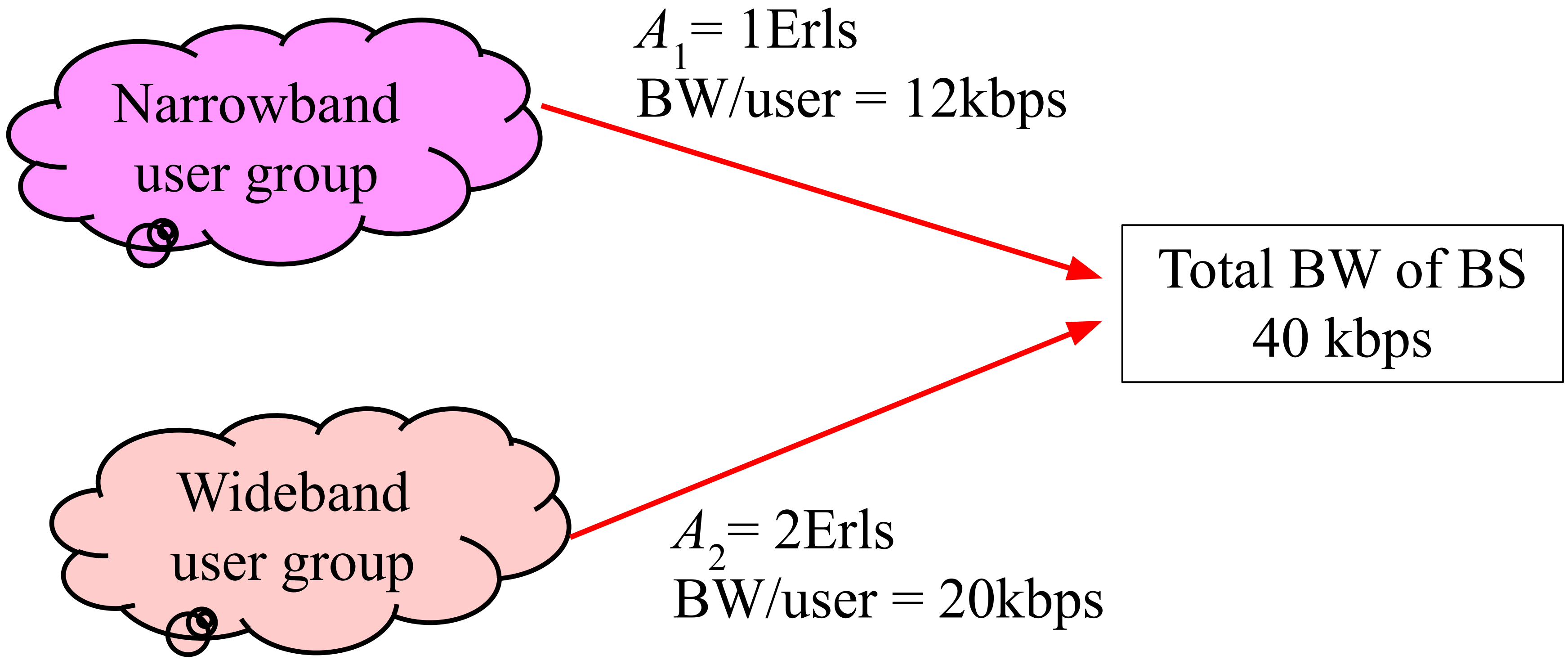
The blocking probability of A_1 traffic,

$B_1 = (1/24+1/3+2+4/3)/13.375 = 3.708/13.375 = 0.277 = 27.7\%$

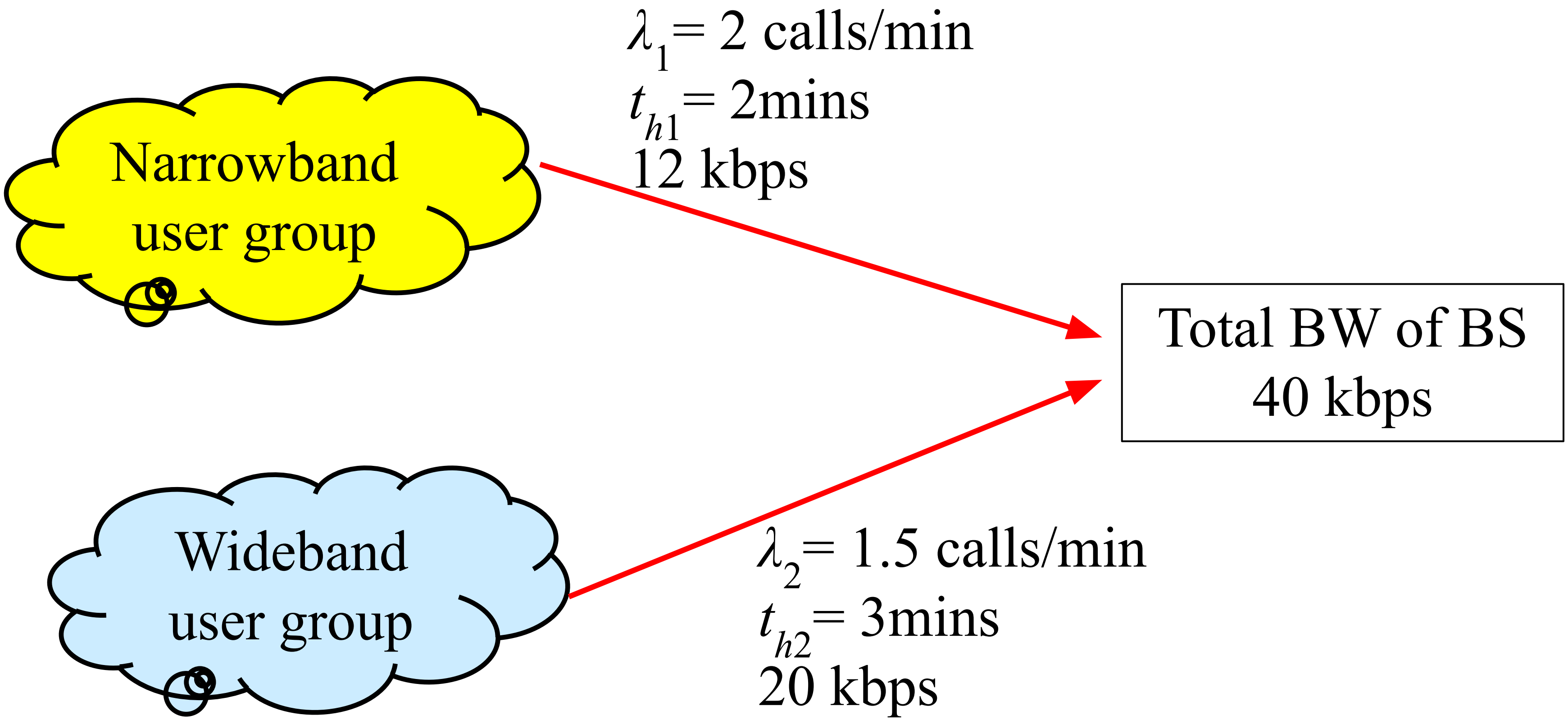
The blocking probability of A_2 traffic,

$B_2 = (1/24+1/3+1+2+4/3)/13.375 = 4.708/13.375 = 0.352 = 35.2\%$

Q1. For the traffic of dissimilar bandwidth of figure below determine the blocking probability of individual traffic.

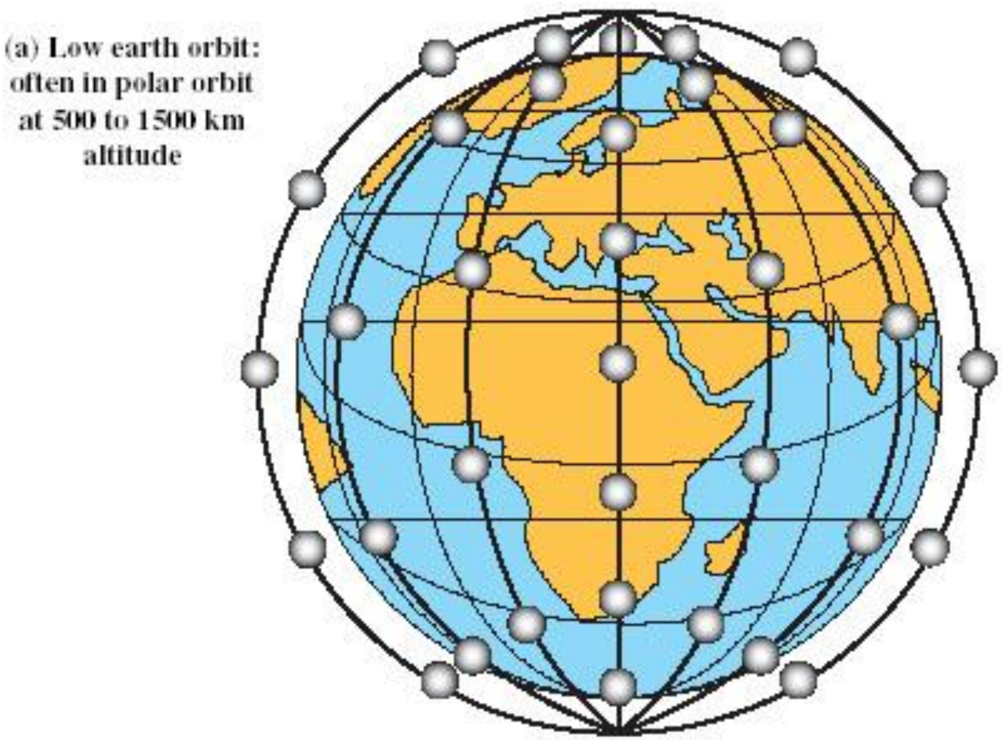


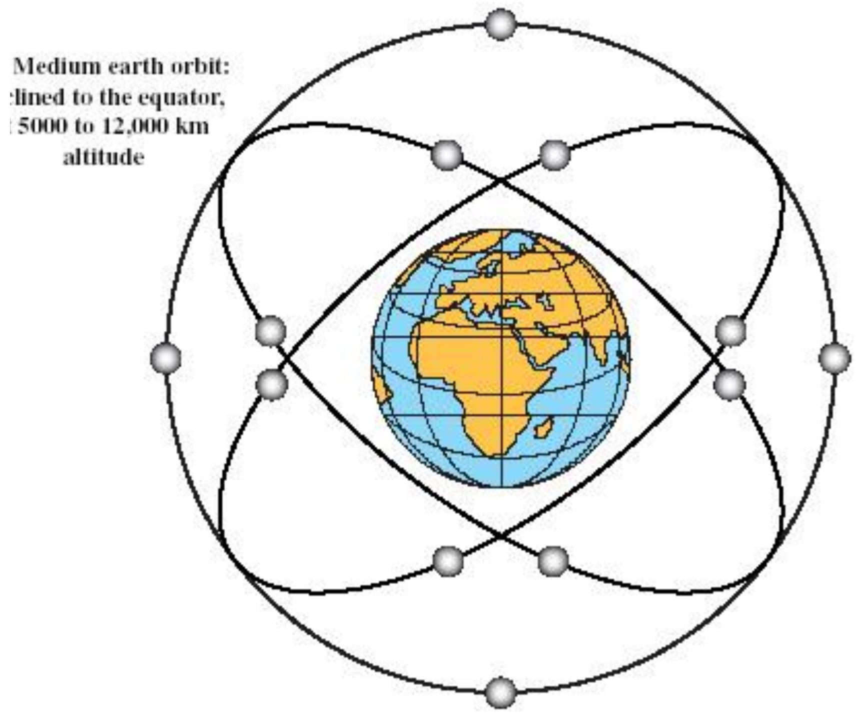
Q2. For the traffic of dissimilar bandwidth of figure below determine the blocking probability of individual traffic.



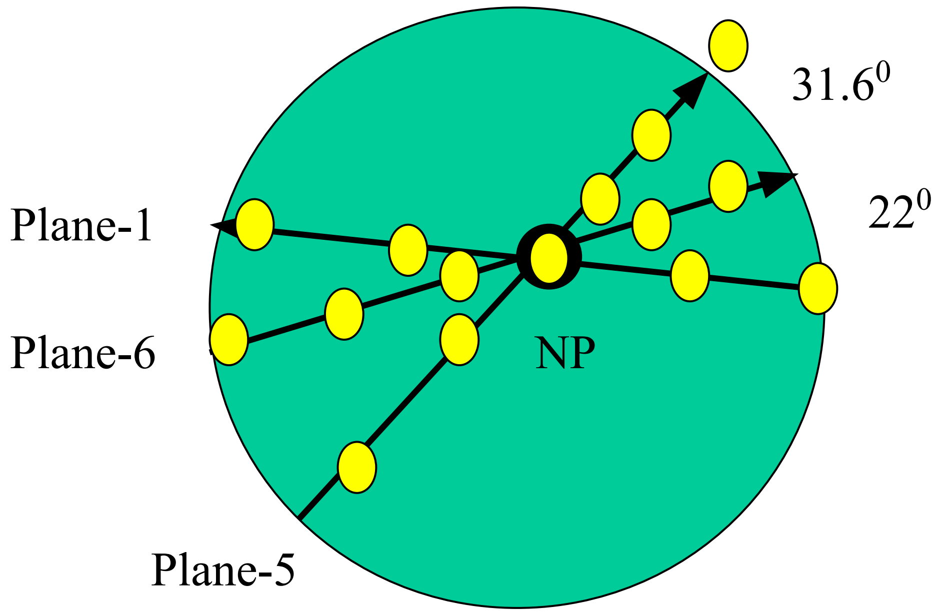
Global Mobile Satellite System

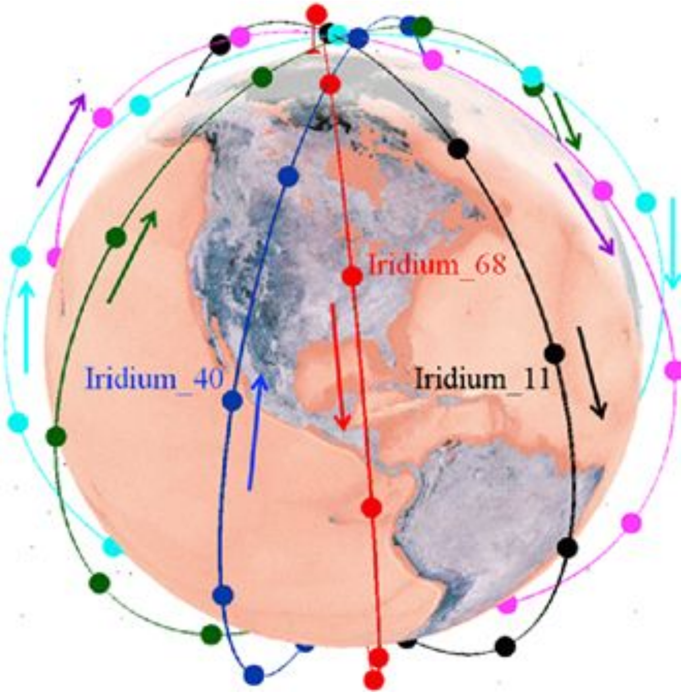
To support a wide range of services and provide superior service quality comparable to that available from **terrestrial** wireless and wired networks, constellations of satellites operating in low earth orbit (LEO) or medium earth orbit (MEO) are considered more suitable.

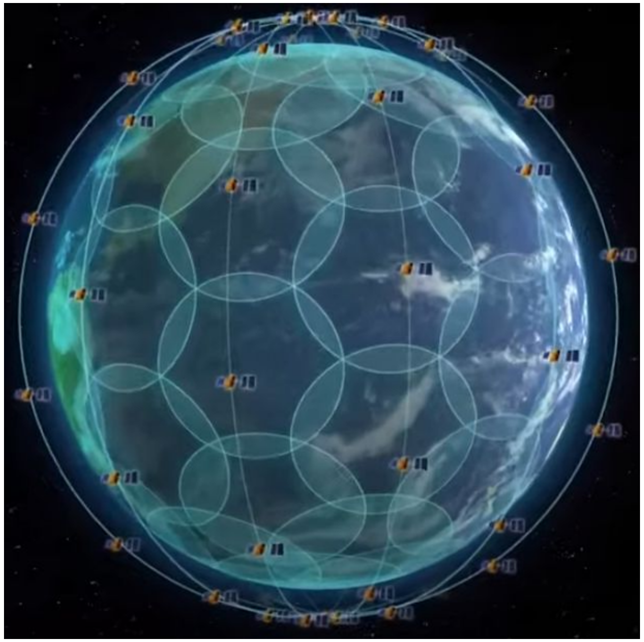




The Iridium System

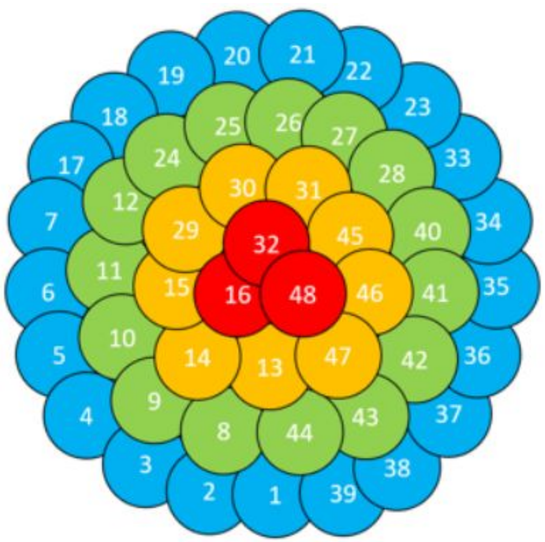


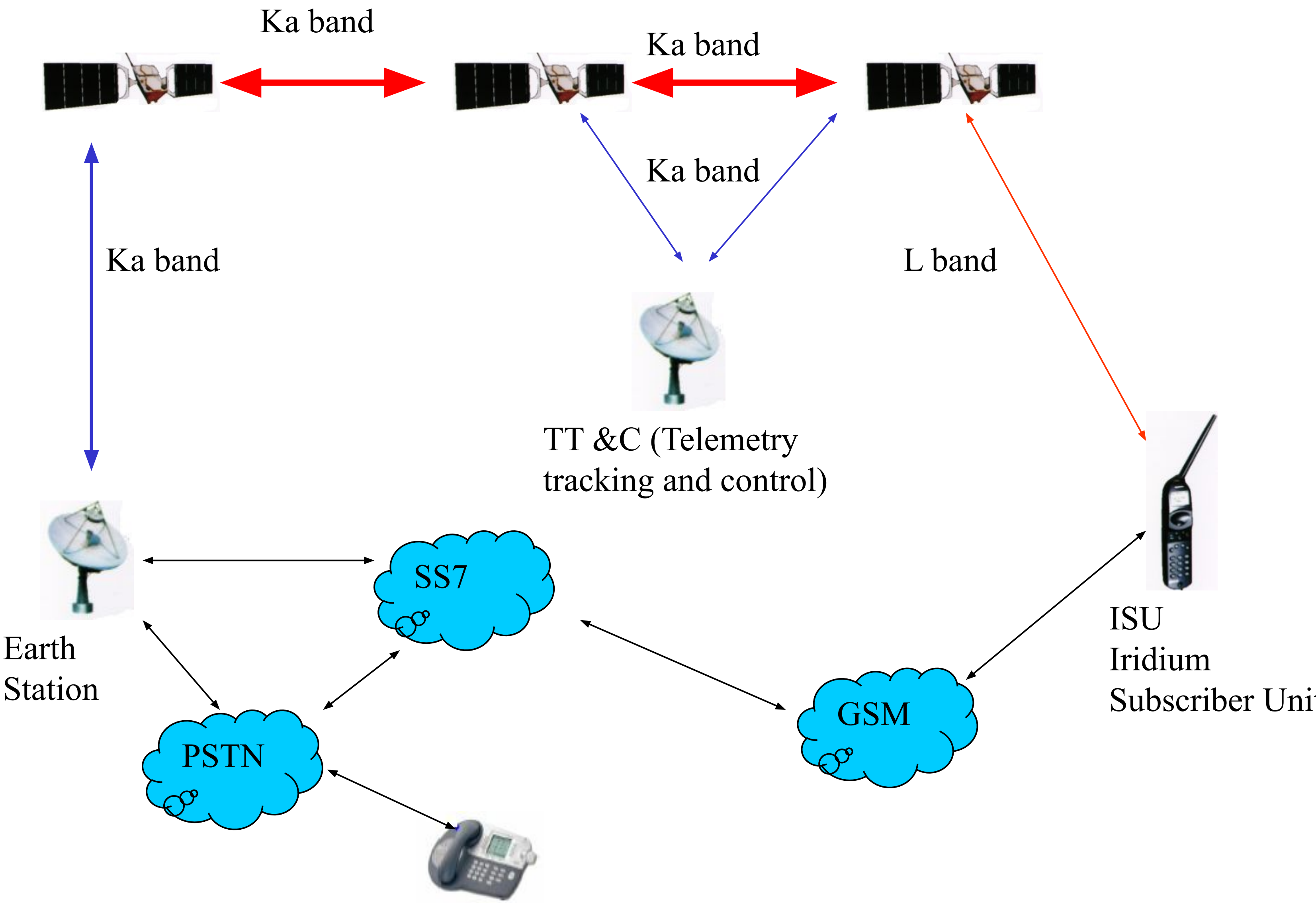




- In **Iridium** 66 satellites are grouped in six orbital planes; there are 11 active satellites in each plane with uniform nominal spacing 31.6^0. The satellites have circular orbits at an altitude of 783Km.

- Each Iridium satellite uses antenna array provides a 48 overlapping spot beam, and each beam has minimum diameter of 600 Km, can be individually switched.
- Only about 2/3 of the beams will be active at any given time because some of the beams will be switched off when the satellites are in the vicinity of the poles, where beam patterns tend to overlap, or when the satellites are over countries or region in which, Iridium does not have regulatory management to operate.





High level network architecture for **Iridium** system

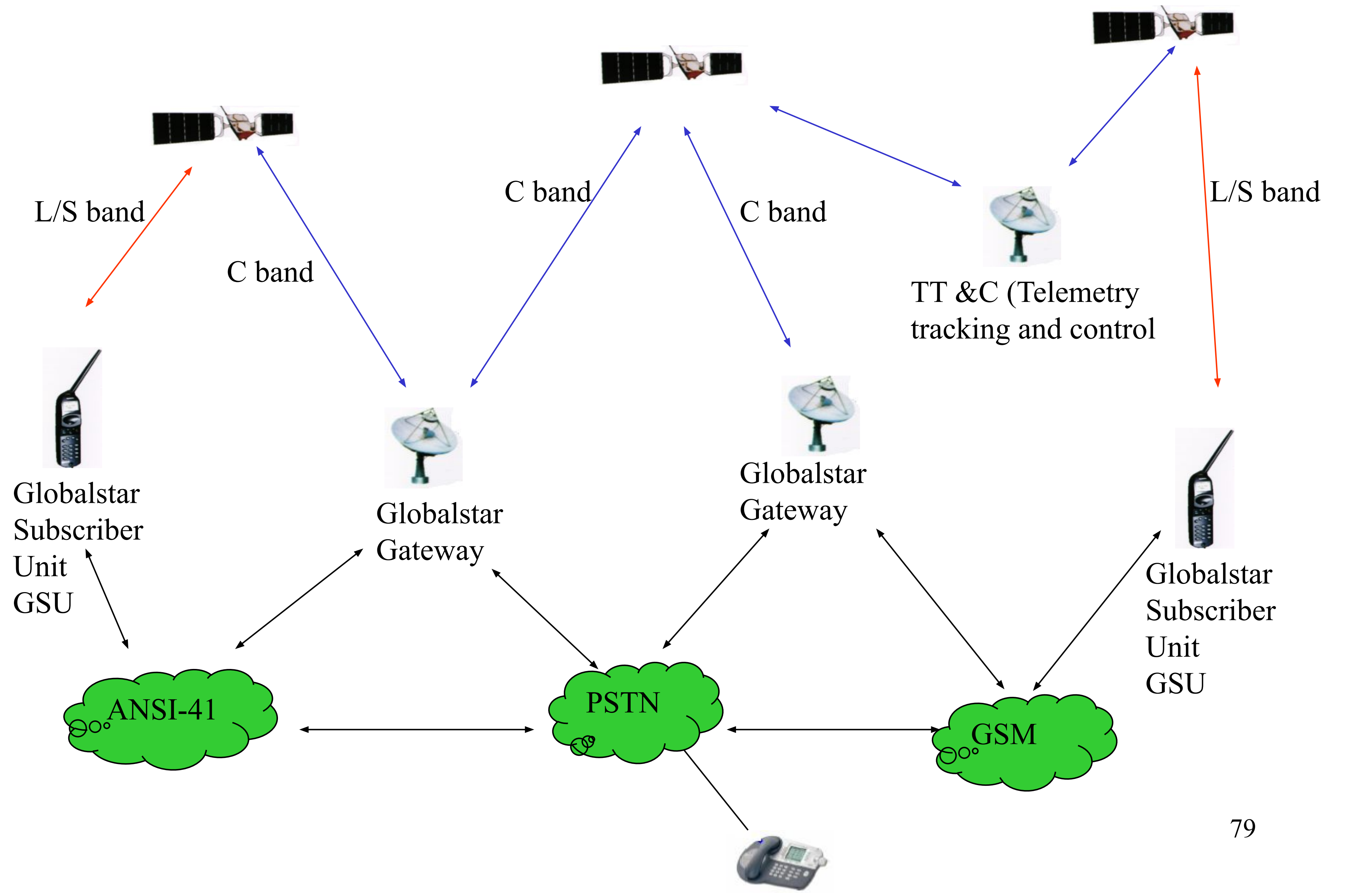
The Globalstar System

- Globalstar is a global mobile satellite system based on constellation of 48 LEO satellites. Unlike the Iridium system, Globalstar system does not use inter-satellite links but rather depends on a large number of interconnected earth stations or gateway for efficient call routing and delivery over the terrestrial network.

- Globalstar's constellation of 48 LEO satellites is designed to orbit at an altitude of 1414 Km above earth's surface in eight orbital planes inclined at 52^0.

- Each satellite supports a 16-beam antenna patterns with an average beam diameter of 2250km. To mitigate blocking and shadowing, Globalstar will deploy path diversity, whereby multiple satellites may be used to complete a call.

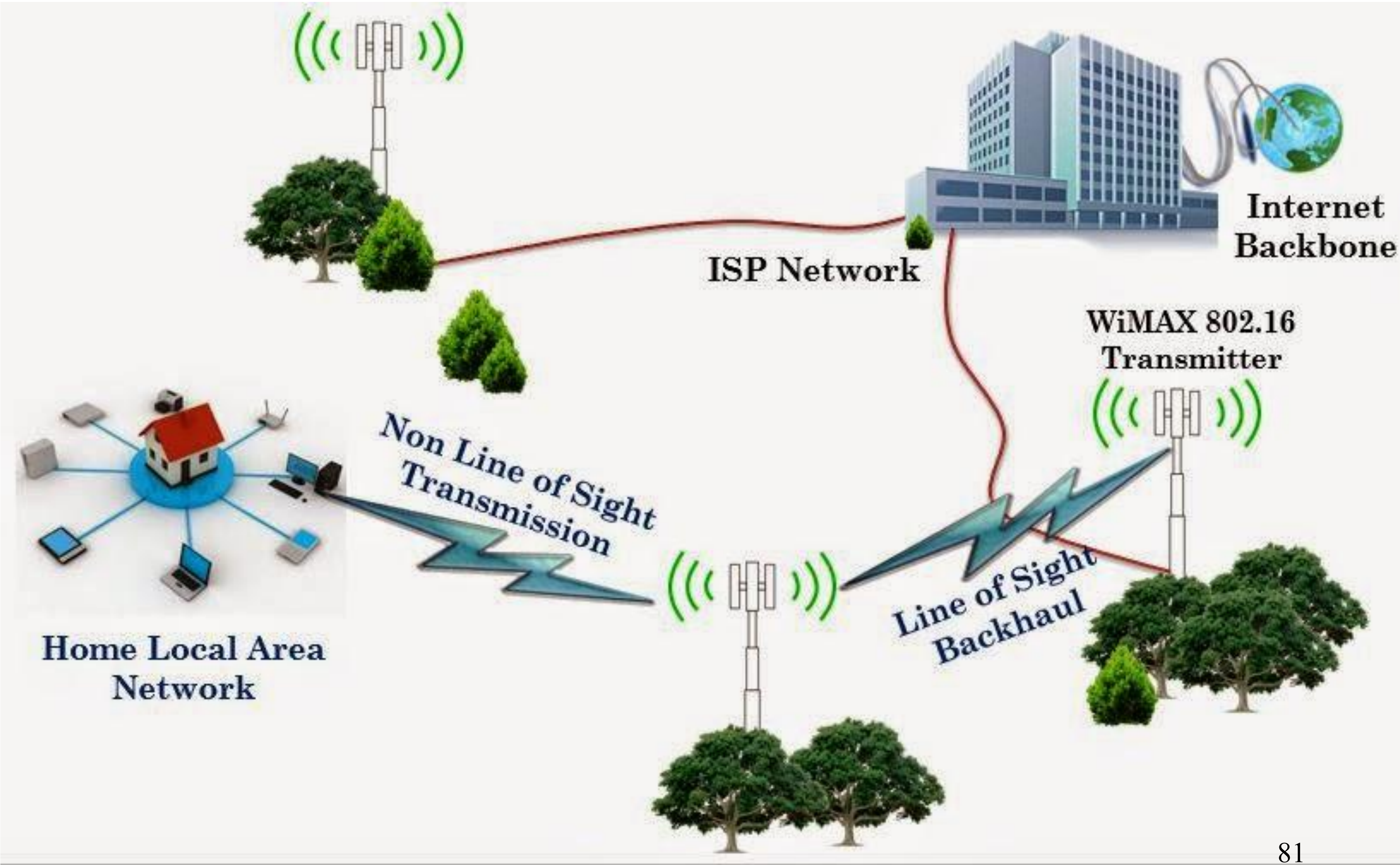
- The network architecture for the **Globalstar mobile** satellite system shown below is very similar to the Iridium system except that Globalstar does not support ISLs.



Wireless MAN

WiMAX (Worldwide Interoperability for Microwave Access) i.e. IEEE 802.16 is an IP based, wireless broadband access technology that provides performance similar to 802.11/Wi-Fi networks with the coverage and QoS (quality of service) of cellular networks.

WiMAX and WiFi integration (example-1)



- An architecture based on WiMAX and WiFi networks as shown in Fig. 1. In this architecture there is a WiMAX base station encompassing multiple WiFi hotspots (access point).

- The WiMAX BS connects to the Internet and acts as the backbone to the WiFi network. The WiMAX subscribers referred to as subscriber stations (SS), communicate directly to the WiMAX Base Station (BS) whereas the WiFi users have to use a special access point (AP) to communicate with the WiMAX BS called ***WiFi - WiMAX Bridge***.

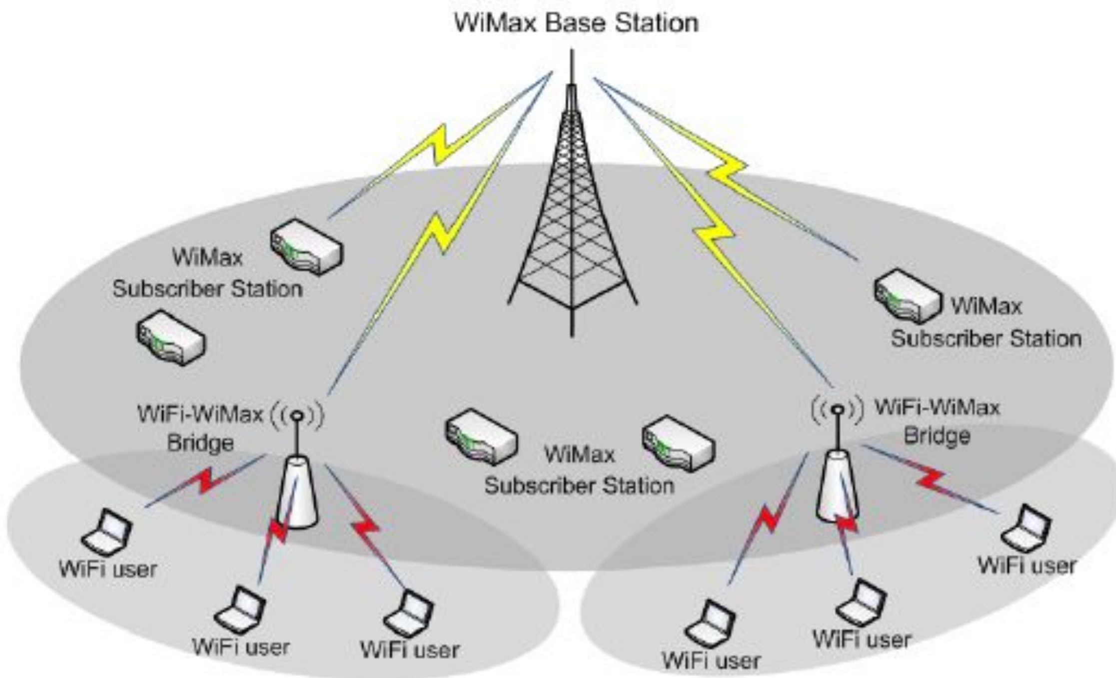


Fig.1 Model of WiMAX and WiFi integration

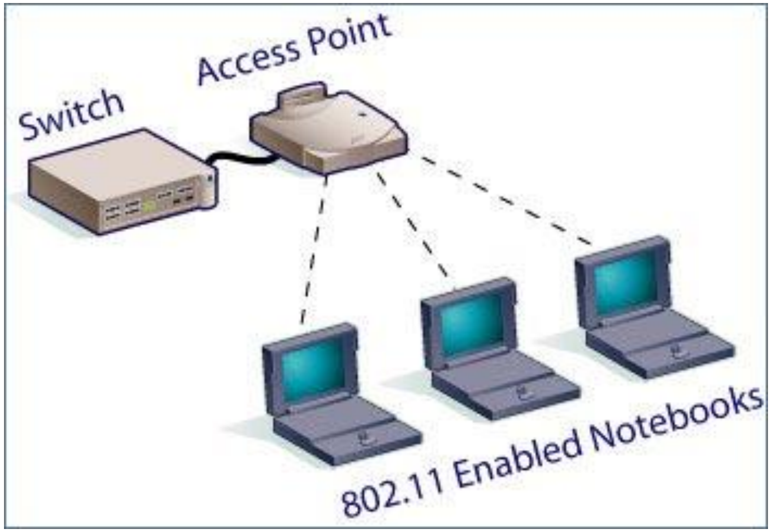
Wireless LAN

Several devices (computers, smart phones, cameras, printers, TABs etc.) are connected by wireless link within a small coverage area, usually a room or building.

Benefits of Wireless LANs

- People can access the network from where they want; they are no longer limited by the length of the cable.

- Some cities have started to offer mobile Wireless LANs. This means that people can access the internet even outside their normal work environment, for example when they ride the train.



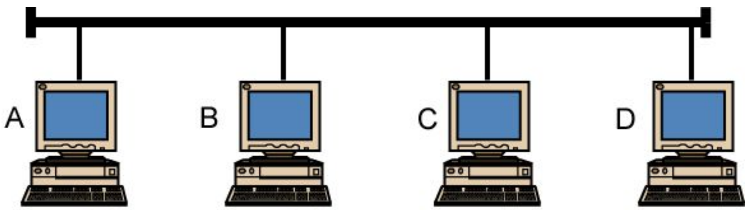
- ✔Easy installation and we don’t need extra cables for installation.
- ✔It is easier to add or remove workstation or node.
- ✔Access points can serve a varying number of computers using DHCP i.e. no need to set IP on the node while entering the coverage of the Wi-Fi.

Major Problems with Wireless network

- ✔ Hidden terminal problem
- ✔ Multipath propagation
- ✔ Handoff
- ✔ Low data rate
- ✔ Security

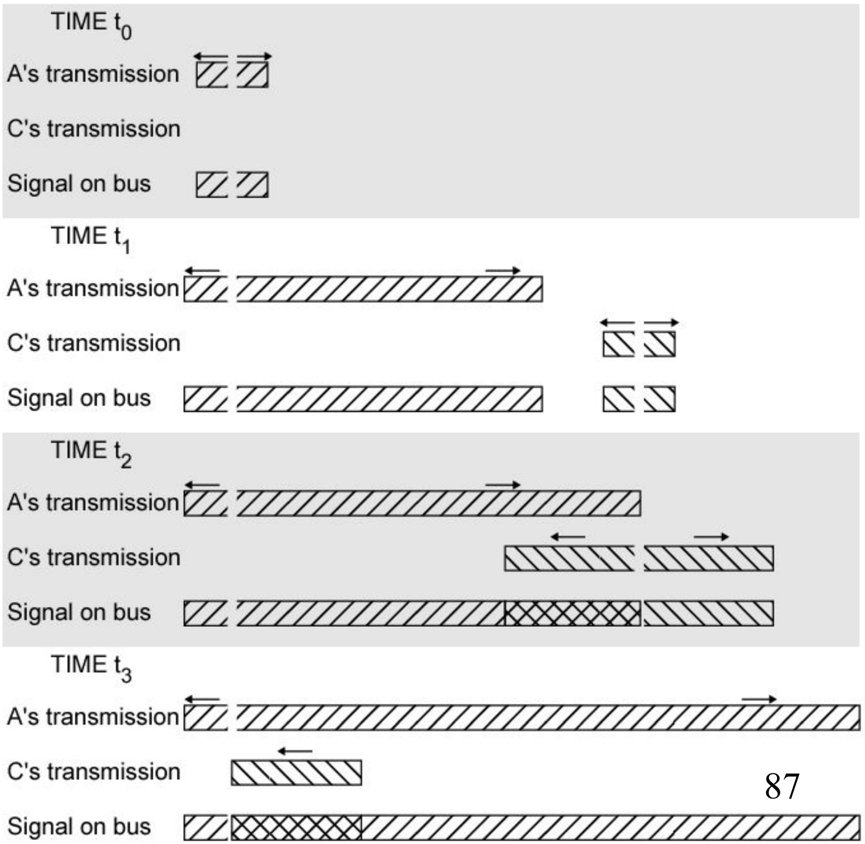
First Problem: A computer on Ethernet always listen to the ether before transmitting. Only if the ether is idle does the computer begin transmitting. With wireless LANs, that idea does not work so well.

CSMA/CD Operation (Wired Communication)



In wireless communication the situation is cumbersome since the received power is too small compared to transmitted power.

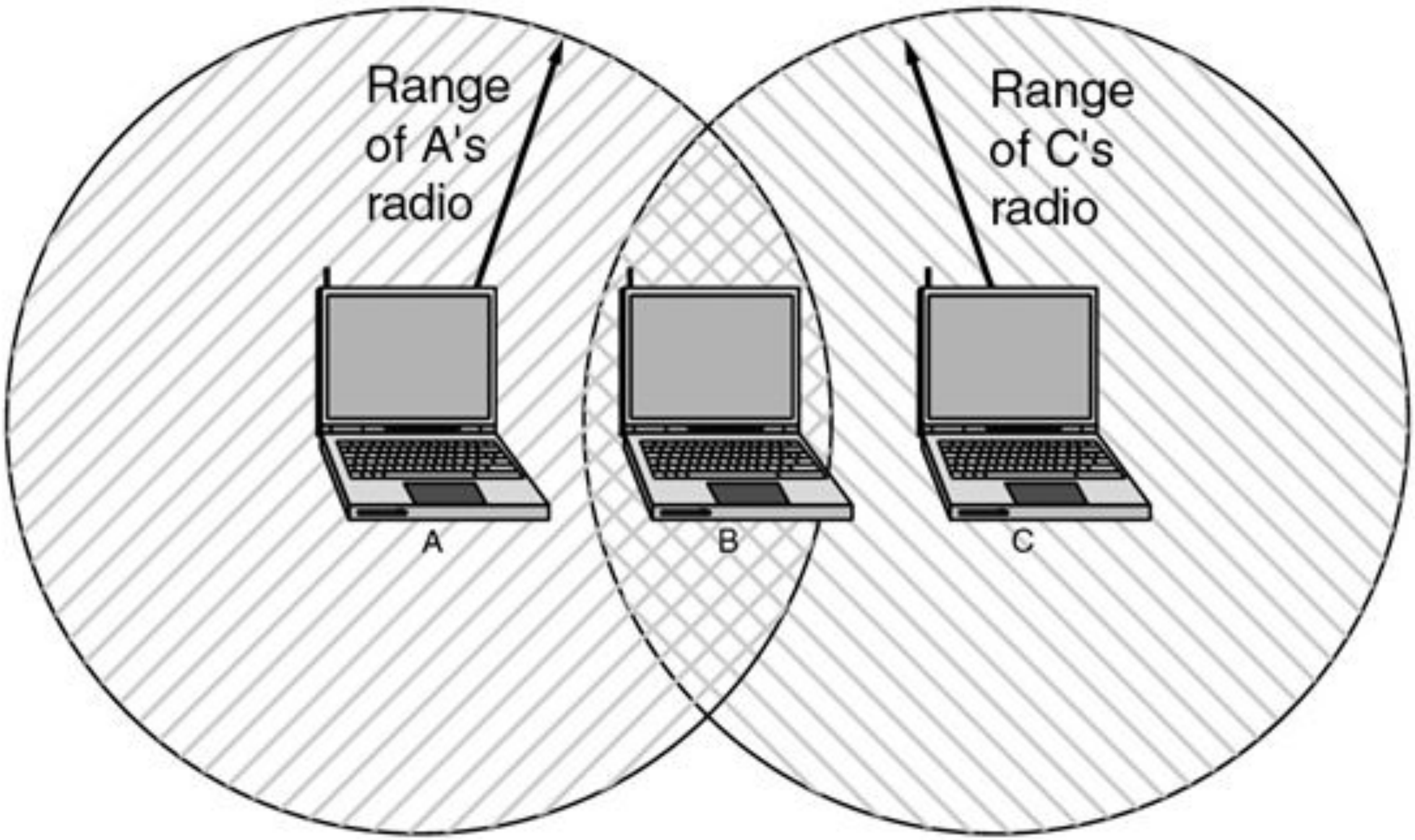
CSMA/CA Operation (wireless communication)



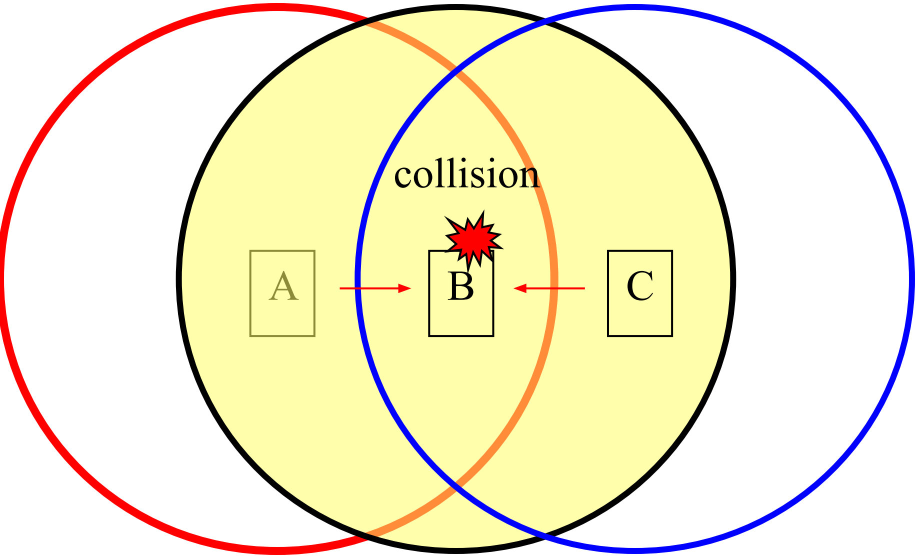
C detects a collision!

A detects a collision!

❖Suppose that computer A is transmitting to computer B (one way), but the radio range of A is too short to reach computer C. If C wants to transmit to B it can listen to the ether before starting, but the fact that it does not hear anything (since it is outside the coverage of A) does not mean that its transmission will succeed. The 802.11 standard had to solve this problem (CSMA/CA).



The problem we mentioned here is called Hidden Terminal Problem



A and B can hear each other. B and C can hear each other. But A and C cannot hear each other.

When A is sending data to B, C cannot sense this activity and hence C is allowed to send data to B at the same time. This will cause a collision at B.

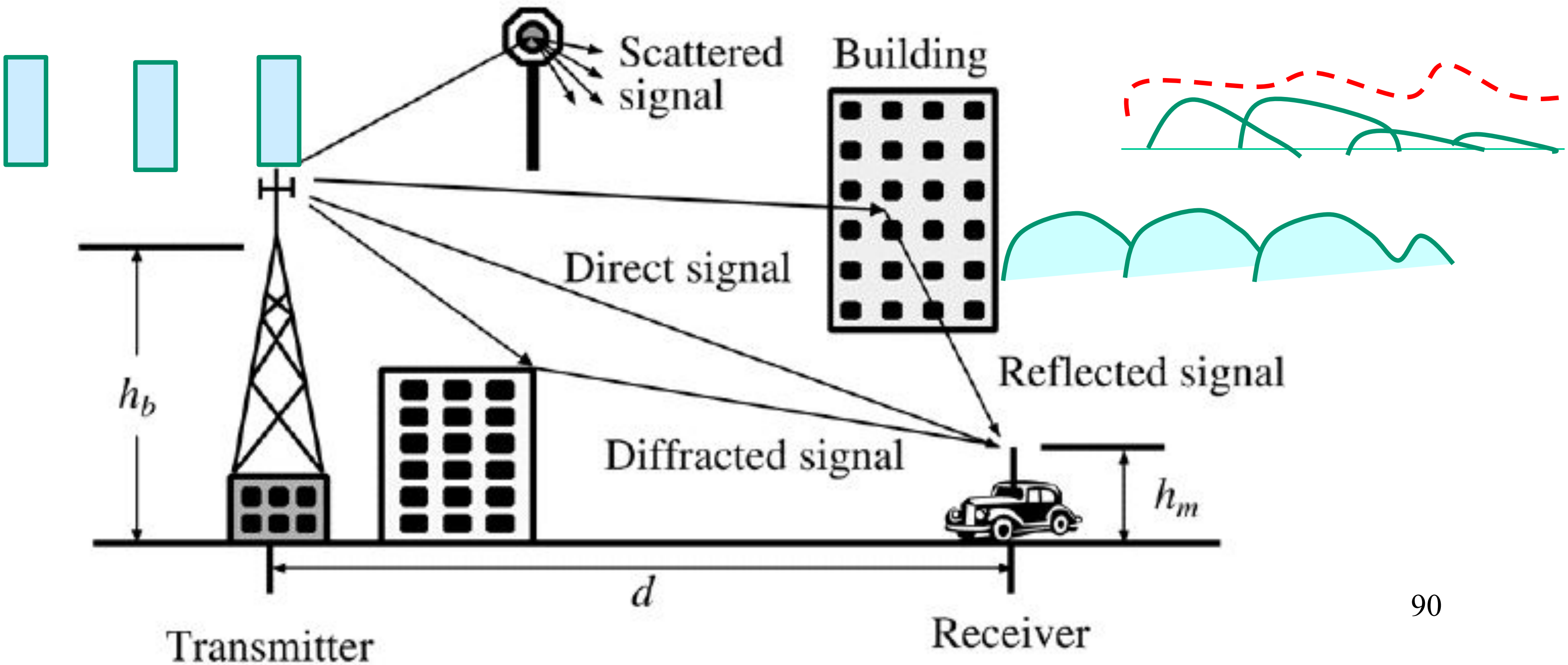
Consider the effect of RTS/CTS:

RTS alerts all stations within range of source (i.e., A) that exchange is under way;

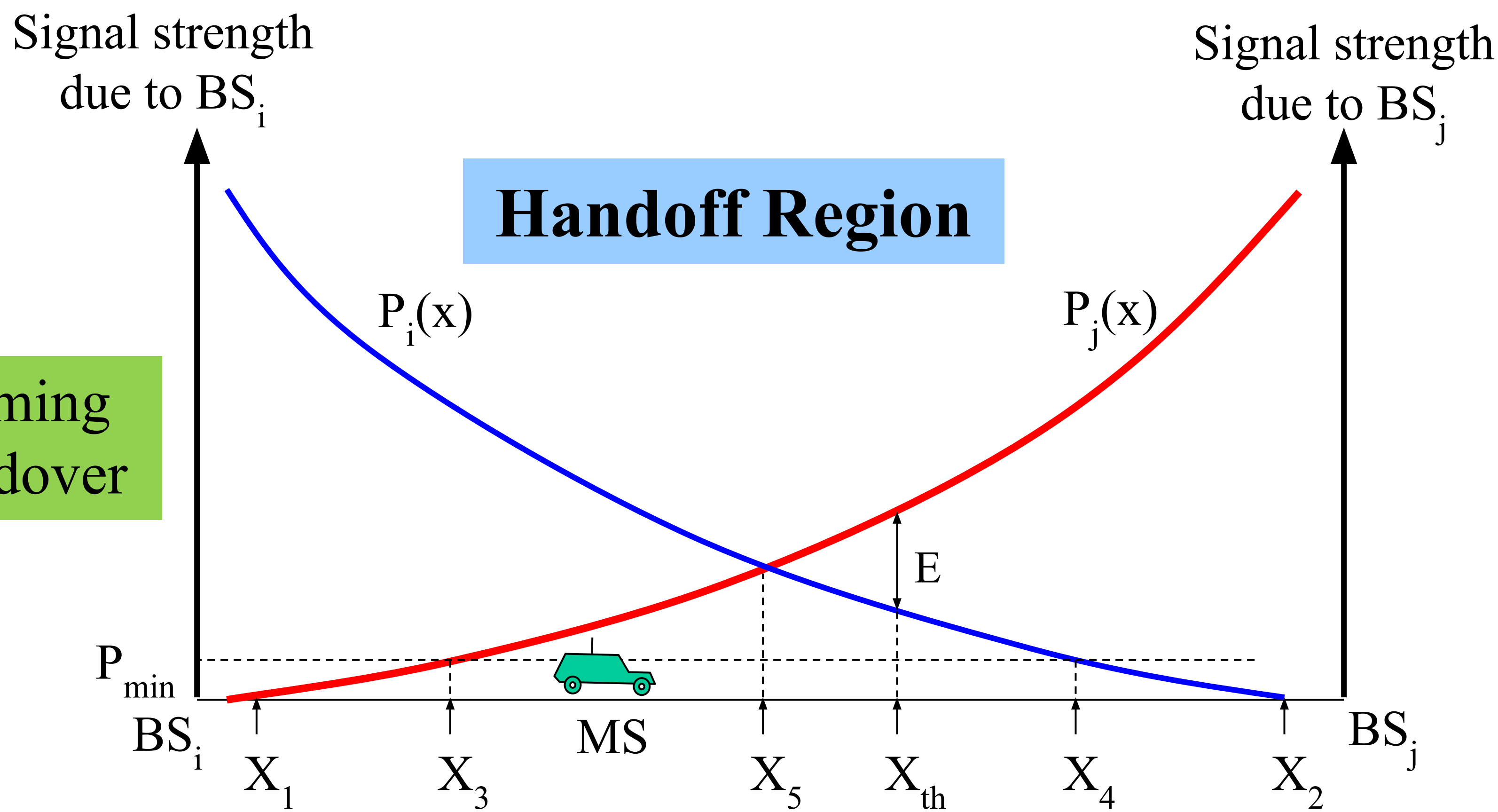
CTS alerts all stations within range of destination (i.e., B).

❖Second Problem: Multipath propagation (Multipath fading) due to presence of reflecting, diffracting and scattering objects hence cause multiple versions of the signal arrive at the receiver. In this case pulses at receiving end are smoothen and broadened hence overlapped with adjacent pulses. Such phenomenon creates Inter Symbol Interference (ISI).

With small variation of distance and time cause wide variation of received signal called **small scale fading** experienced in a dense city.



❖Third Problem: Handoff (handover) is necessary like mobile communications



- ✔ Roaming
- ✔ Handover

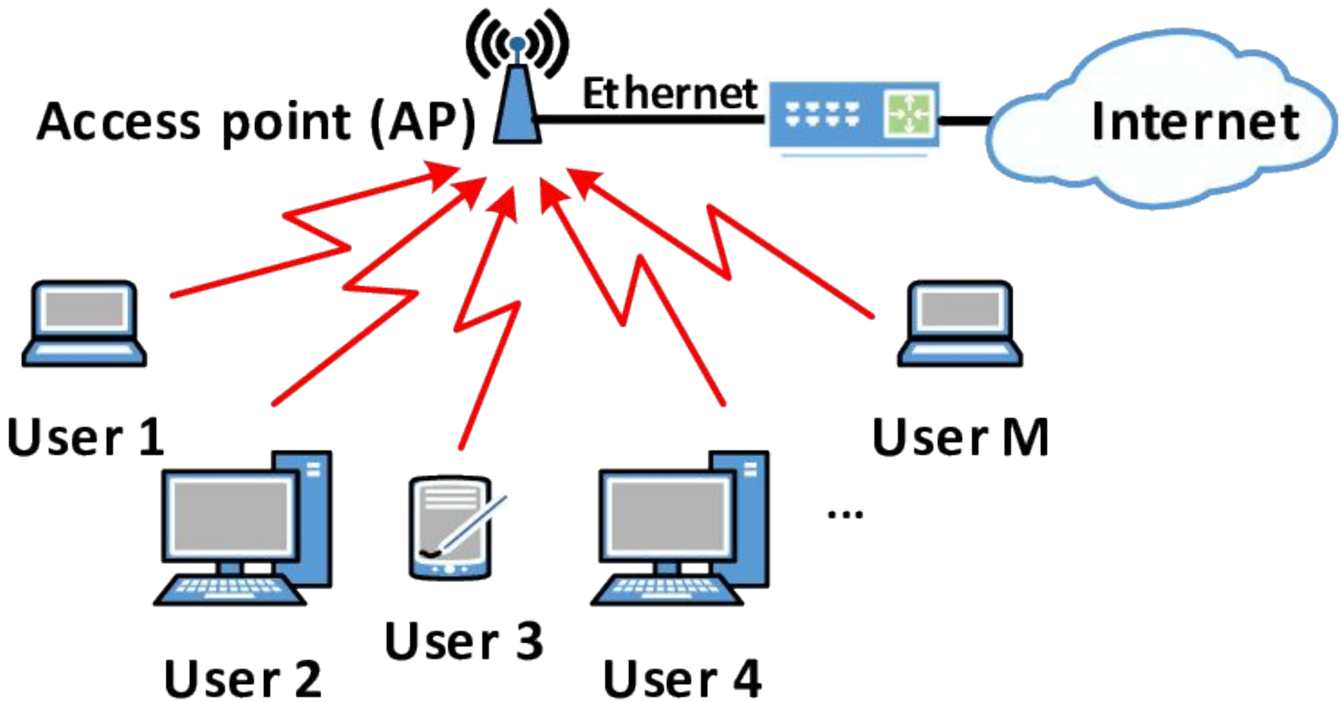
By looking at the variation of received signal strength of an MS from either base station, it is necessary to decide the appropriate location to initiate the handoff.

Forced Termination

- ✔ Delay in initiation of handoff.
- ✔ Going outside of the cover area.
- ✔ If no traffic channel is available.

❖Fourth Problem: Low data transfer rate than wired connection because of multipath propagation and fading channel. If the number of connected devices increases then data transfer rate also decreases.

❖Fifth Problem: Wireless transmission is more exposed to attack by unauthorized users, so you must pay particular attention to security (encryption at different layers of OSI).



Wireless LAN requirements:

- ❖Throughput
- ❖Number of nodes
- ❖Connection to backbone LAN
- ❖Battery power consumption
- ❖Transmission robustness and security
- ❖License free operation
- ❖Handoff/roaming
- ❖Dynamic Configuration

Wireless LAN Technologies

Wireless LAN technologies can be classified into five types according to physical transmission medium and access techniques:

- Infrared LAN
- Li-Fi
- Narrowband RF
- Wi-Fi (Spread-spectrum and OFDMA)
- Mobile Ad-hoc network (network without access point)

Infrared LANs

Each signal-covering cell in an infrared LAN is limited to one room. Coverage is small, since the infrared rays cannot penetrate through wall and other opaque obstacles. Three alternative transmission techniques are used for infrared data transmission:

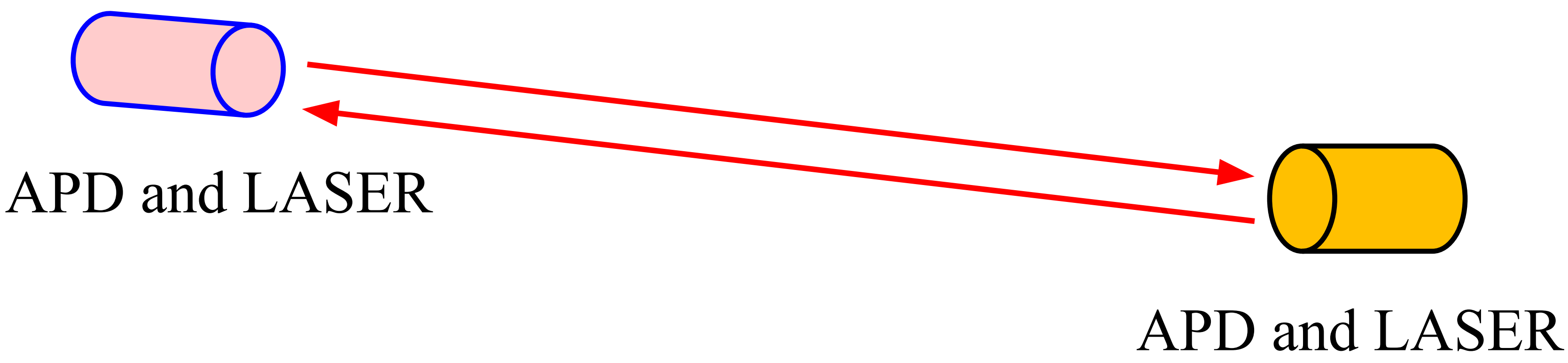
✔Directed beam infrared radiation.

✔Diffuse infrared radiation

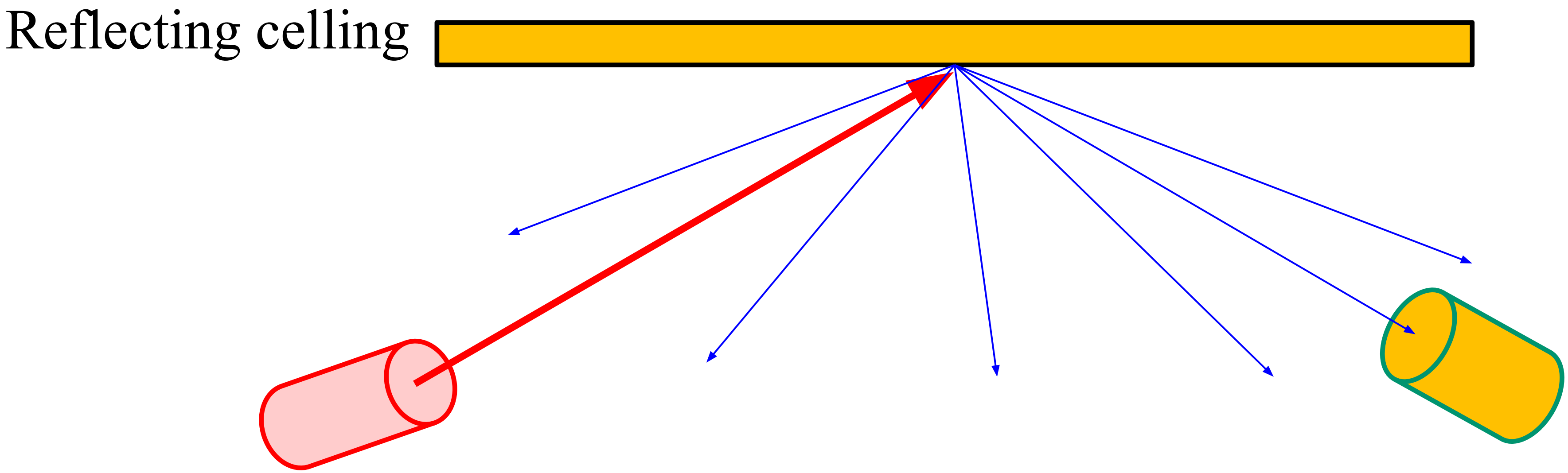
✔Omnidirectional configuration

Optical transmitter → LED, LASER
Optical Receiver→ APD, p-i-n diode

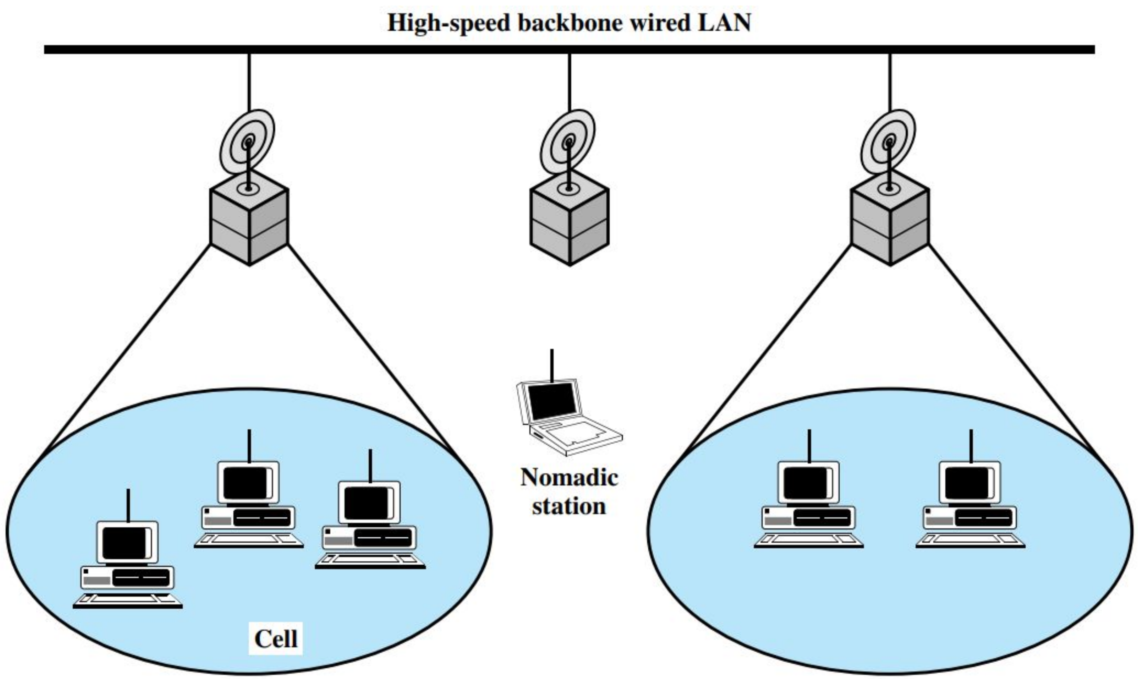
- In **DBIR** (Directed beam infrared radiation) system, the optical beam travels directly without any reflection from the transmitter to the receiver.
- Heavily directional beam of light (low beam aperture angle) between Tx and Rx is possible. The main drawback of this technique is the lack of mobility, and susceptibility (lacking the ability to resist) to blocking by personnel and machines.



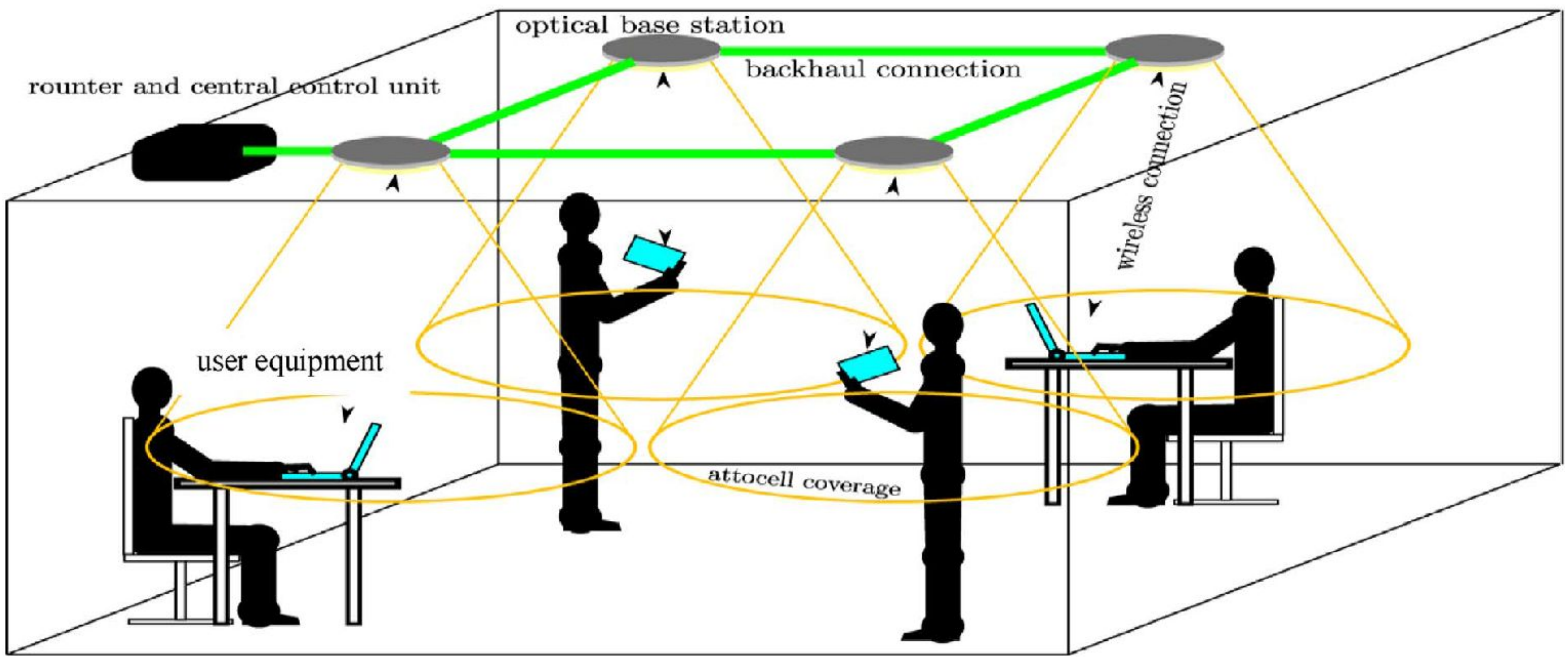
- In **DFIR** (Diffuse infrared radiation) system, the transmitters send optical signals in a wide angle to the ceiling and after one or several reflections the signals arrive at the receivers.
- This technique have a higher path loss than DBIR, requiring higher transmitter power levels and receivers with larger light collection area.
- Another challenging problem in this technique is the multipath dispersion causes inter-symbol- interference (ISI) at higher data rates or in larger cell system.



✔**Omnidirectional configuration** consists of a single BS that is normally mounted on ceilings. Here IR transceivers transmit with directional beam aimed at ceiling base unit, where ceiling transmitter broadcasts signal received by IR transceivers.



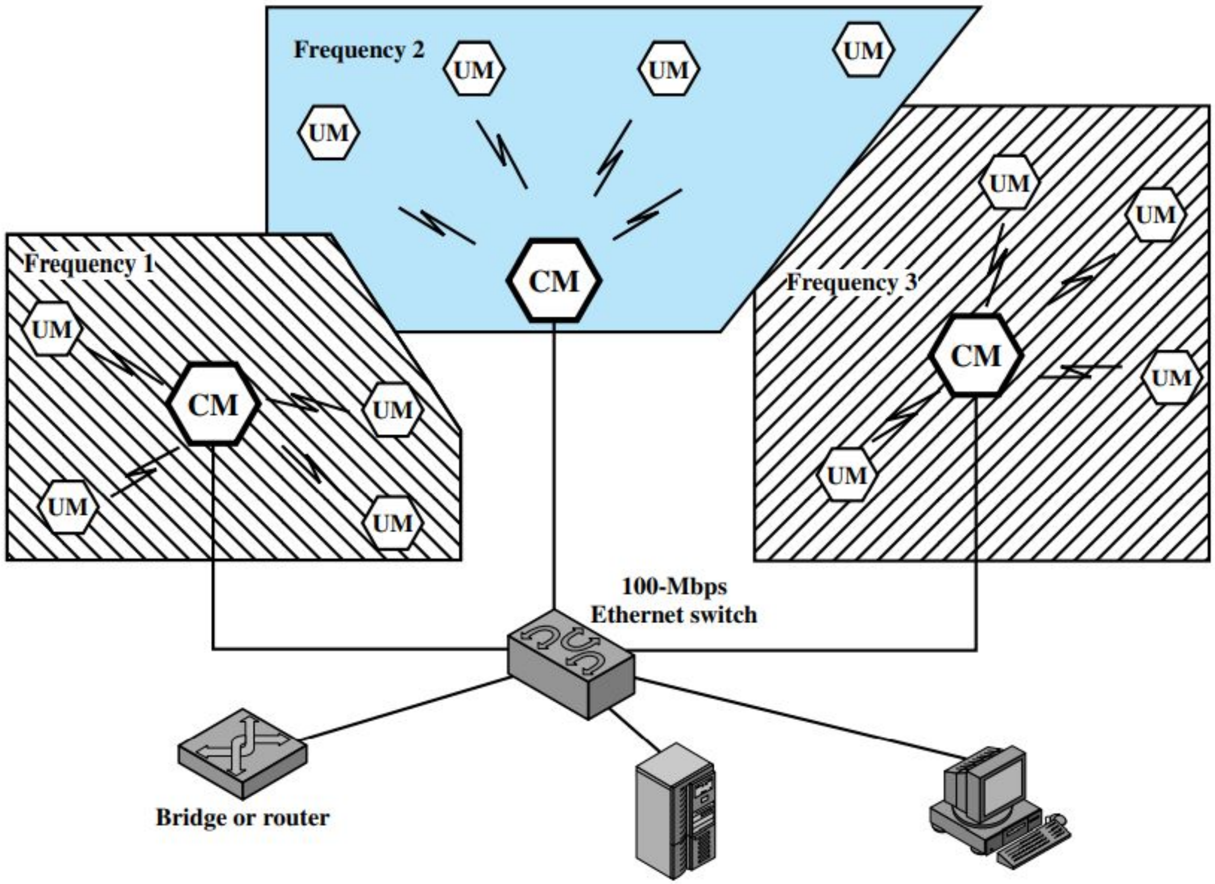
Light-Fidelity(LiFi) networking



- ✔ Li-Fi is a light communication system that is capable of transmitting data at high speeds over the visible light and infrared spectrums using LED lamp. Each light is driven by a LiFi modem also serves as an optical base station or access point (AP).
- ✔ The **optical base stations** are connected to the core network by high speed backhaul connections.

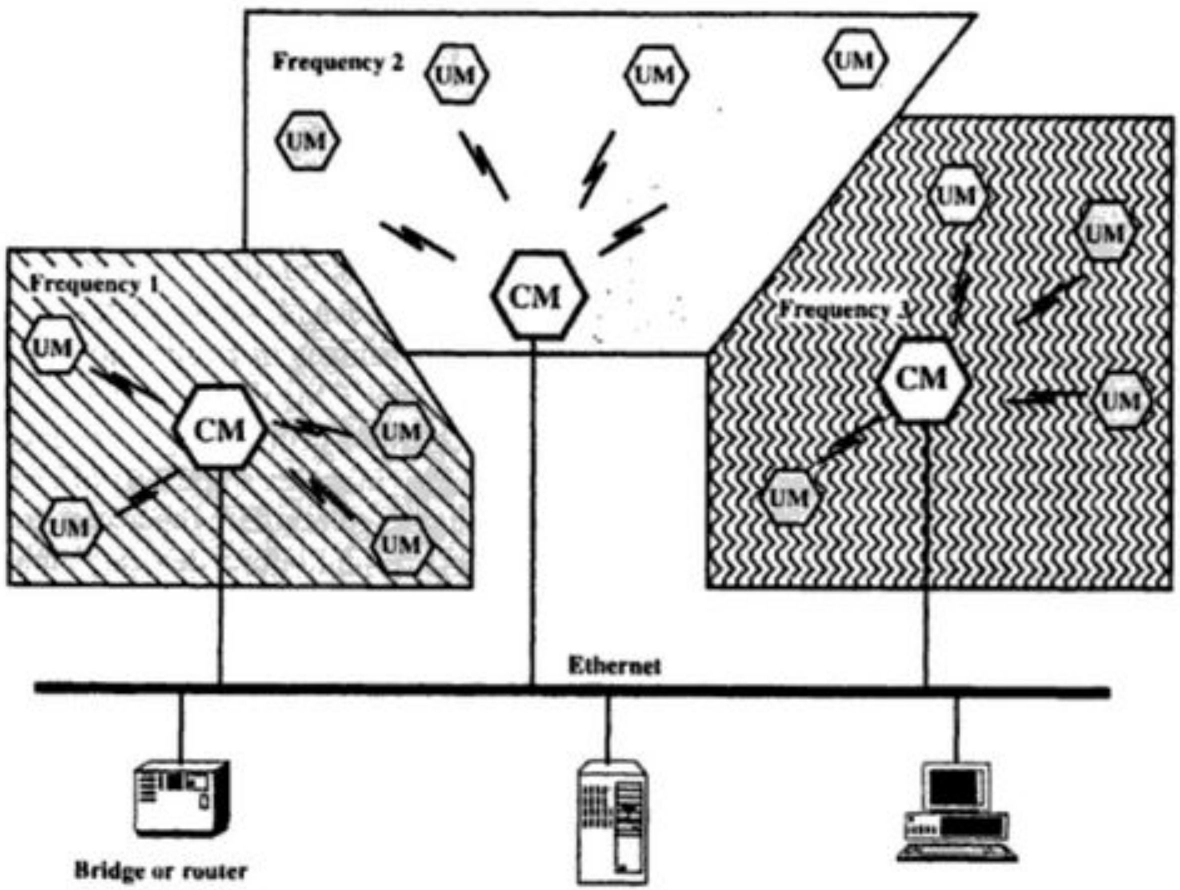
Narrowband RF LANs

Narrowband RF LANs use very narrow bandwidth. Adjacent cells use different frequency bands. The transmissions are encrypted to prevent attacks.



UM→ User Module

There are several **Control Module** (**CM**) to interface wireless LAN to the backbone Ethernet (wired LAN).In previous figure each cell has its own CM and connected by a switch to route traffic among them.



UM→ User Module

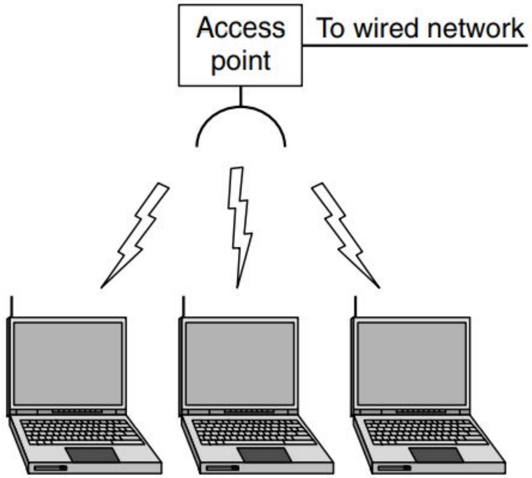
CM → Control Module

IEEE 802 Activities and Wi-Fi (IEEE 802.11)

✔ The **Institute of Electrical and Electronics Engineers (IEEE**, read *I-Triple-E*) is the largest technical professional association headquartered in New York City that is dedicated to advancing technological innovation and excellence. It has more than 400,000 members in more than 160 countries, about 51.4% of whom reside in the USA.

The IEEE 802 LAN/MAN Standards Committee (includes personal, local, and metropolitan area networks) innovates and specifies the Ethernet and wireless standards.

- Wired
  - 802.3: Ethernet
  - 802.17: Packet Ring (new)
- Wireless
  - 802.11: Wireless LAN
    - Local Area Network
  - 802.15: Wireless PAN
    - – Personal Area Network (e.g. Bluetooth™)
  - 802.16: WirelessMAN™
    - – Metropolitan Area Networks

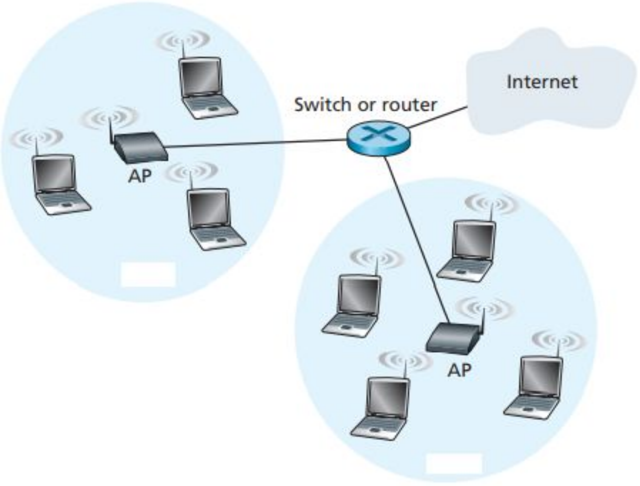


- Wireless LAN access based on IEEE 802.11 technology, is known as Wi-Fi, is now just about everywhere—universities, business offices, cafes, airports, homes, and even in airplanes.

- 802.11 networks are made up of clients, such as laptops and mobile phones, and infrastructure called APs (access points) that is installed in buildings.

- Access points are sometimes called base stations. The access points connect to the wired network, and all communication between clients goes through an access point.

- The services and protocols specified in IEEE 802 map to the lower two layers (data link and physical).



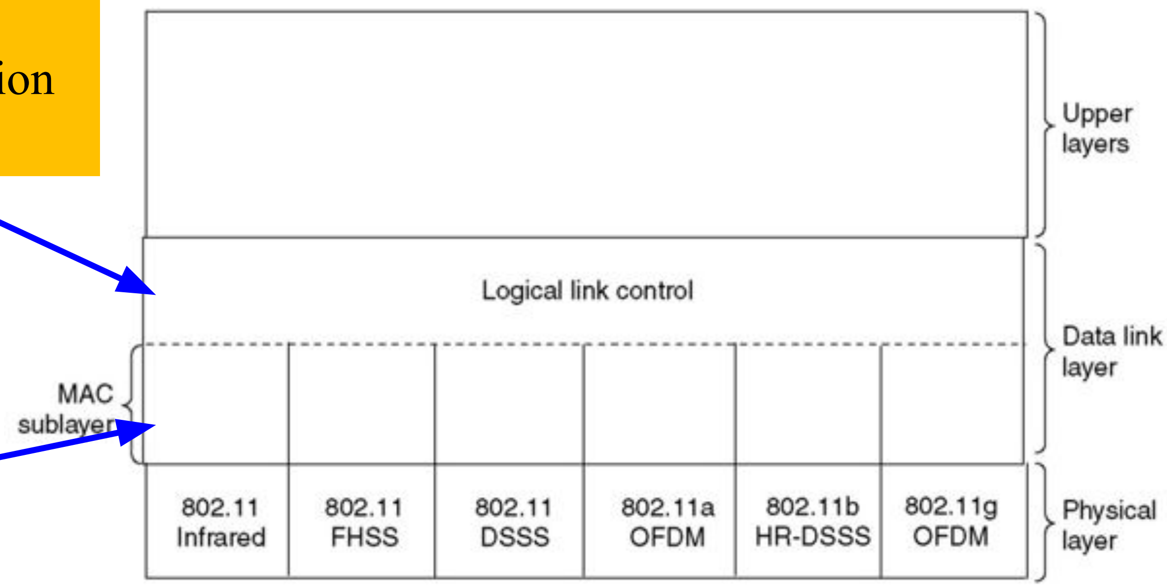
The 802.11 (Wi-Fi) Protocol Stack

- The physical layer corresponds to the OSI physical layer fairly well, but the data link layer in all the 802 protocols is split into two or more sub-layers.

- In 802.11, the MAC (Medium Access Control) sub-layer determines how the channel is allocated, that is, who gets to transmit next. Above it is the LLC (Logical Link Control) sub-layer, whose job it is to hide the differences between the different 802 variants as far as the network layer is concerned.

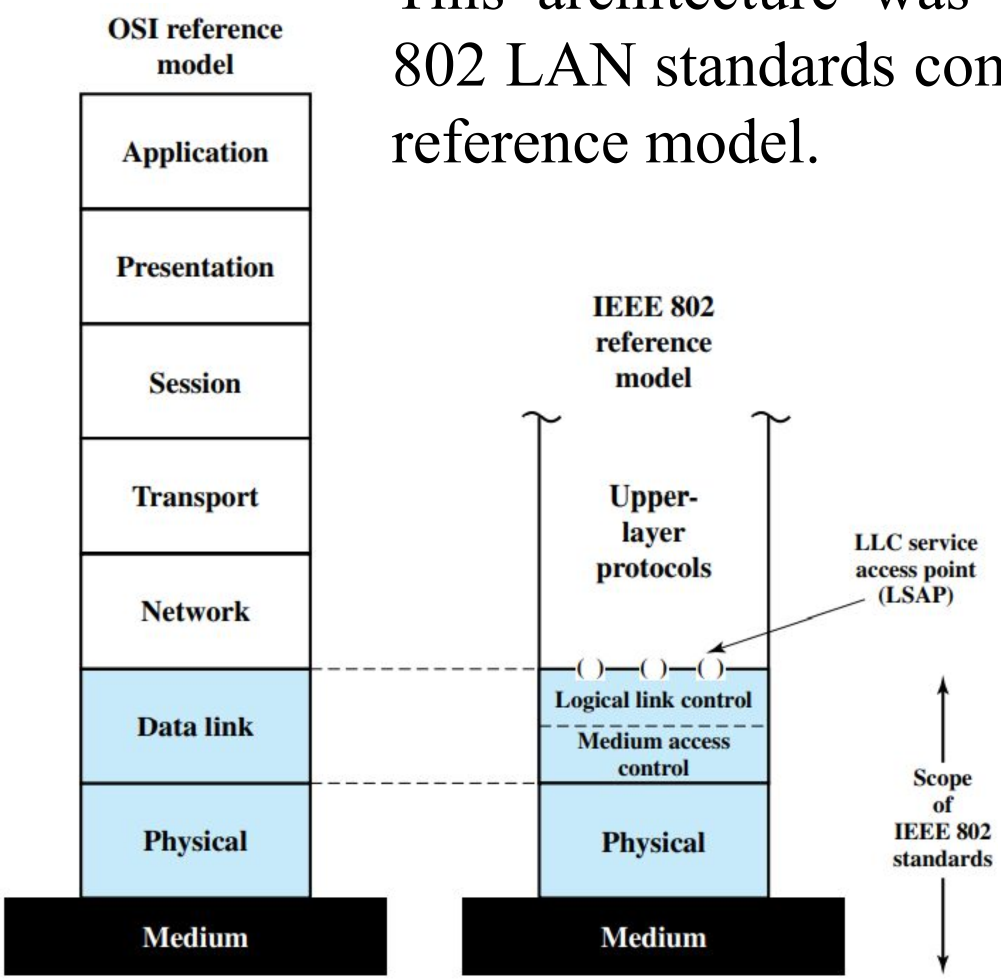
Framing
Error correction/detection
Flow control

Channel allocation

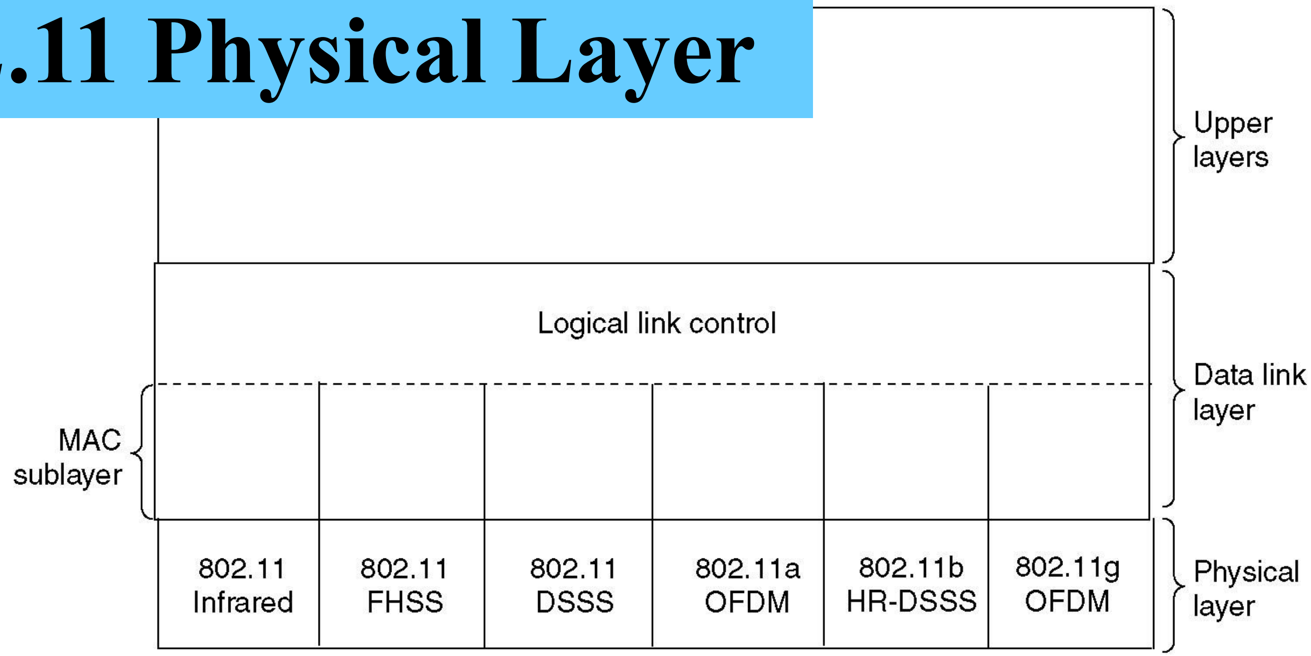


IEEE 802 v OSI

This architecture was developed by the IEEE 802 LAN standards committee called IEEE 802 reference model.



The 802.11 Physical Layer



The signal at **physical layer** has the following forms:

Infrared with PPM scheme

FHSS (Frequency Hopping Spread Spectrum)

DSSS (Direct Sequence Spread Spectrum)

OFDM (Orthogonal Frequency Division Multiplexing)

HR-DSSS (High Rate DSSS)

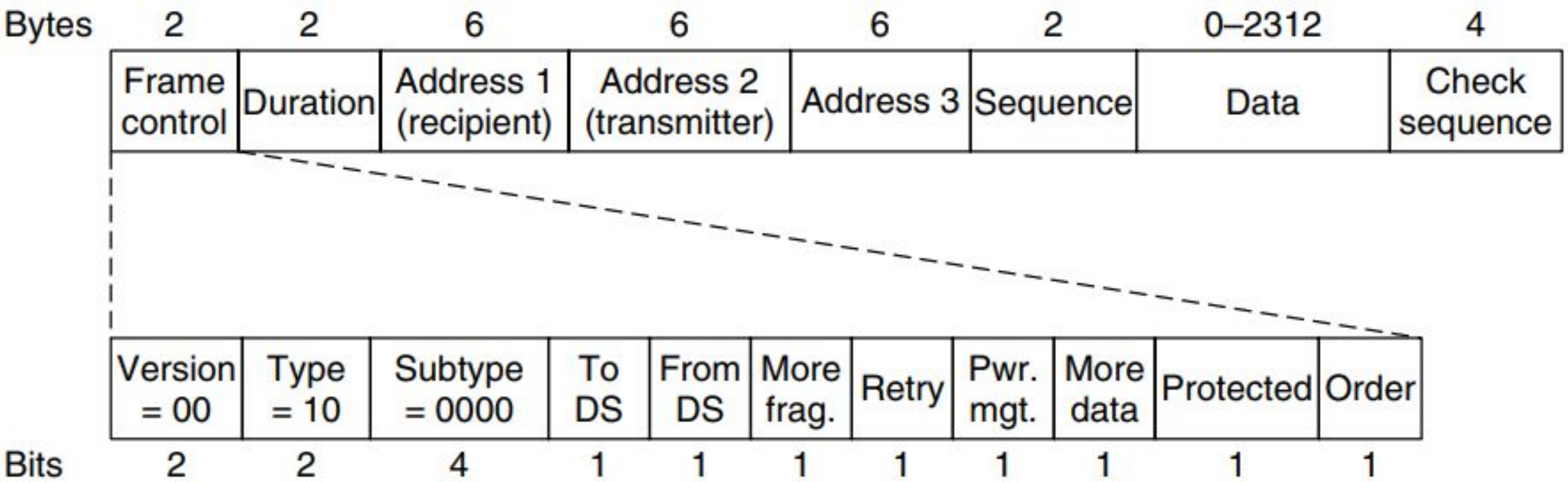
Physical Layer

IEEE	*Technique*	*Band*	*Modulation*	*Rate (Mbps)*
802.11	FHSS	2.4 GHz	FSK	1 and 2
	DSSS	2.4 GHz	PSK	1 and 2
		Infrared	PPM	1 and 2
802.11a	OFDM	5.725 GHz	PSK or QAM	6 to 54
802.11b	DSSS	2.4 GHz	PSK	5.5 and 11
802.11g	OFDM	2.4 GHz	Different	22 and 54

Data Link Layer of IEEE 802.11

The **Data Link** layer of IEEE 802.11 has two parts: LLC (creation of frame, error control and flow control) and MAC (allocation of channel) sublayer. This layer provides several key functionalities: reliable data delivery, media access control and security features.

The frame format of IEEE 802.11

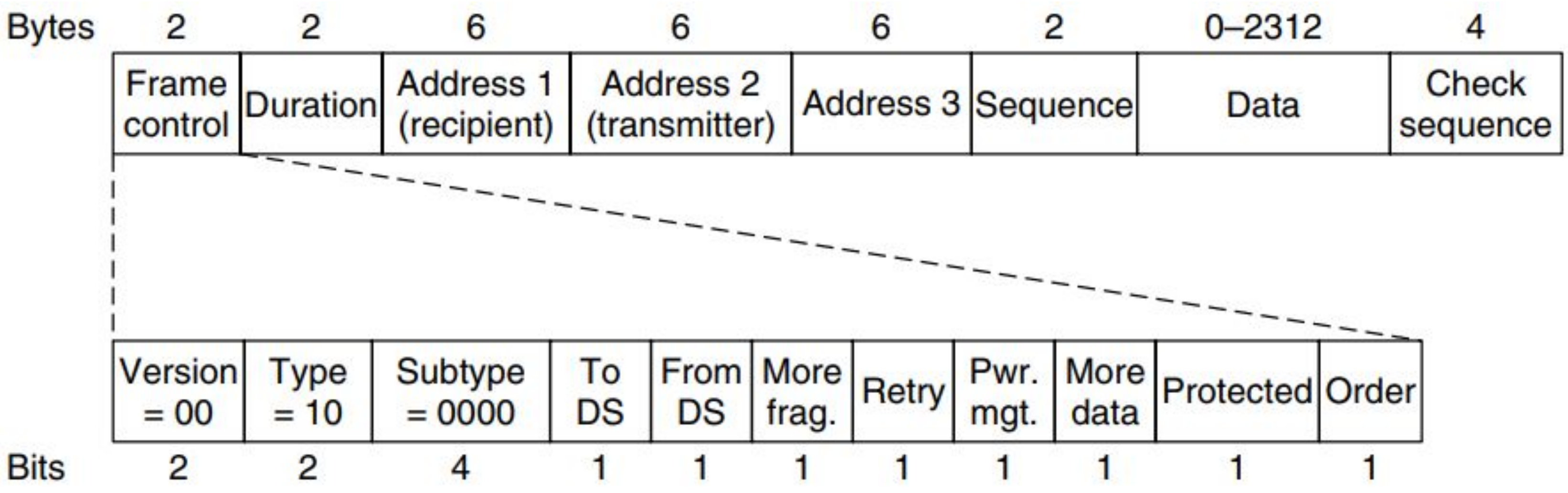


- The frame control (FC) field provides information on the type of frame has 11 subfields.
- The first of these is the Protocol **version**, two bits representing the protocol version. Currently used protocol version is zero. Other values are reserved for future use.
- Then come the **Type** of 2 bits (data, control, or management); where 'Type of information: management (00), control (01), or data (10)'.

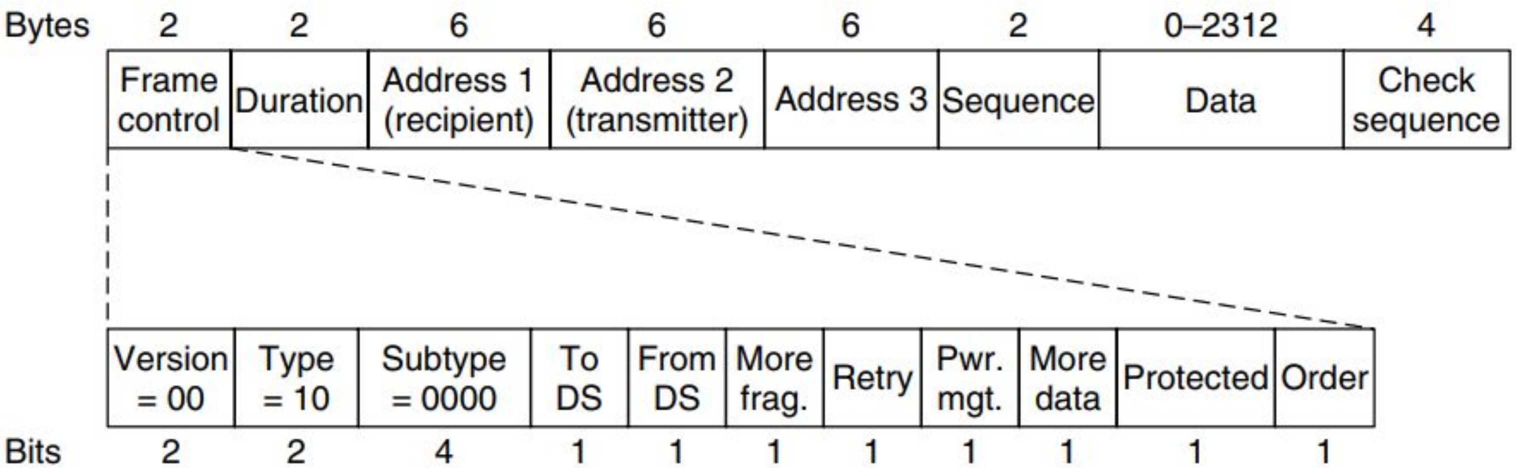
- A node wishing to send data initiates the process by sending a Request to Send frame (RTS). The destination node replies with a Clear To Send frame (CTS). **Subtype** fields indicates like the following category.
- It is a 4 bit subfields states whether the field is a Request to send (RTS) or Clear to send(CTS) control frame. For a regular data frame the value is set to 0000

Values of subfields in control frames

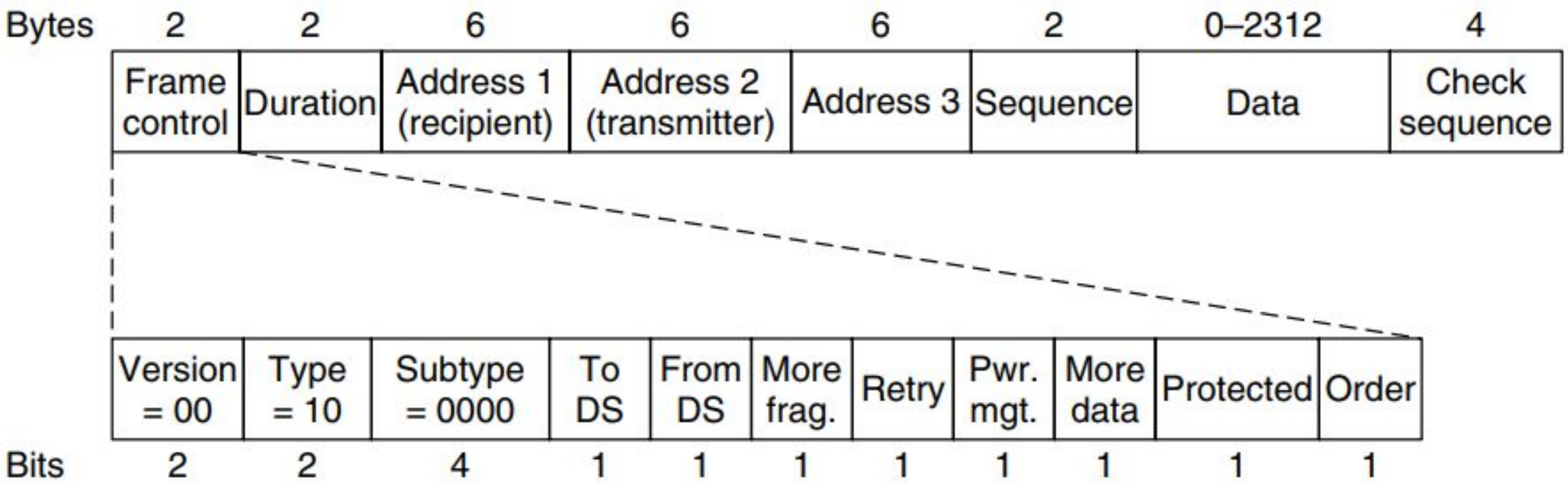
Subtype	*Meaning*
1011	Request to send (RTS)
1100	Clear to send (CTS)
1101	Acknowledgment (ACK)



- ✔ The flags To DS and From DS bits are set to indicate whether the frame is going to or coming from the network connected to the AP(access point).
- ✔ Data frames sent to or from an AP have three addresses, all in standard IEEE 802 format. The first address is the receiver, and the second address is the transmitter. The third address is address of distant endpoint (for example AP of receiver).



- The More fragments bit means that more fragments will follow.
- The single bit subfield which when set to 1 specifies a retransmission of the previous frame. The Retry bit marks a retransmission of a frame sent earlier.
- The Power management bit is used by the base station to put the receiver into sleep state or take it out of sleep state. If the bit set to 1 that means the sender is operating power saving mood.
- The More data bit indicates that the sender has more data frames for the receiver.
- The **Protected** bit specifies that the frame body has been encrypted using the WEP (Wired Equivalent Privacy) algorithm. Finally, the **Order** bit tells the receiver that a sequence of frames with this bit on must be processed strictly in order.



- the Duration field, tells how long the frame and its acknowledgement will occupy the channel, measured in microseconds.
- The address field denotes the 6-byte (MAC address of 48 bits) source, destination and distant end point (BS of end node).
- The sequence control (SC) field consists of 2 bits reserved for fragmentation and reassembly . It detects duplicate frames and determines the order of frames for higher layer.
- Data field contains the packet of maximum 2312 bytes.
- The 32-bit CRC field is used for error detection.

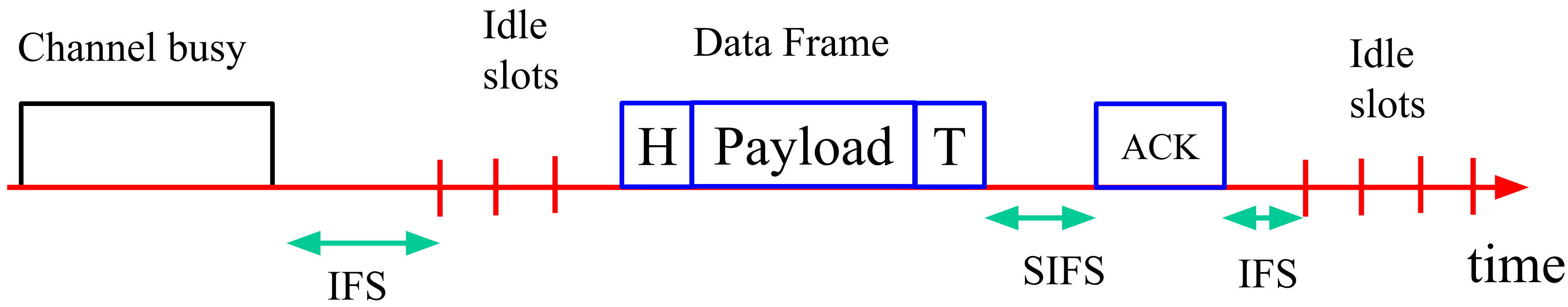
To DS	*From* DS	*Address* 1	*Address* 2	*Address* 3	*Address* 4
0	0	Destination	Source	BSS ID	N/A
0	1	Destination	SendingAP	Source	N/A
1	0	Receiving AP	Source	Destination	N/A
1	1	Receiving AP	SendingAP	Destination	Source

Carrier Sense Multiple Access/ Collision Avoidance (CSMA/CA)

- The wireless LAN system cannot detect collisions because the power of the transmitting device is much stronger than the its received signal power.
- In this situation collision detection is not practical, it makes sense to try to devise a system that can help prevent collisions. Thus the CA is CSMA/CA refers to 'collision avoidance'.
- In wired communication both the transmitted and received signals are equally strong therefore collision can be detected easily, whereas situation is not favorable in wireless link, there collision avoidance (CA) is used in wireless network instead of collision detection (CD).

- Sender usually uses **binary exponential backoff** to avoid collision.

Binary Exponential Backoff Algorithm



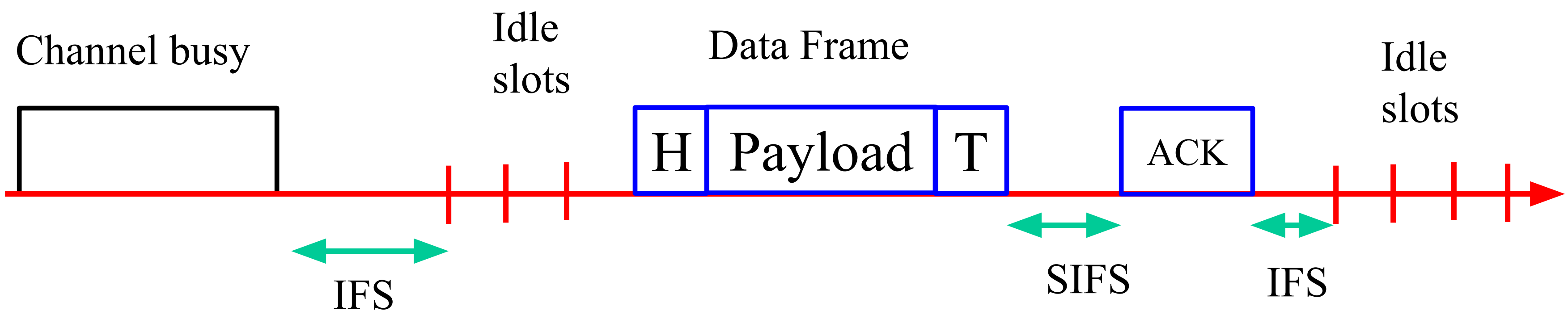
To reduce the packet dropping probability or to enhance throughput of wireless LAN exponential binary backoff algorithm is widely used. The access method of MAC protocol of IEEE 802.11 based on exponential binary backoff algorithm can be explained with the following steps.

Step: 1

The transmitting node first senses the status of the channel. If the channel is found busy then the Tx node continues to monitor the channel.

Step:2

If the channel is found idle for a fixed duration know as IFS (Inter-frame Space), the Tx chooses a random number according to the binary exponential back off algorithm. The random number is used as a back off timer.

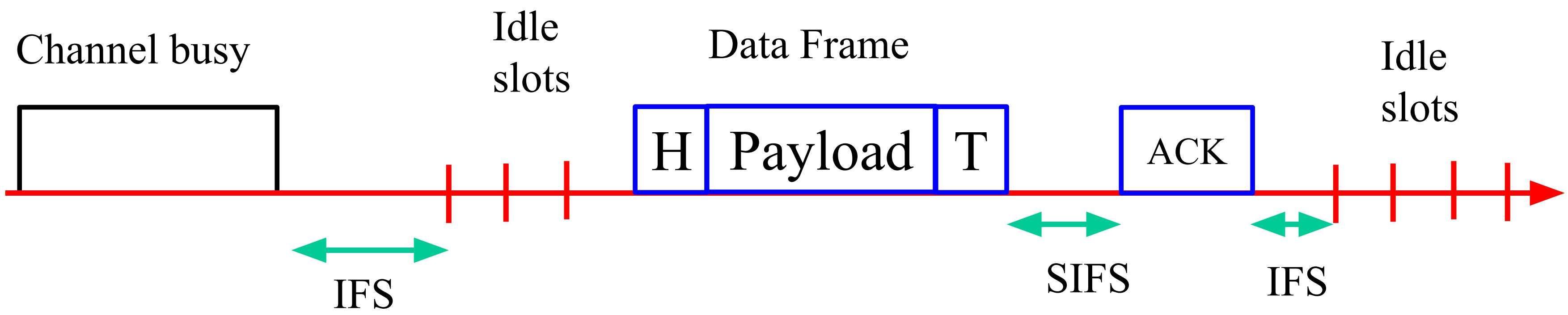


Step:3

Time immediately after the IFS (Inter-frame Space) is slotted known as idle slots where the duration of a slot is considered as the sum of the time required to sense a station and to switch the Tx from sensing / listening mode to transmitting mode.

Step:4

Elapsing of each idle slot the back off timer is decreased by one. If the channel is found busy before the back off timer reaches to zero then repeat the steps 1 to 3. The transmission of data from begins only if the back off timer reaches to zero.



Step:5

✔To determine whether a data frame transmission is successful, after its completion, a positive acknowledgement (ACK) is transmitted by the receiver. ACK is transmitted after a short interframe space (SIFS) period upon receiving the entire data frame successfully.

✔If ACK is not detected within an SIFS period after the completion of the data frame transmission, the transmission is assumed to be unsuccessful, and a retransmission is required.

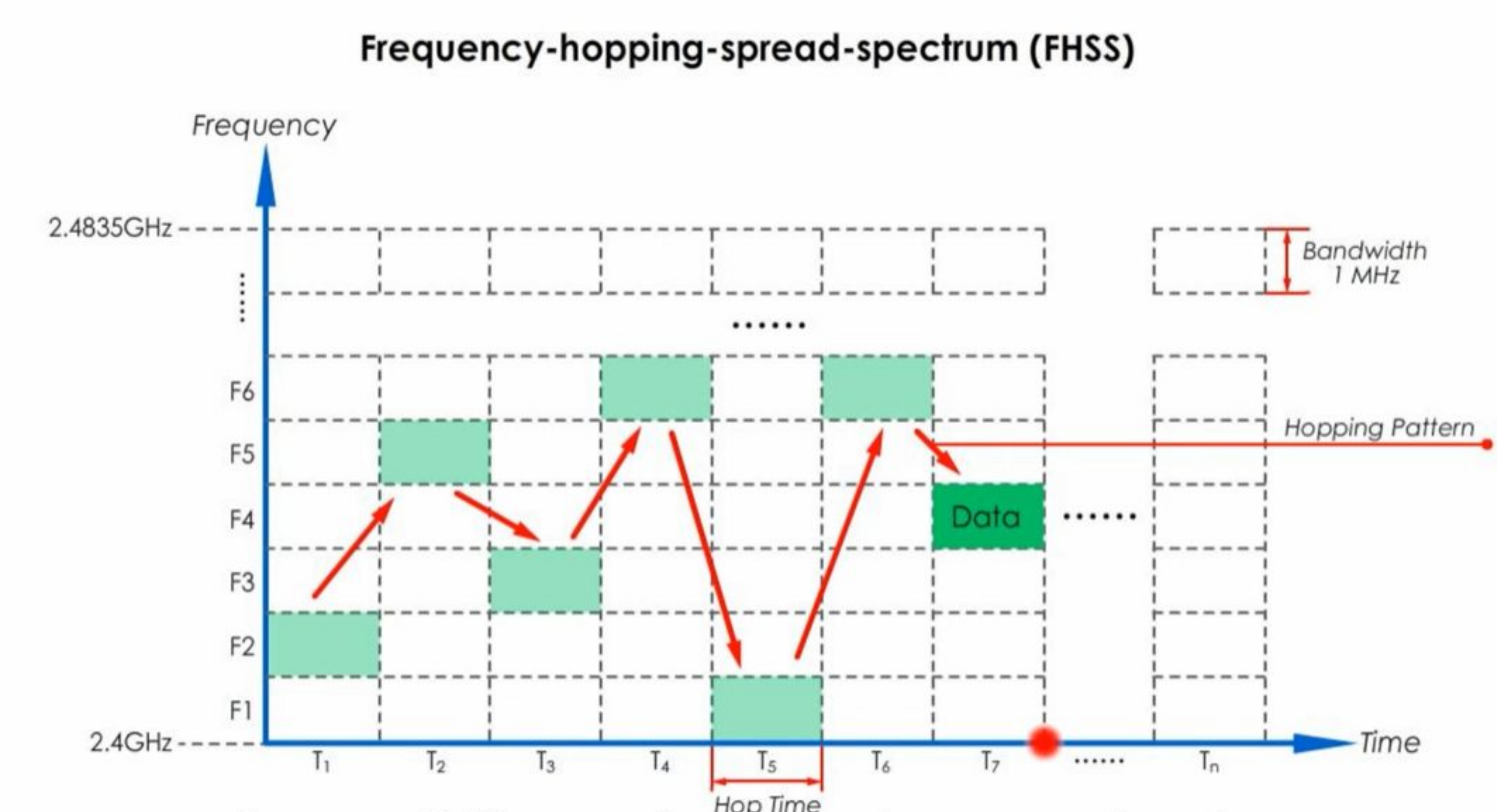
Access Techniques of WLAN (Wi-Fi) CDMA and OFDMA

Spread Spectrum (CDMA) LANs

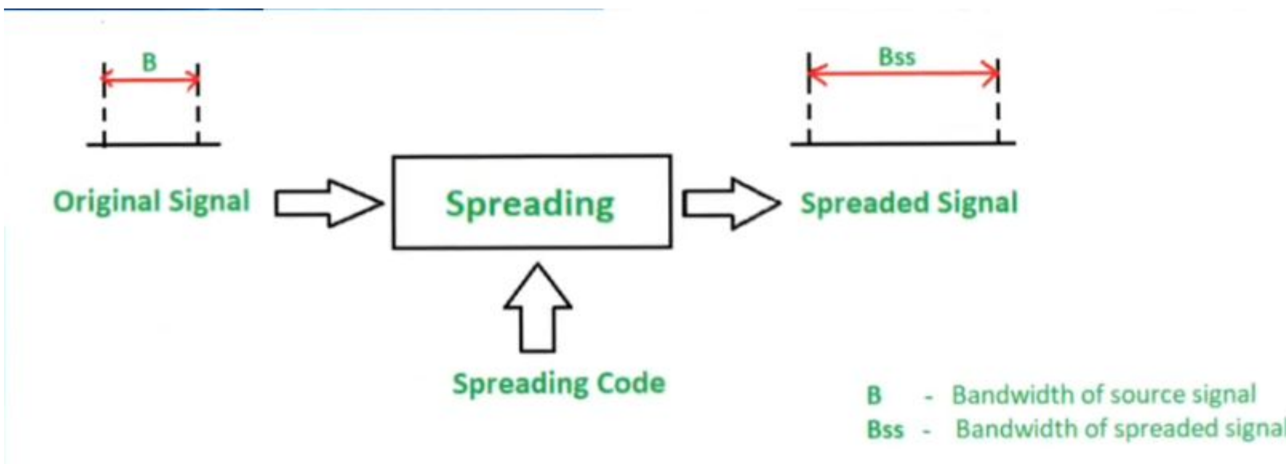
- ✔ The idea behind **spread spectrum** is to spread the signal over a wider frequency band than normal in such a way as to minimize the impact of interference from other devices. ***Frequency hopping*** is a spread spectrum technique that involves transmitting the signal over a random sequence of frequencies, that is, first transmitting at one frequency, then a second, then a third, and so on.

- ✔ For example, **802.11** (wireless LANs) provides 1 or 2 Mbps transmission in the 2.4 GHz band using either frequency hopping spread spectrum (FHSS) which uses 2 or 4 level FSK or direct sequence spread spectrum (DSSS) which uses BPSK or QPSK.

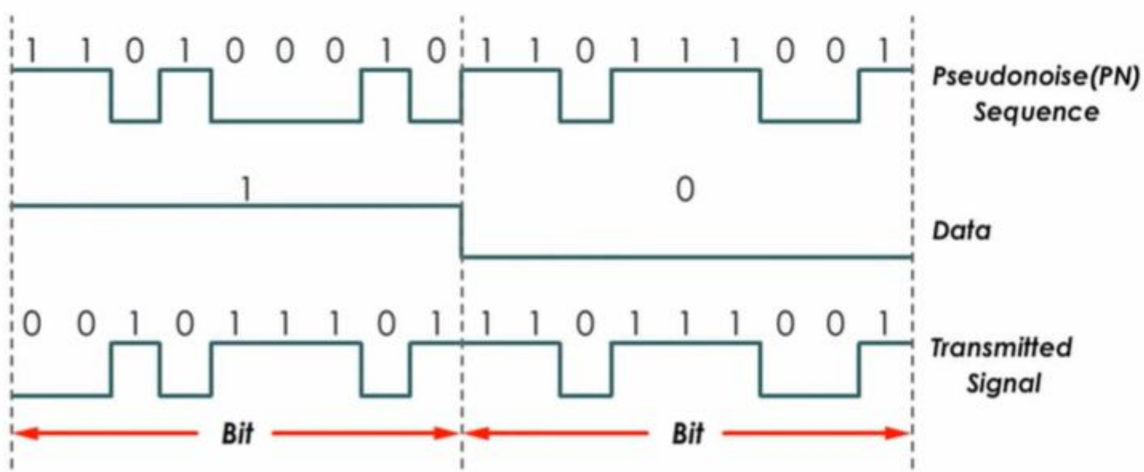
- ✔ It forms the basis of 3G mobile phone networks and is also used in GPS (Global Positioning System).



Frequency-hopping-spread-spectrum (FHSS)
Frequency
2.4835GHz
F6
F5
F4
F3
F2
F1
2.4GHz
Bandwidth
1 MHz
Hopping Pattern
Data
Time
T1
T2
T3
T4
T5
T6
T7
Tn
Hop Time

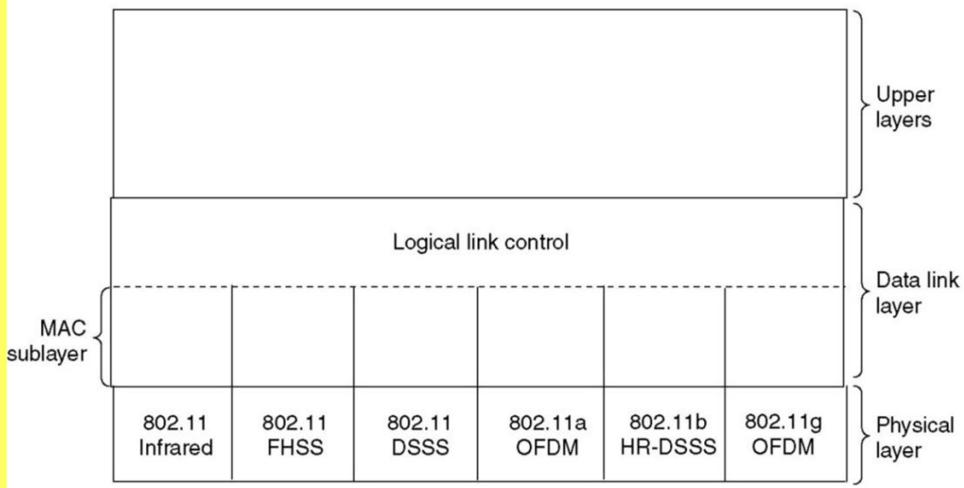


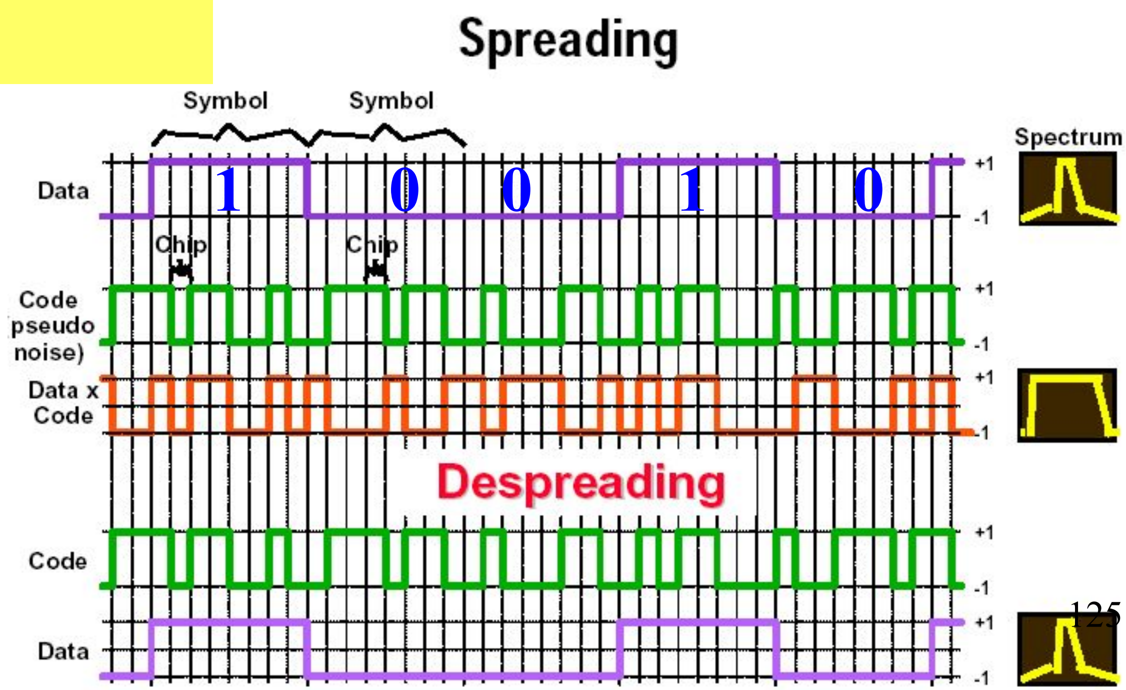
Inputs		Outputs
X	Y	Z
0	0	0
0	1	1
1	0	1
1	1	0

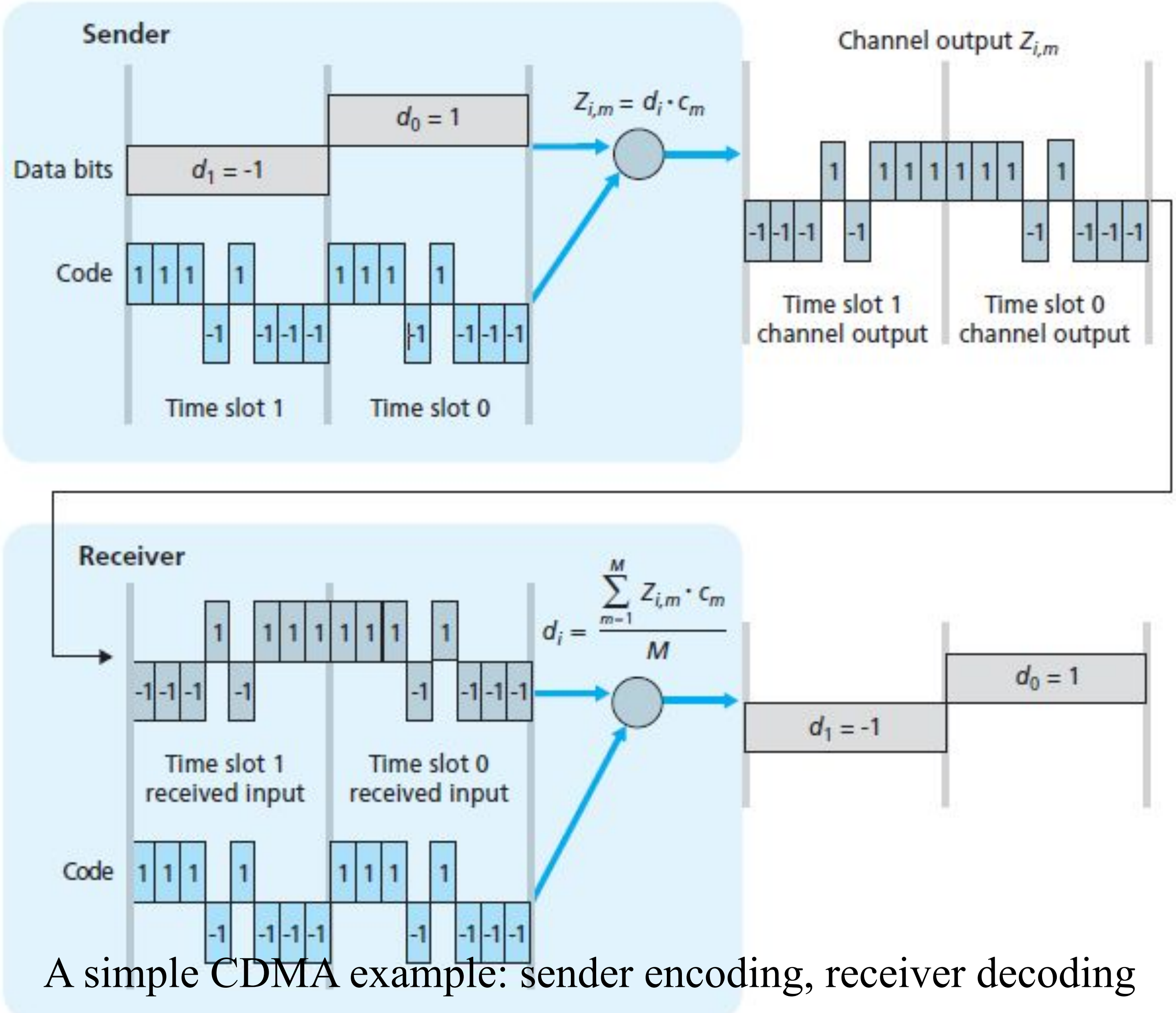


DS Spectrum Spreading Technique

- Spread spectrum involves the use of a much wider BW than actually necessary to support a given data rate. The result of using wider BW is to minimize interference and drastically reduce BER. It operates in 2.4GHz band at data rate of 1Mbps and 2Mbps.

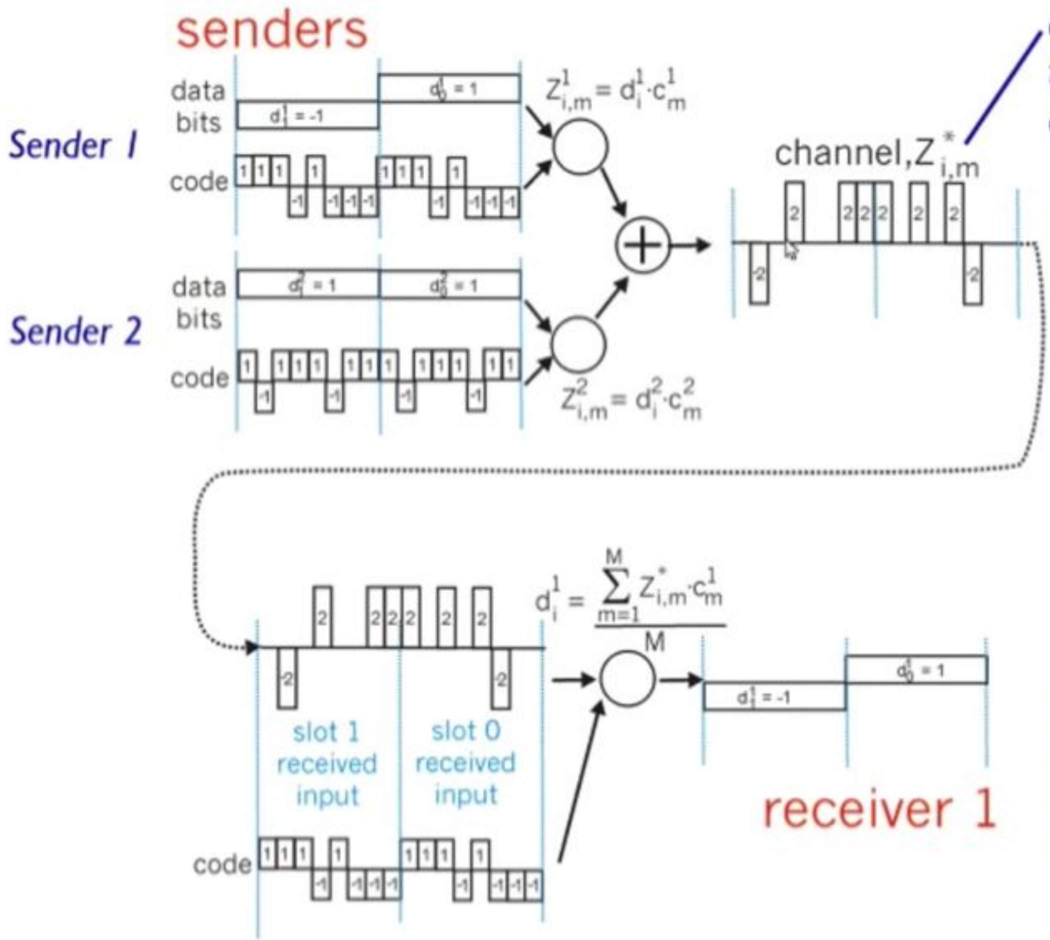






A simple CDMA example: sender encoding, receiver decoding

CDMA: two-sender interference



channel sums together transmissions by sender 1 and 2

using same code as sender 1, receiver recovers sender 1's original data from summed channel data!

Orthogonal Frequency Division Multiplexing IEEE 802.11a and 802.11g

- ***Orthogonal Frequency Division Multiplexing*** (OFDM) is a multi-carrier modulation scheme that transmits data over a number of **orthogonal subcarriers**. A conventional transmission uses only a single carrier modulated with all the data to be sent.
- OFDM breaks the high speed data string into low speed data string using S/P converter. Each sub-data stream is modulated by individual orthogonal sub-carrier and the data is sent in parallel. As illustrated in Figure 1, this can be compared with a transport company utilizing several smaller trucks (multi-carrier) instead of one large truck (single carrier).

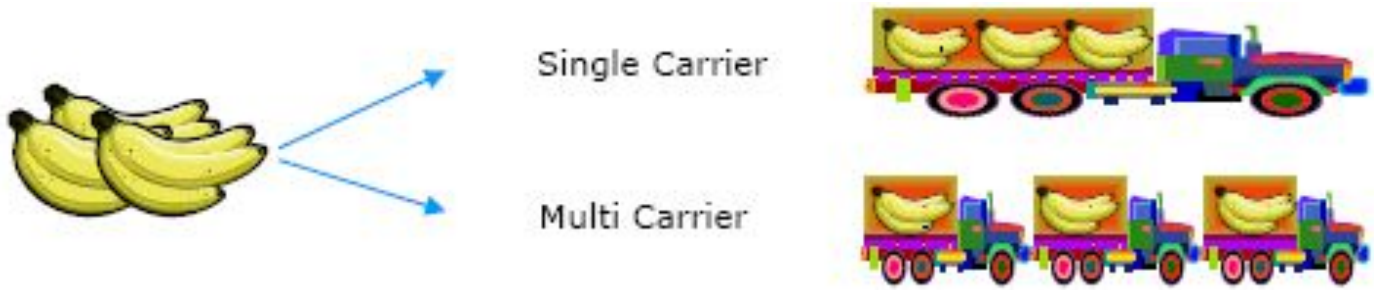
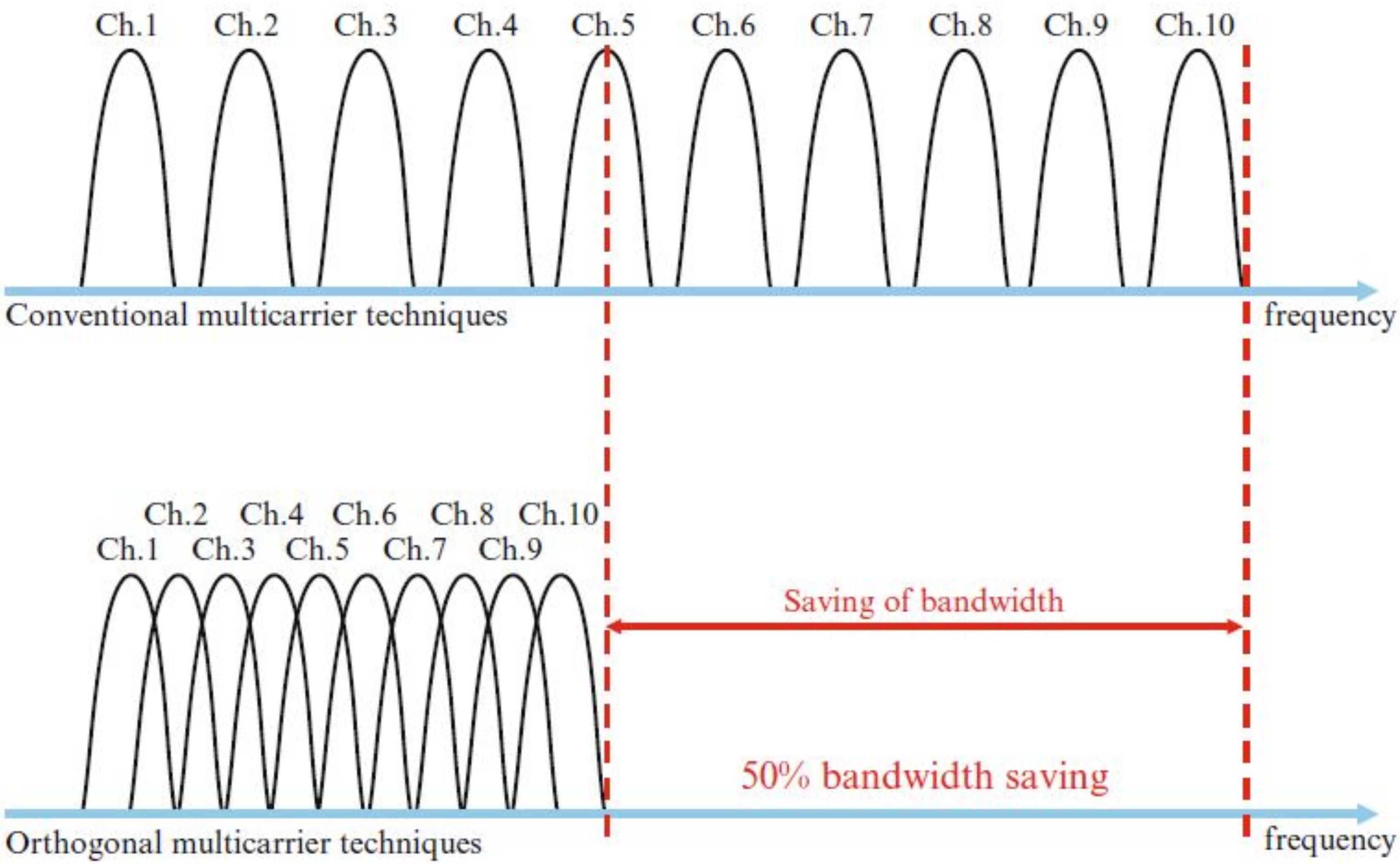
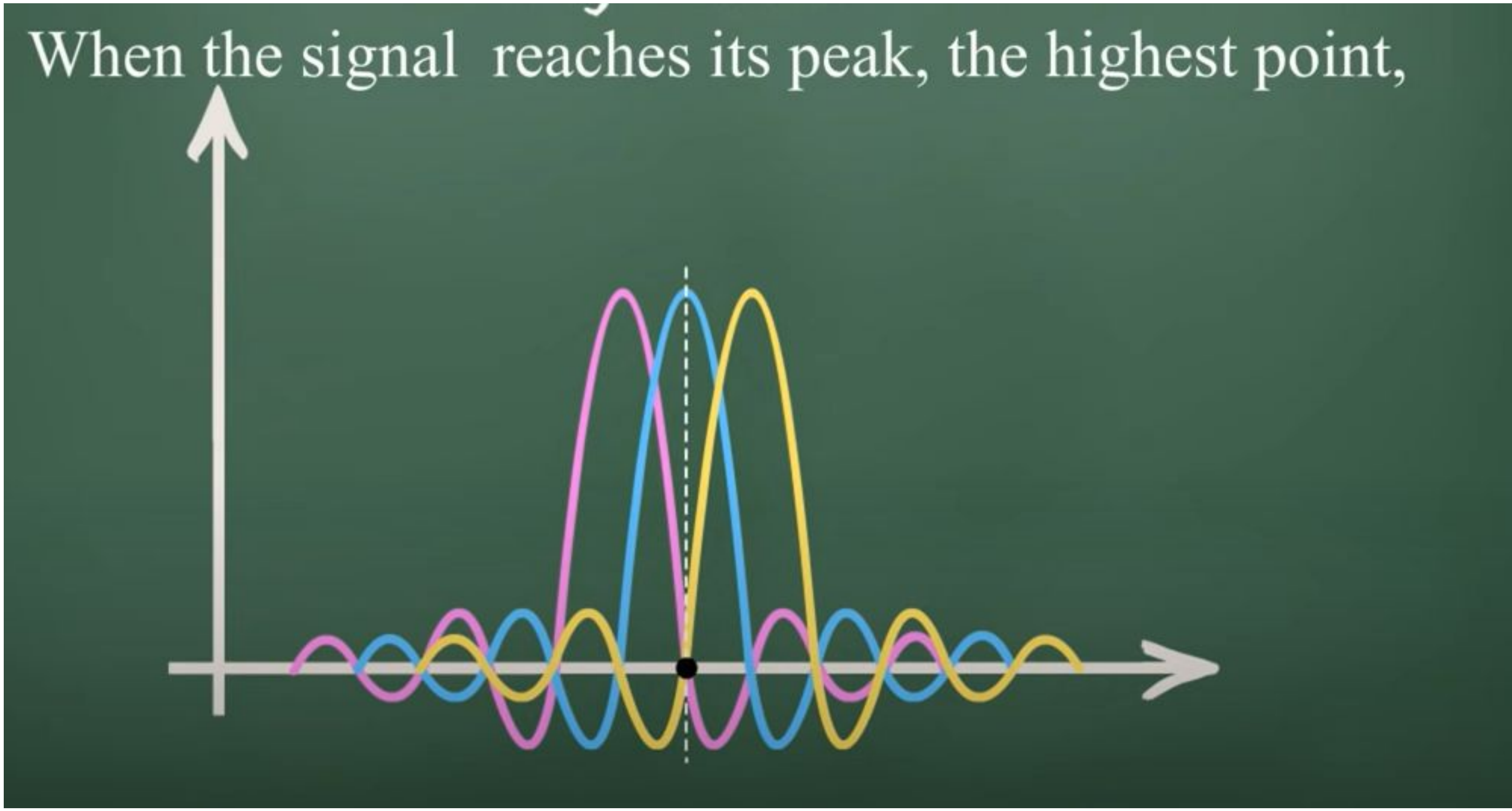


Fig.1 Single carrier vs. multi-carrier transmission

OFDM Versus FDM




When the signal reaches its peak, the highest point,

OFDM offers many advantages over single-carrier modulations:

1. It elongates the symbol period so that the signal is more robust against inter symbol interference caused by channel dispersions and multipath interference.

2. It divides the entire frequency band into narrow bands so that it is less sensitive to wide-band impulse noise and fast channel fades.

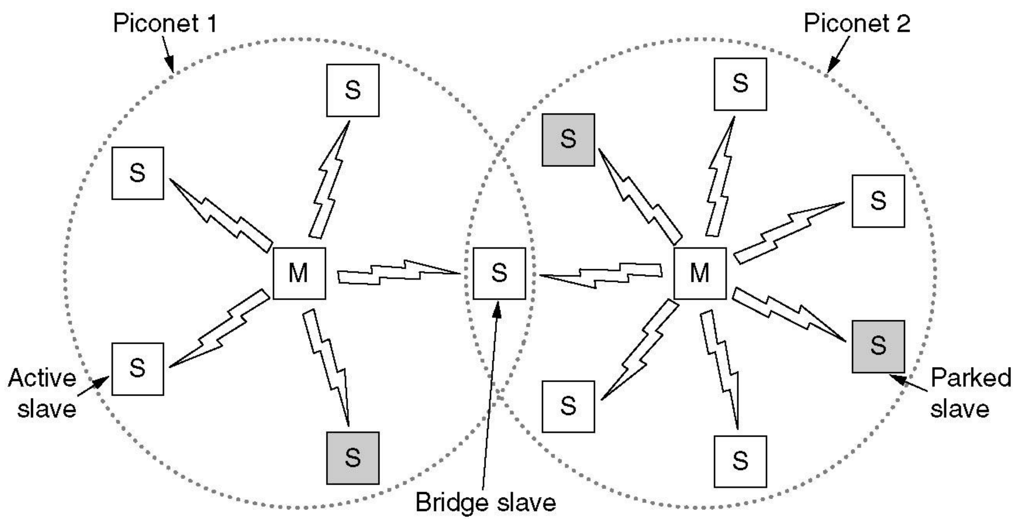
3. Splitting the channel into narrowband channels enables significant simplification of equalizer design in multipath environments.

4. Different modulation formats and data rates can be used on different subcarriers depending on the noise level of individual sub-bands (the symbol periods are kept the same). In serial transmission, certain types of noise (such as time varying tone interference) may cause an entire system to fail; the parallel OFDM system can avoid this problem by adaptively reducing the data rate of the affected sub-bands or dropping them.

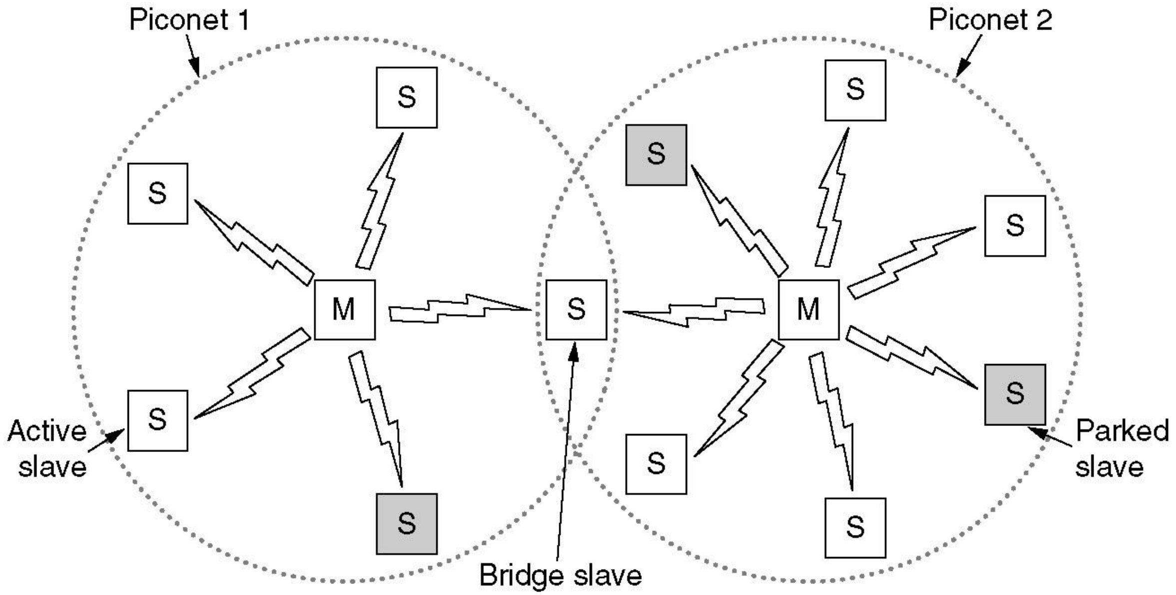
5. OFDM can be implemented digitally using an inverse discrete Fourier transform and discrete Fourier transform (IDFT/DFT) pair (via the efficient fast algorithm IFFT/FFT pair), which greatly reduces the system complexity.

Wireless Personal Area Networks (Bluetooth, Zigbee)

Bluetooth is a PAN under IEEE 802.15.1 standard, which operates over a short range, at low power, and at low cost for interconnecting notebooks, camera, DVD player, smartphones. It operates in the rage of 2.4 GHz to 2.483 GHz (unlicensed BW) and uses GFSK modulation.

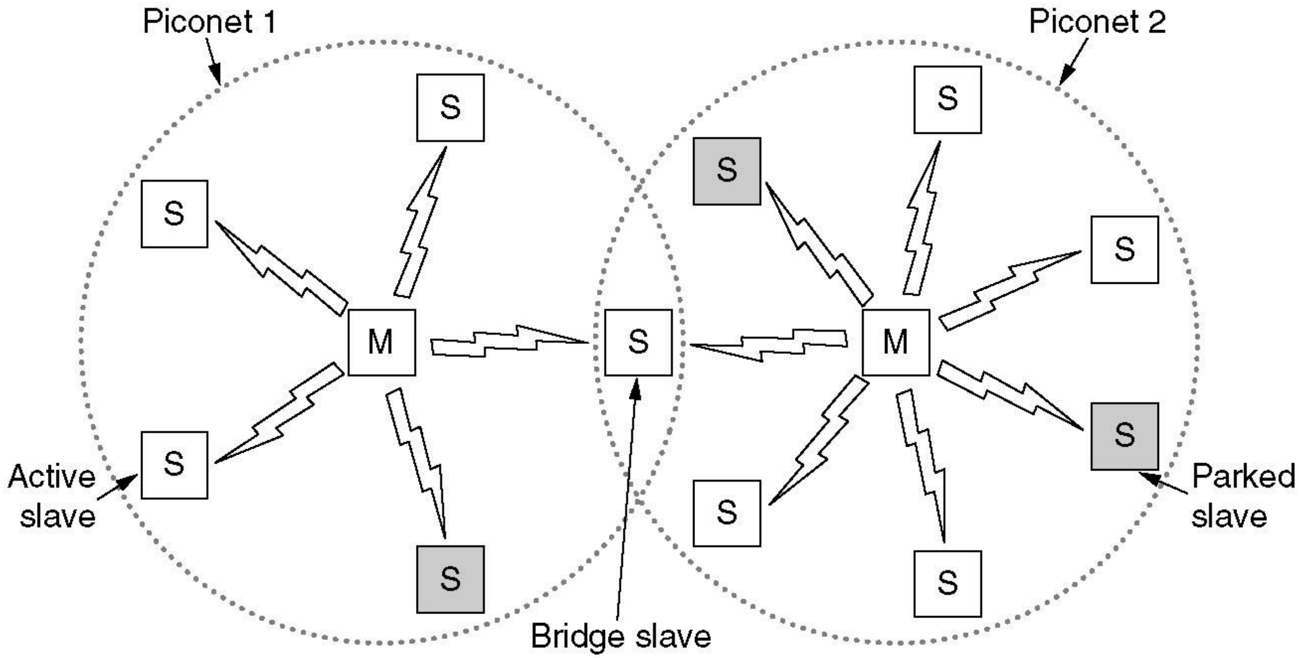


The basic Bluetooth network configuration, called a ***piconet***, consists of a master device and up to seven slave devices, as in Fig. above. Any communication is between the master and a slave; the slaves do not communicate directly with each other. A Bluetooth device has a built-in short range radio transmitter.



✔ All the secondary/slave stations synchronize their clocks and hopping sequence with the primary. Bluetooth uses frequency-hopping spread spectrum (FHSS) in the physical layer to avoid interference from other devices or network.

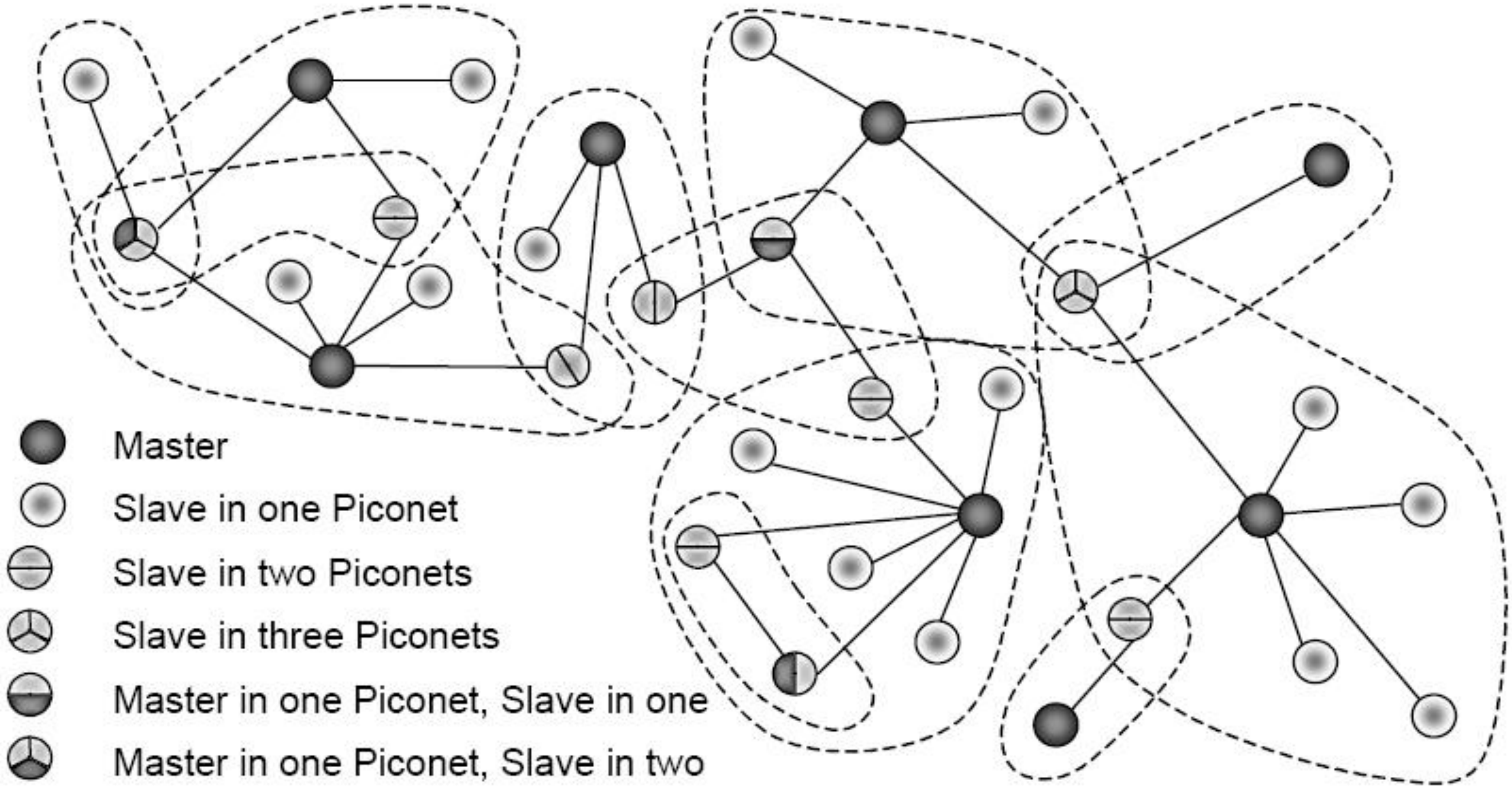
✔ Bluetooth defines two types of networks called: piconet and scatternet.



- Piconets can be combined to form **scatternet** where a secondary user of one piconet acts as bridge to another piconet. The bridge secondary/slave acts as a primary in receiving packets from the original primary of first piconet then deliver the packet to secondaries of the second piconet.

- Although a piconet can have maximum 7 slaves/secondary , additional secondaries can be in **parked state**. A secondary in parked state is synchronized with the primary, but can not take part in communication until it is removed from parked state to the active state.

✔No central network structure: "Ad-hoc" network.



Two types of links can be created between primary and secondary:

A **synchronous connection-oriented (SCO)** link is used when avoiding latency (delay in data delivery) is more important than integrity (error free delivery) for example real time audio. A **physical link** is established between the **primary (master)** and **secondary (slave)** by **reserving specific time slots** at regular intervals.

The **basic unit of connection** is **two slots**:

- ✔ One for transmission from **primary to secondary**,
- ✔ One for transmission from **secondary to primary**.
- ✔ If a packet is damaged it is never retransmitted.

- ✔ An **asynchronous connectionless link (ACL)** is used when data integrity is more important than avoiding latency.
- ✔ Example: **File transfers, control messages**.
- ✔ **No fixed slots reserved**; the link is dynamically established as needed.
- ✔ **Retransmissions:** If the payload is lost or corrupted, it is **retransmitted** until correctly received.
- ✔ Suitable for **non-real-time data** where slight delay is acceptable.

Layers of Bluetooth

Application Layer
L2CAL layer
Baseband layer
Radio Layer

- Radio layer is like physical layer of Internet. Uses FHSS, GFSK modulation.
- Baseband layer is like MAC sublayer uses TDMA slot as the physical channel.
- Logical Link Control and Adaption Protocol (L2CAP) is like LLC sublayer.

Zigbee

- ✔ A second personal area network standardized by the IEEE is the 802.15.4 known as Zigbee. Zigbee is targeted at lower powered (to give very long battery life), lower-data-rate, lower-duty-cycle applications than Bluetooth.
- ✔ While we may tend to think that "bigger and faster is better," not all network applications need high bandwidth and the consequent higher costs (both economic and power costs).
- ✔ For example, home temperature and light sensors, security devices (alarm), and wall-mounted switches are all very simple, low-power, low-duty-cycle, low-cost devices. Zigbee is thus well-suited for these devices.
- ✔ Home RF is used to interconnect the various home electronic devices such as, desktops, laptops, telephone, notebooks, cameras, printers, lights and fans etc. Zigbee can be used as home automated system (Home RF System).